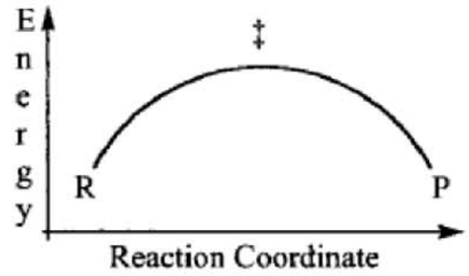
CML103 Lect. slides

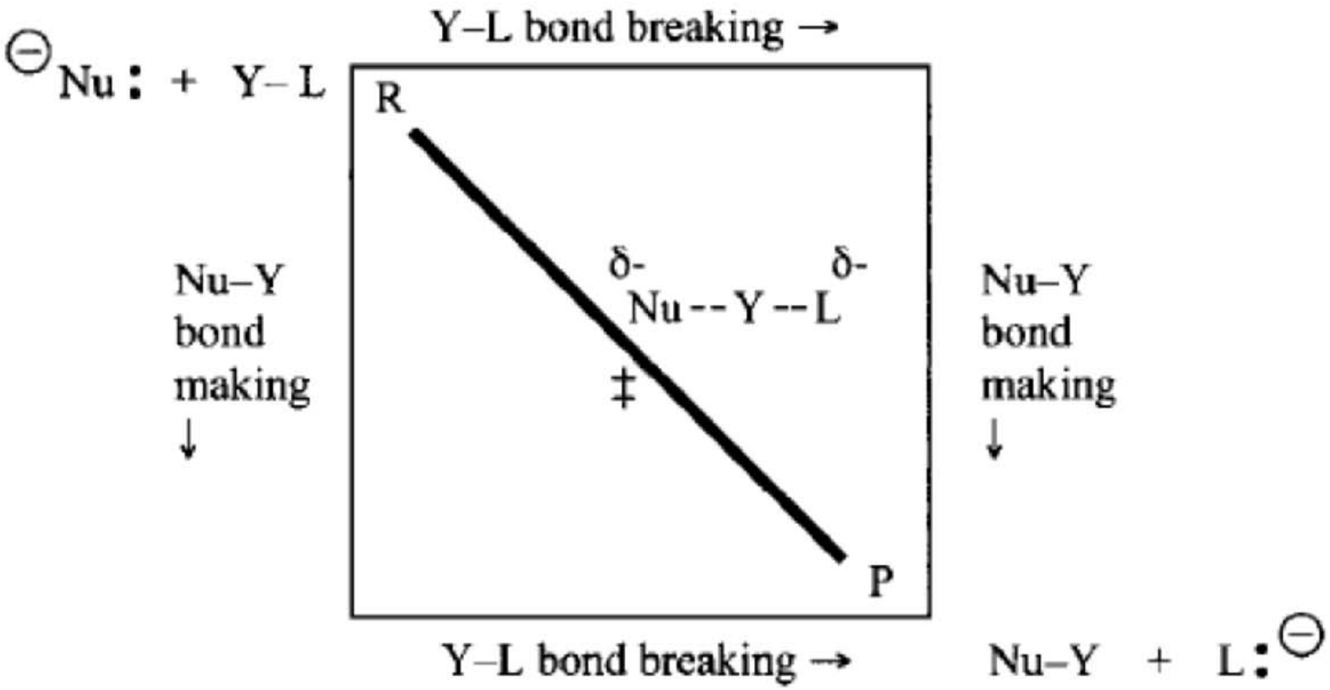
Surface Chemistry (Indian Institute of Technology Delhi)

A simplified energy surface:

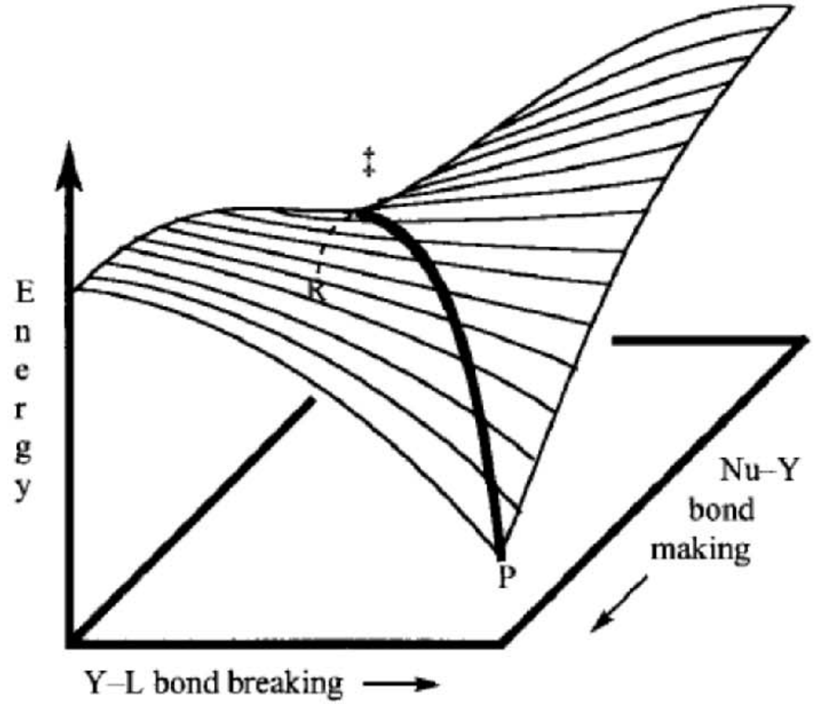
CHEMICAL SPACE

$^{\ominus}$Nu: Y–L ⟶ Nu–Y + L:$^{\ominus}$



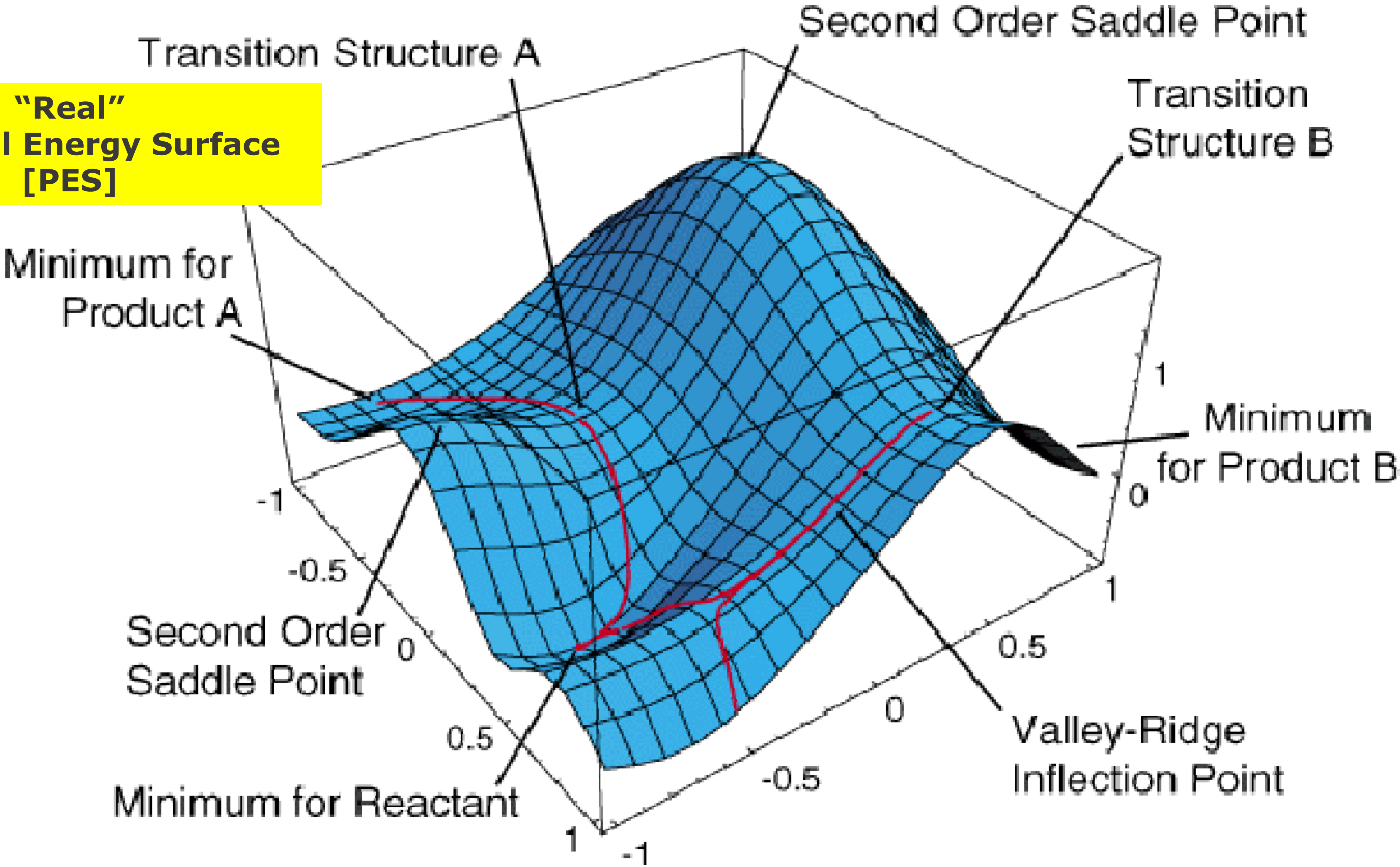


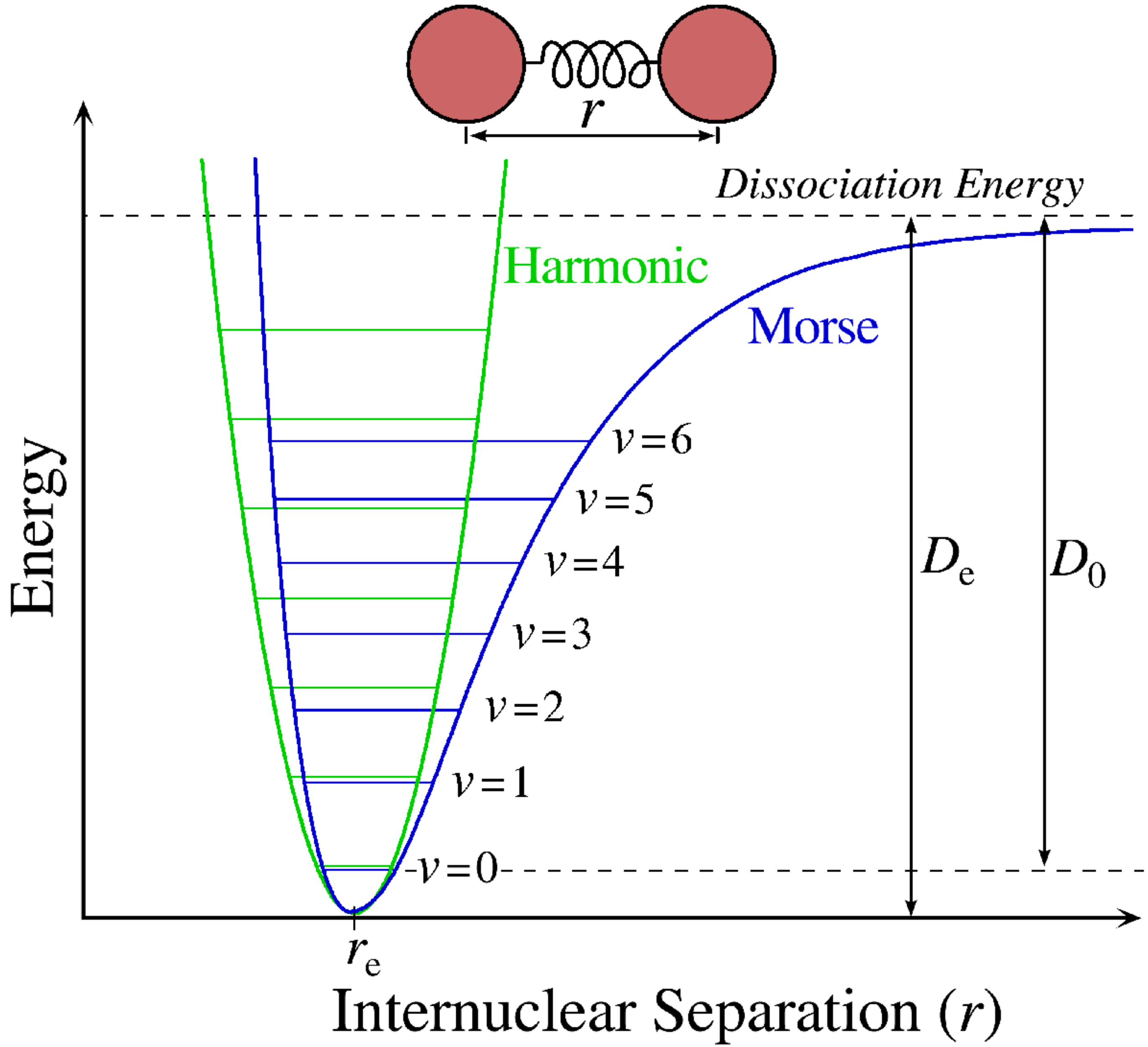
viewed from the top



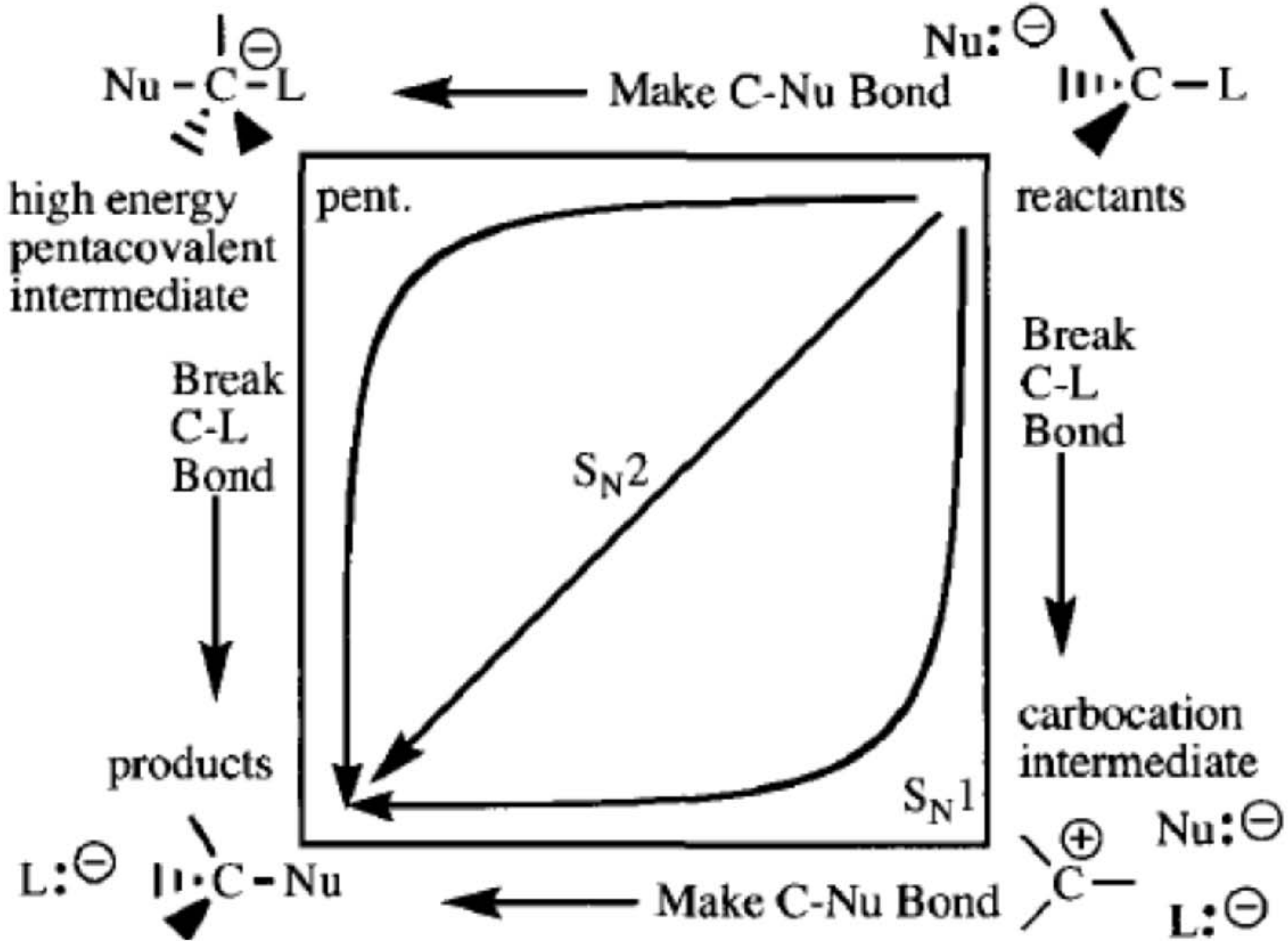
viewed from the side

“Real”
Potential Energy Surface
[PES]



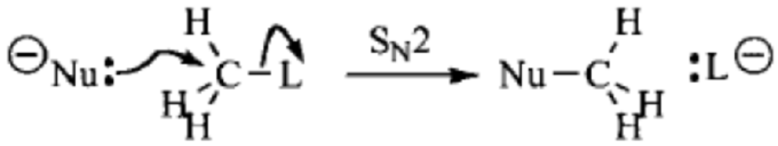

r
Dissociation Energy
Harmonic
Morse
v=6
v=5
v=4
v=3
v=2
v=1
v=0
D_e
D_0
Energy
r_e
Internuclear Separation (r)

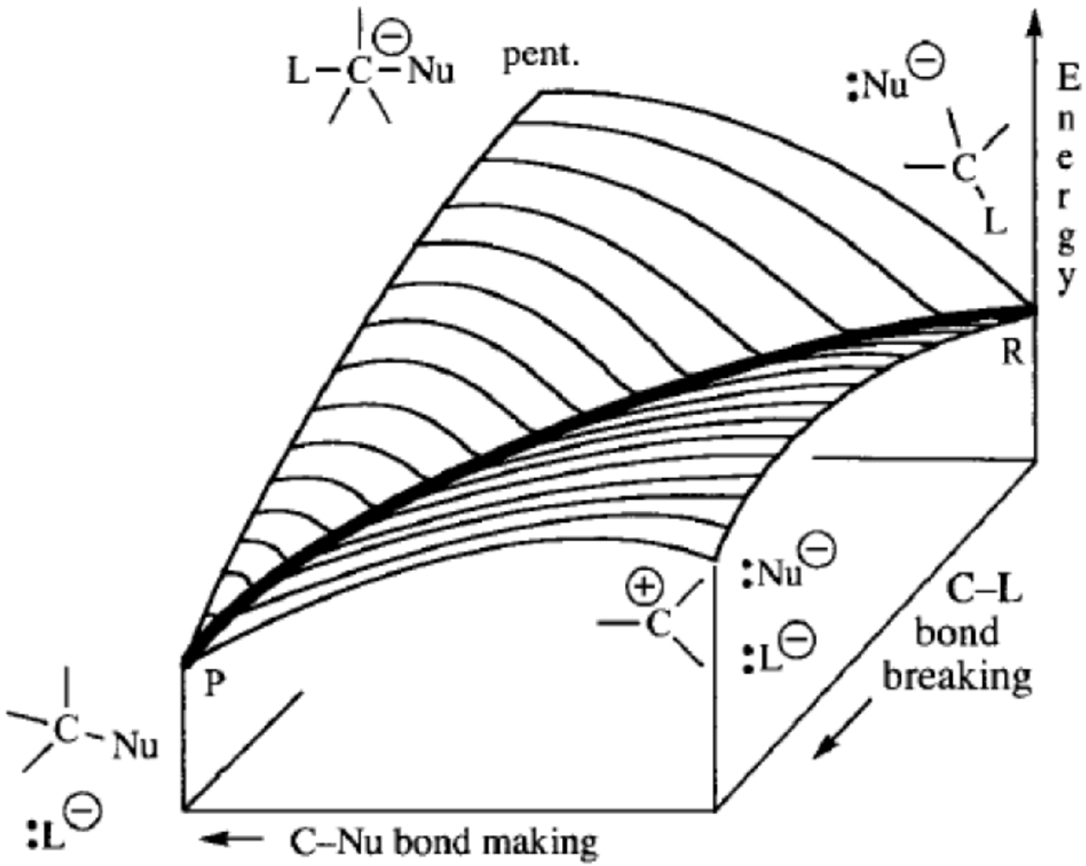
simplified energy surface for substitution at a tetrahedral center

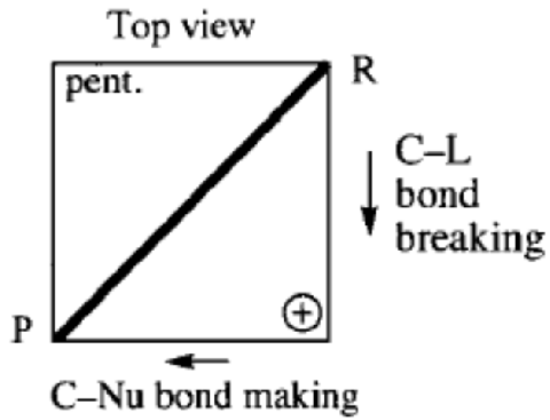


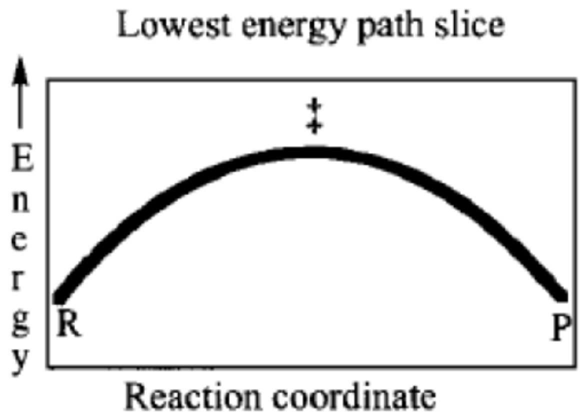
viewed from the top

The S_N2 energy surface; C–L bond breaking and C–Nu bond making are concerted

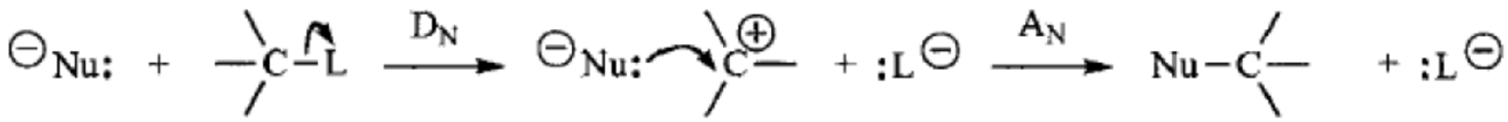


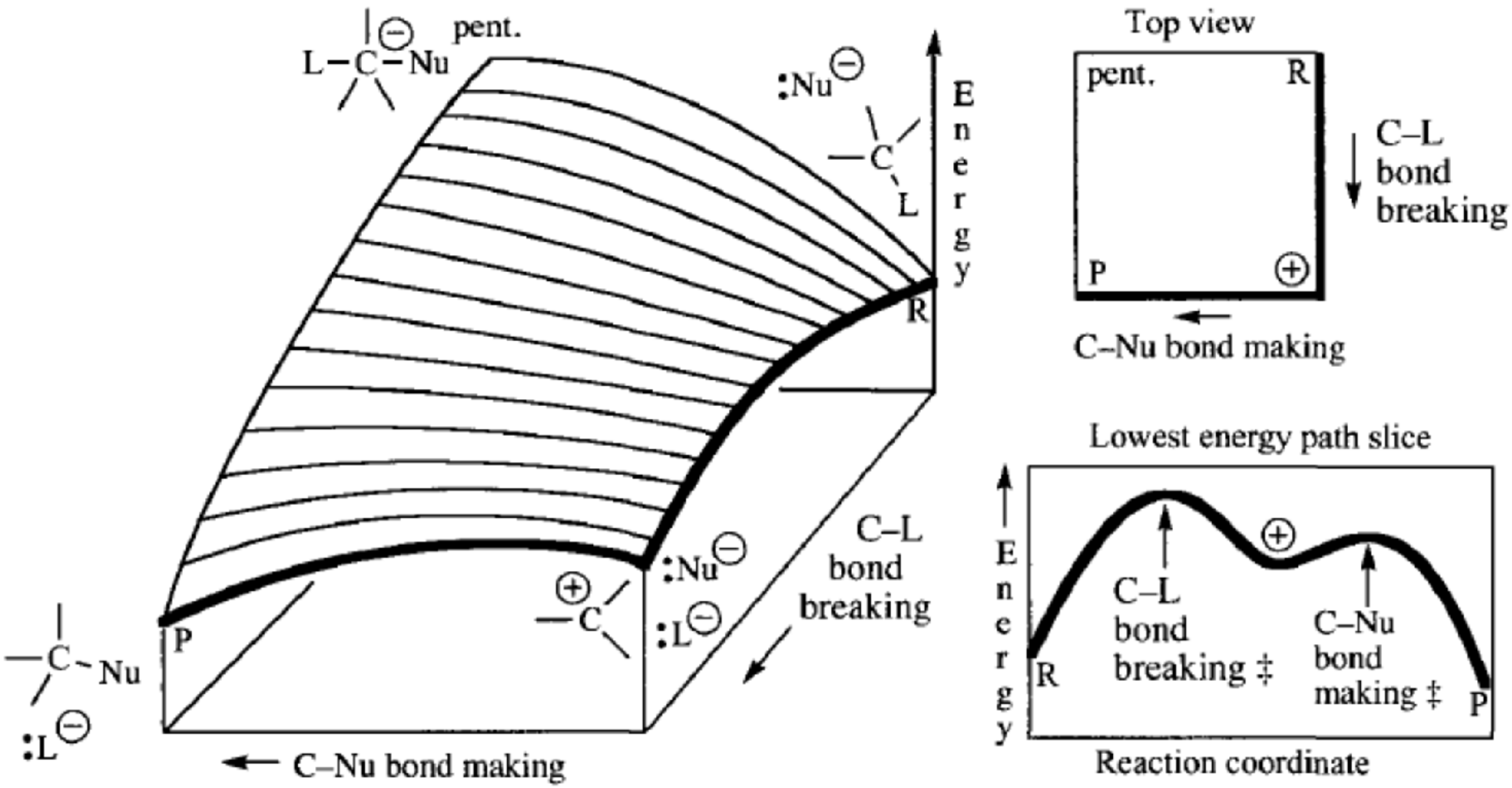




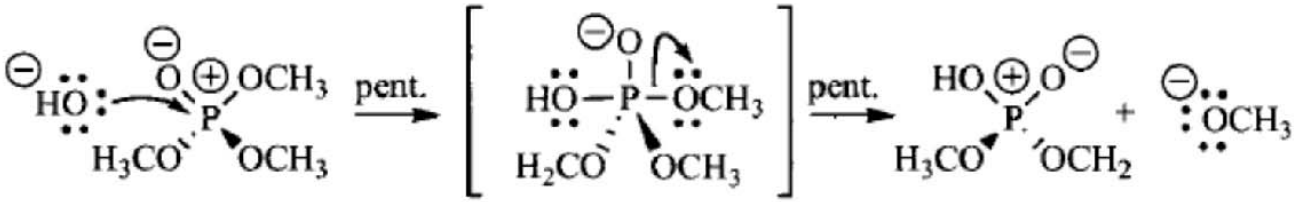


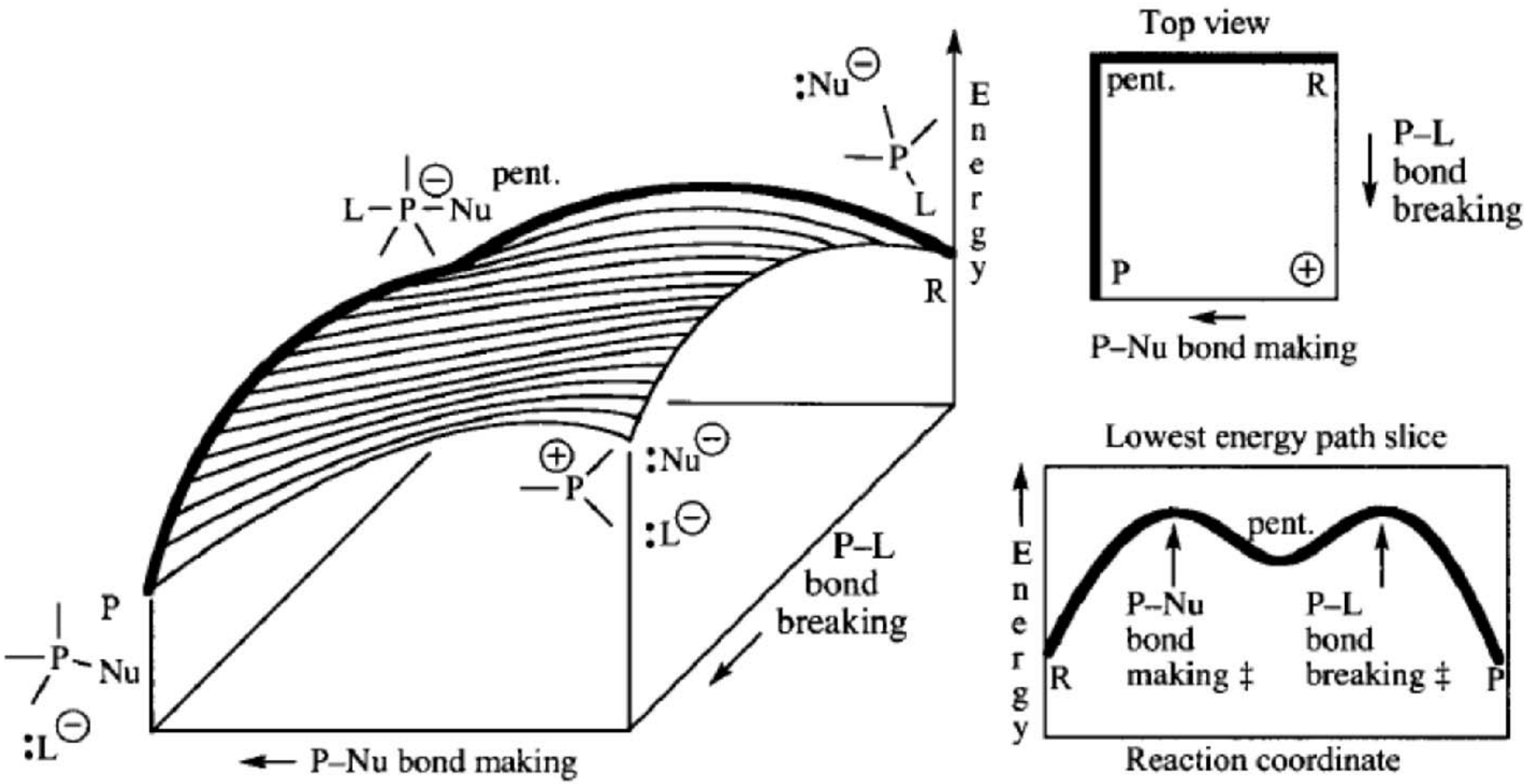
The S_N1 energy surface. The C–L bond breaking occurs first to form a carbocation, and then the C–Nu bond is made.



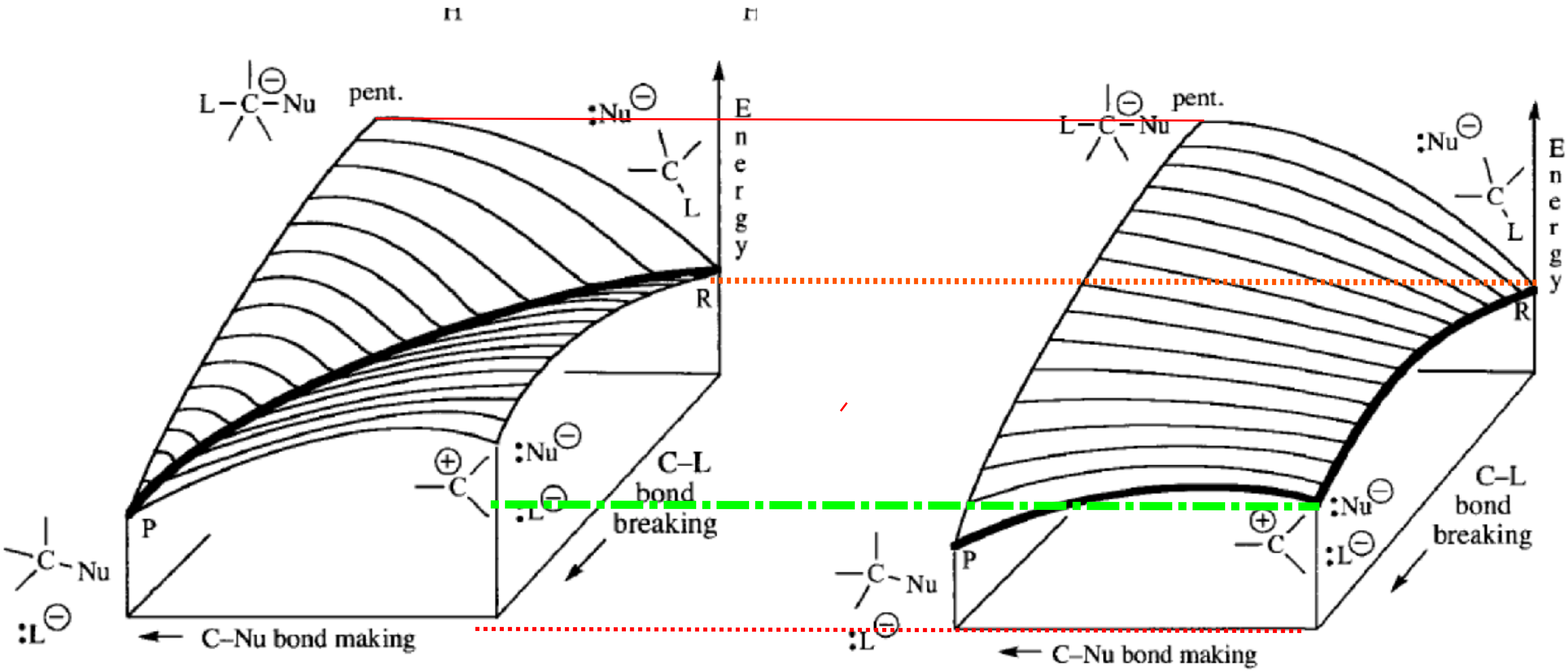


Substitution via a Pentacovalent Intermediate





The energy surface for substitution via a pentacovalent intermediate.



Comparison of S_N2 and S_N1 PES's

THE UNIVERSE OF CHEMICAL TRANSFORMATIONS

(Different "TYPES" of reactions)

(***Changes*** to the ***valence*** of the atom ***where BB/BM is taking place***)

1) **SUBSTITUTION:** One valence group is ***replaced*** by another.
Original valence unchanged, often MW changes

2) **ADDITION/ELIMINATION:** Groups are added of removed.
Original valence changes, MW change certain.

3) **REARRANGEMENTS:** ***Original valence unchanged!***
MW may or may not change.

4) **REDOX:** Most of the time, ***original valence changes***.
There are exceptions.

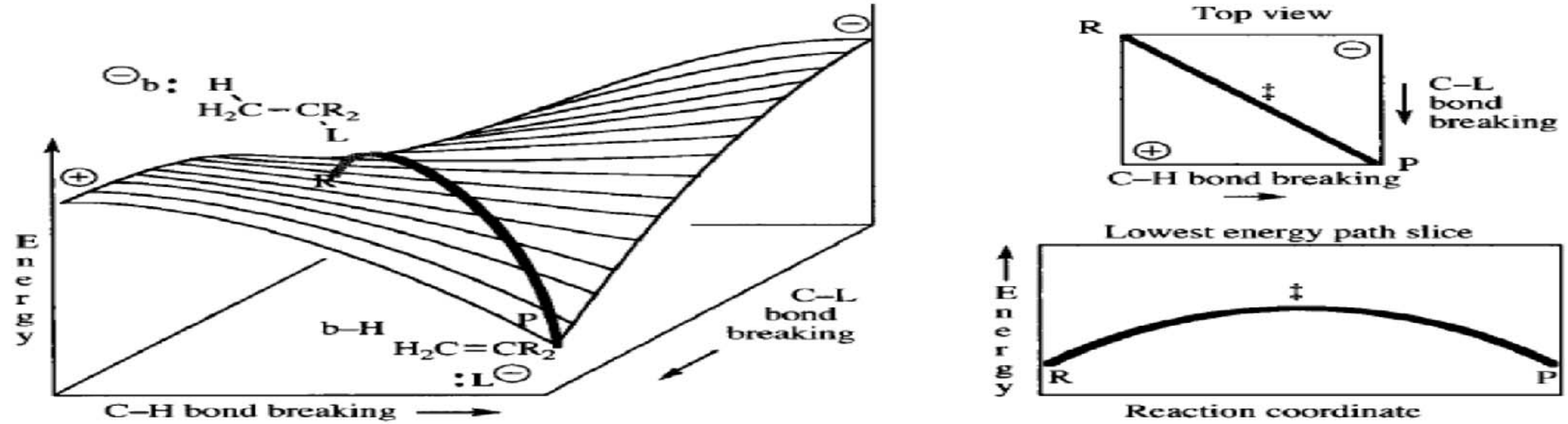


Figure 4.23 The E2 energy surface.

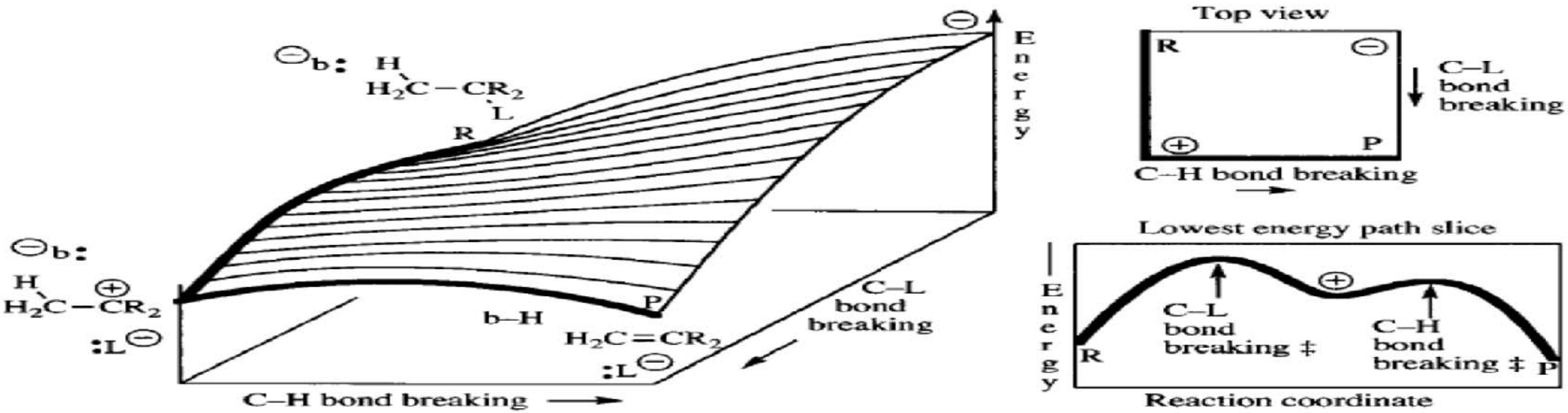
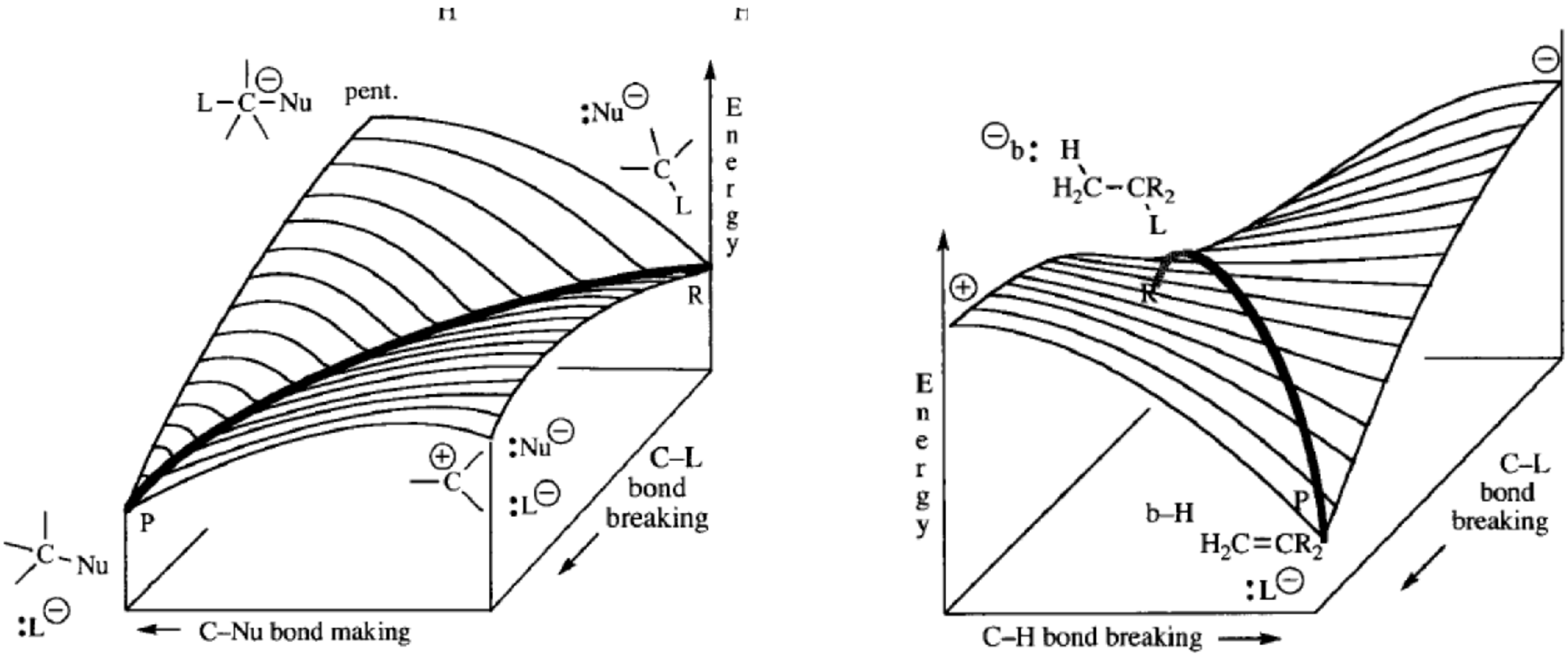
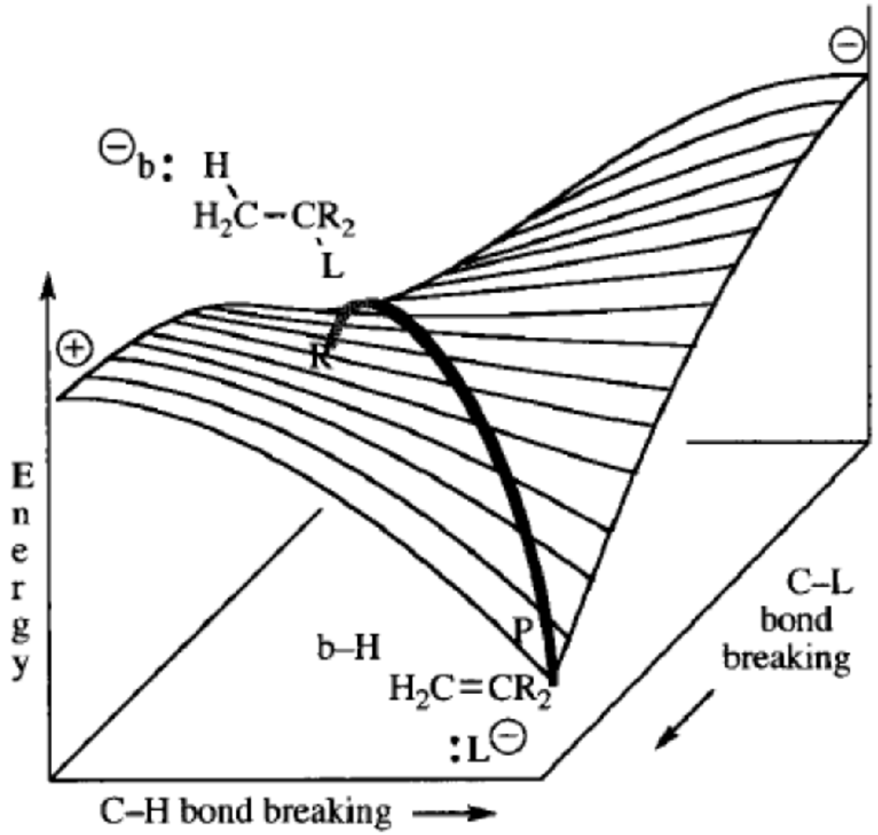


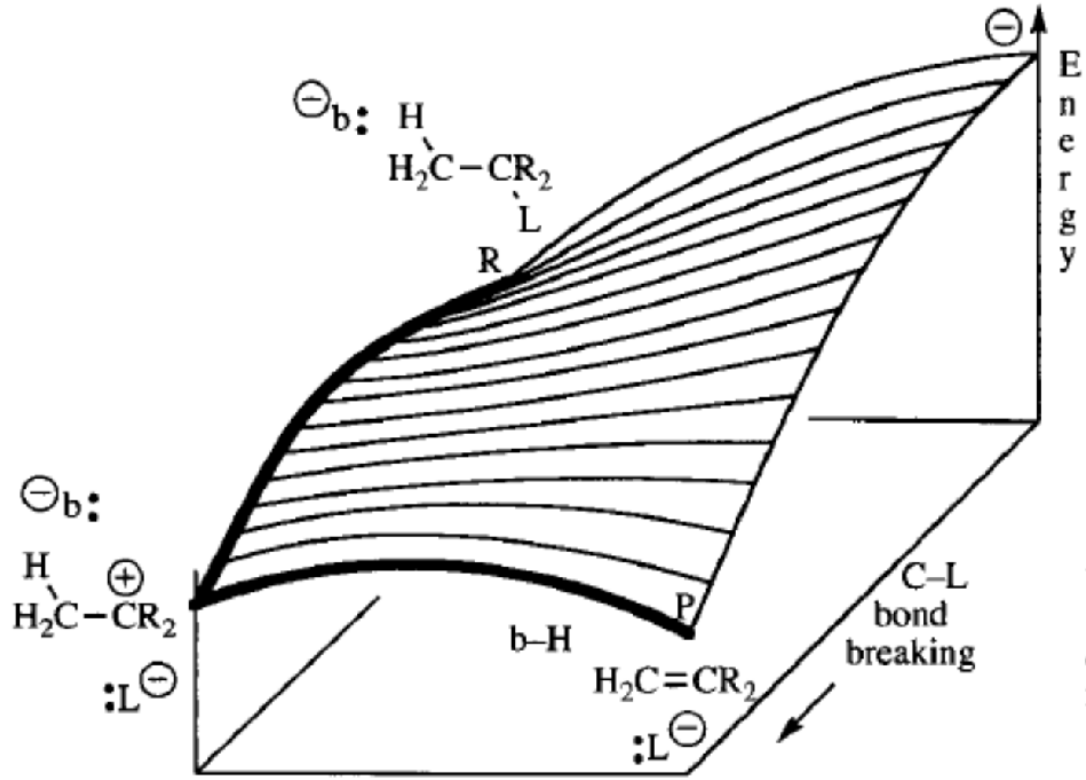
Figure 4.24 The E1 energy surface.

Comparison of E2 and E1 PES's

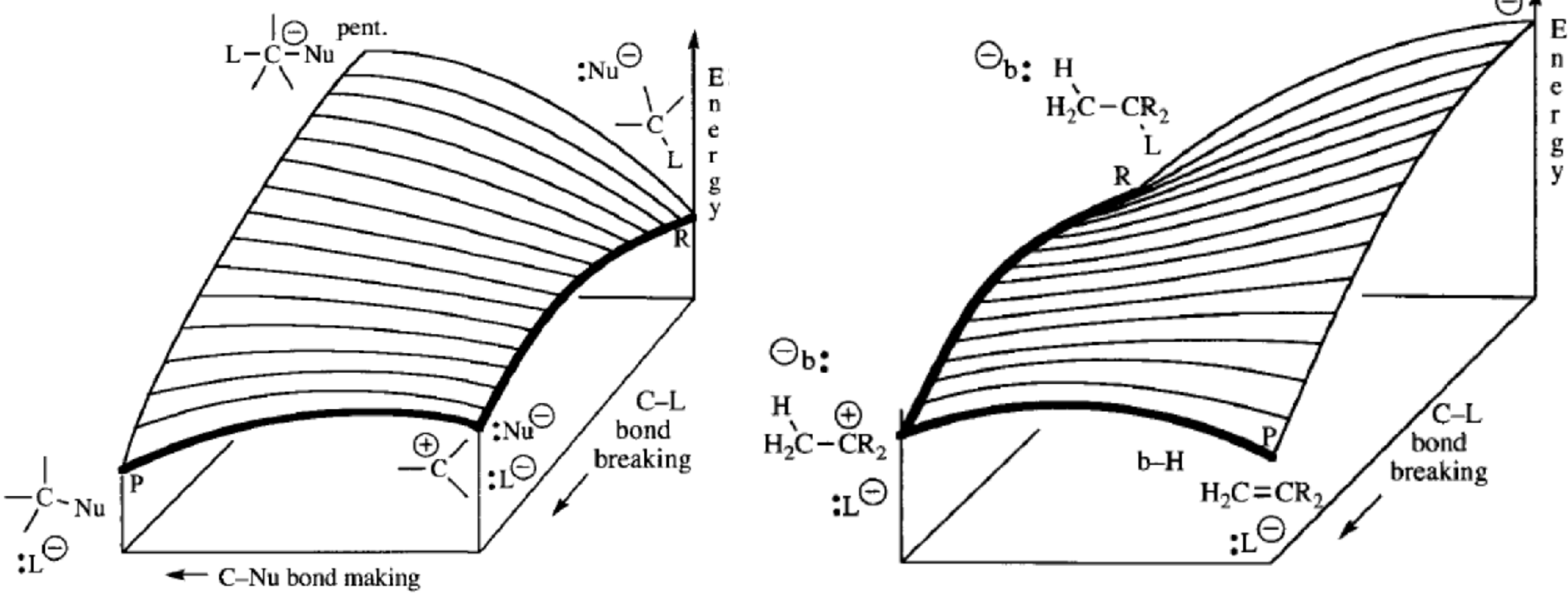


Comparison of S_N2 and E2 PES's





Comparison of E2 and E1 PES’s



Comparison of S_N1 and E1 PES's

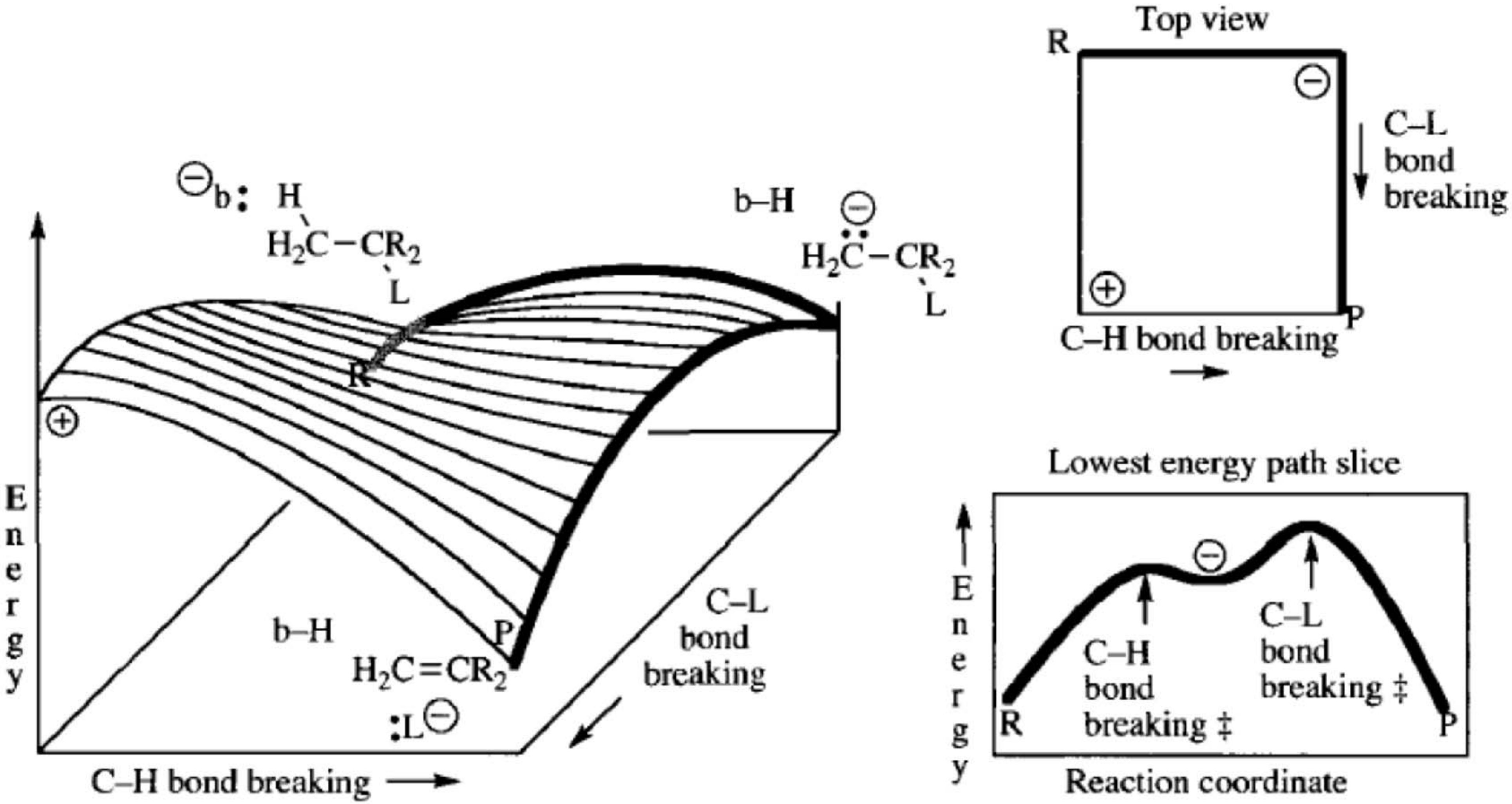


Figure 4.25 The E1cB energy surface.

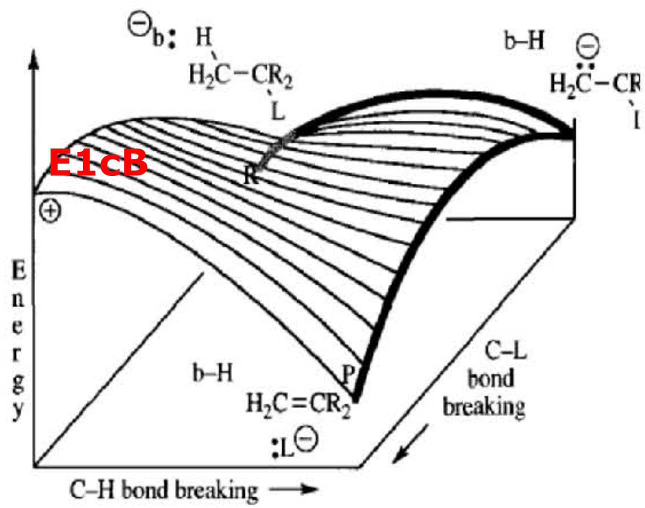
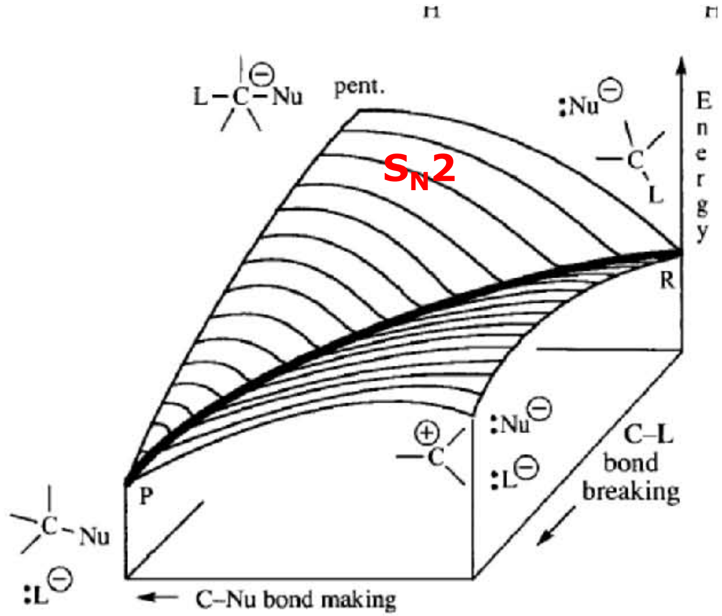
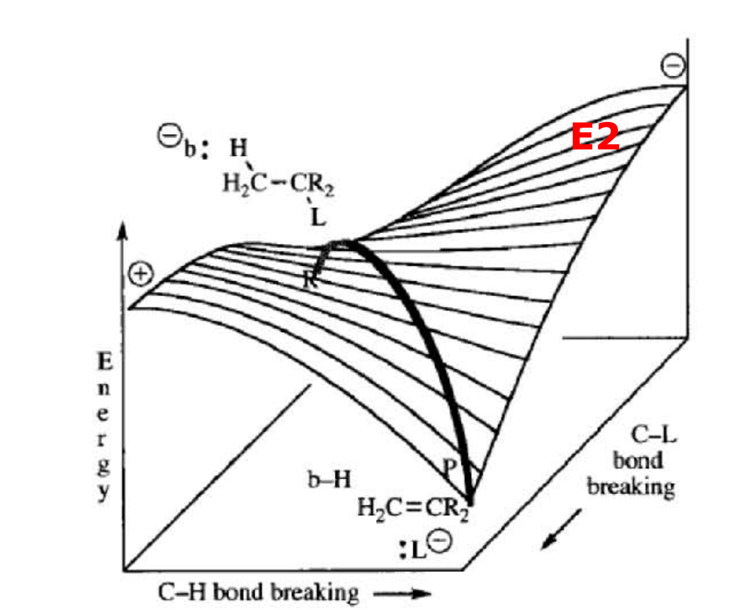
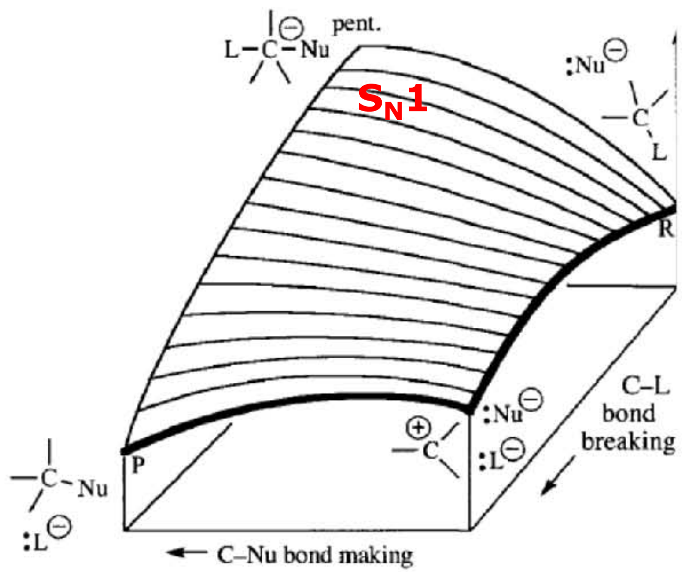
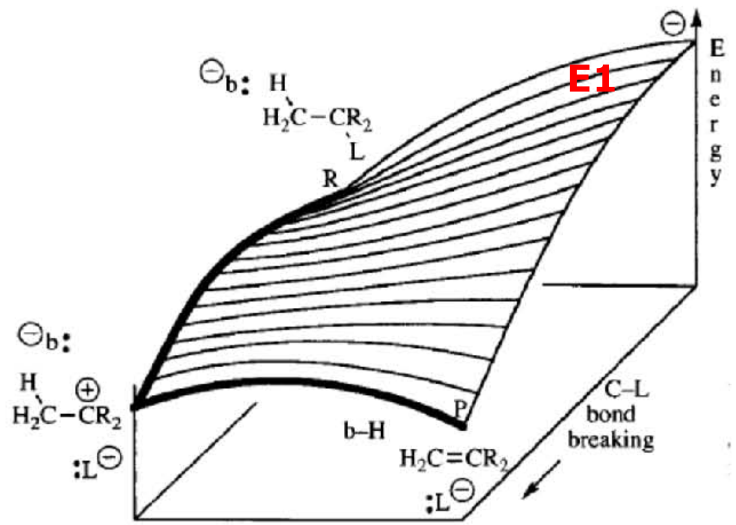


Figure 4.25 The E1cB energy surface.









Where to start?

To stabilize the TS$^{\#}$, FIRST, we NEED to know what is 'happening' in the TS$^{\#}$.

'Happening' = Is electron density increasing or decreasing?

The Hammett Relationship

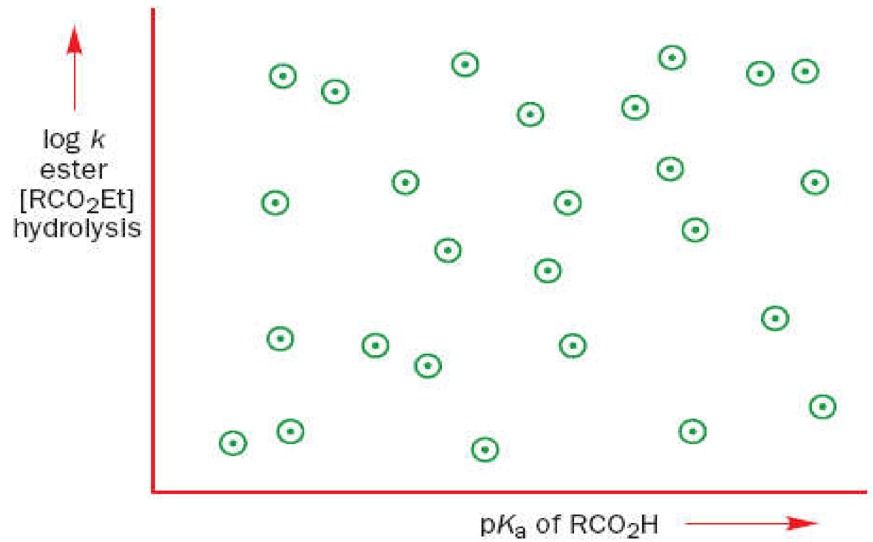
Gives an idea of what the transition state is "*really*" like

Hammett took the arbitrary [NOT REALLY] decision to use the pK_a of an acid as a guide.

For example, the rate of hydrolysis of esters might well correlate with the pK_a of the corresponding acid.

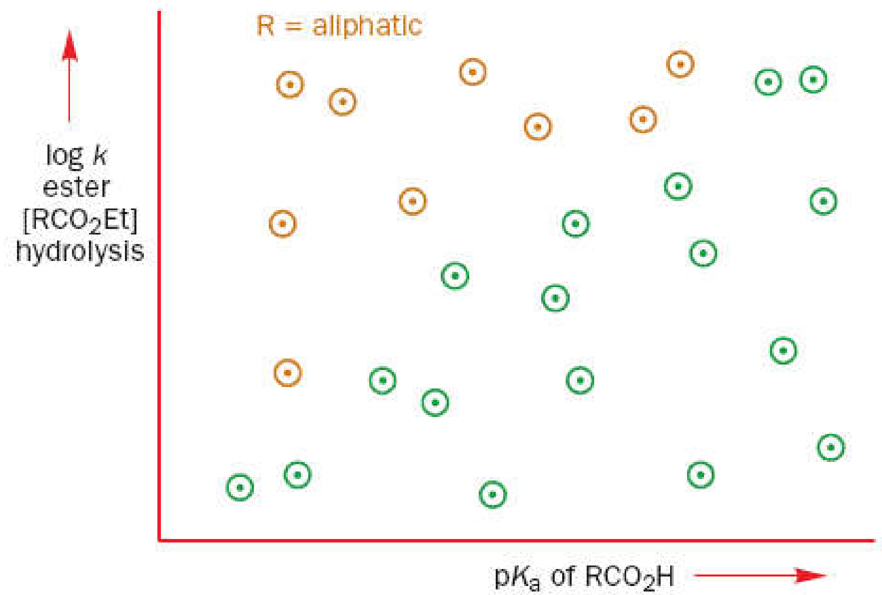
Plotted the rates of ethyl ester hydrolyses (as log k since pKa has a log scale) against
the pK_as of the corresponding acids (both aliphatic and aromatic)

Results NOT very encouraging.



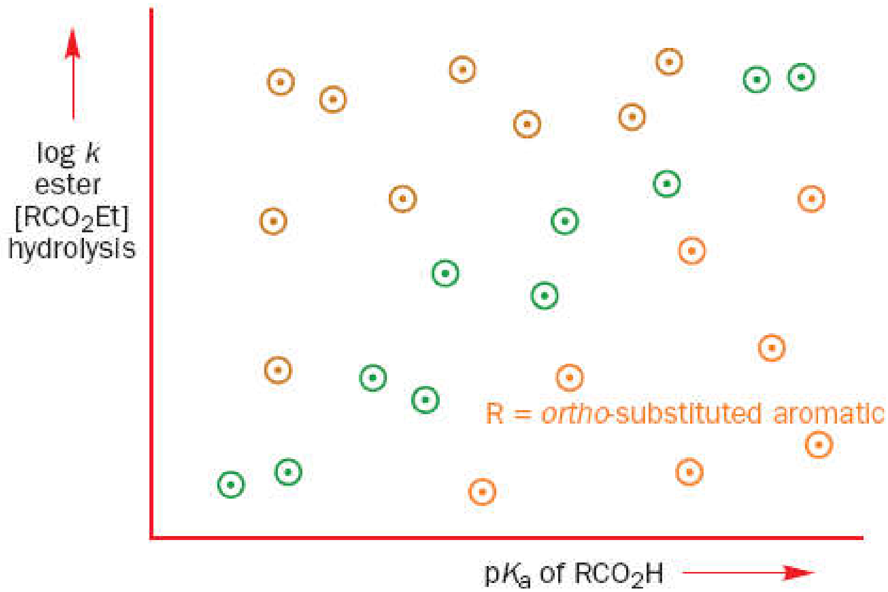
A CLOSER LOOK:

Points towards the top of the graph belonged to the substituted acetic acids



EVEN MORE CLOSER LOOK:

Ortho- substituted esters reacted more slowly than their *meta*- and *para*-isomers and came towards the bottom of the graph (orange points)

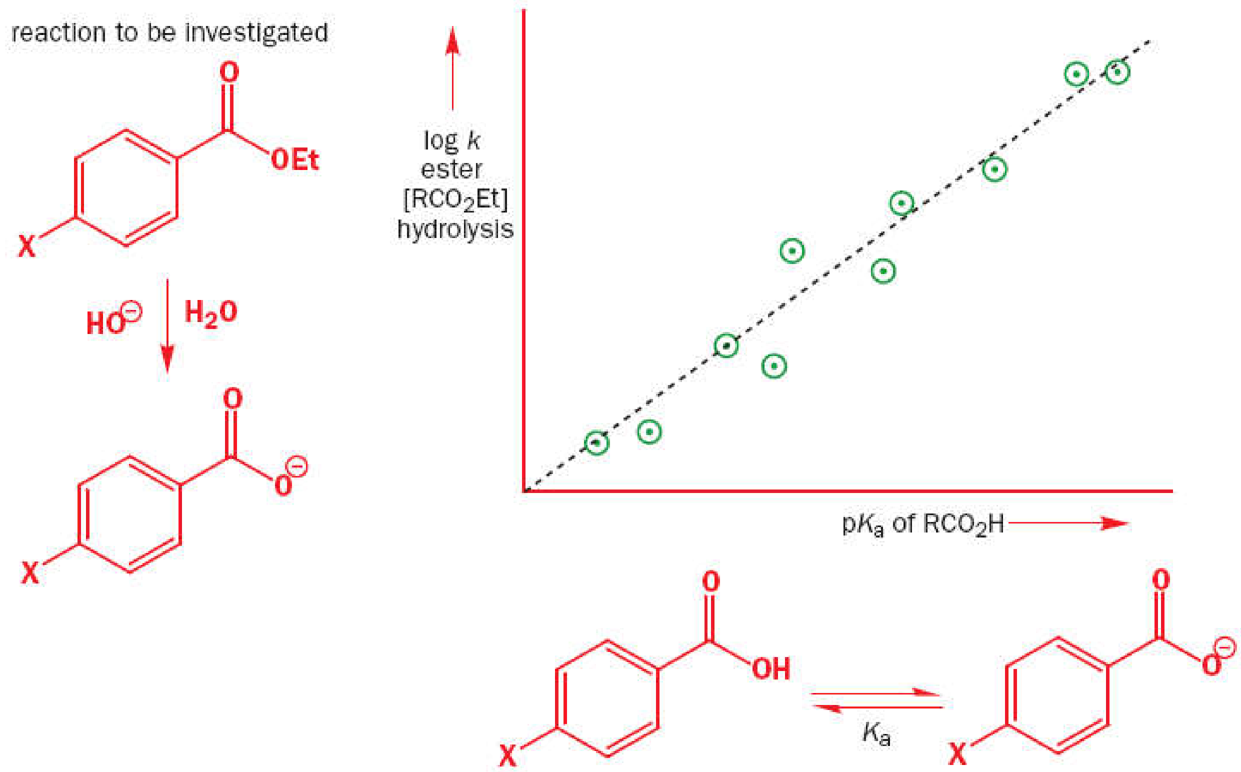


Not a perfect correlation

Hammett had removed the examples where steric hindrance was important ***in the reaction being studied***.

What he had discovered was that in a subset of ALL esterification reactions Substituents had a SIMILAR effect to what they had in the ionization of benzoic acid

For the *para* substituents only:



Notice that the straight line is not perfect.
This graph is an invention of the human mind.
It is a correlation between things that are not directly related.

If you determine a rate constant by plotting the right function of concentration against time and get an imperfect straight line, that is your fault because you haven't done your measurements carefully enough.

If you make a Hammett plot and the points are not on a straight line (**and they won't be**) then that is *not* your fault. The points really don't fit on a perfectly straight line.

The Hammett substituent constant σ

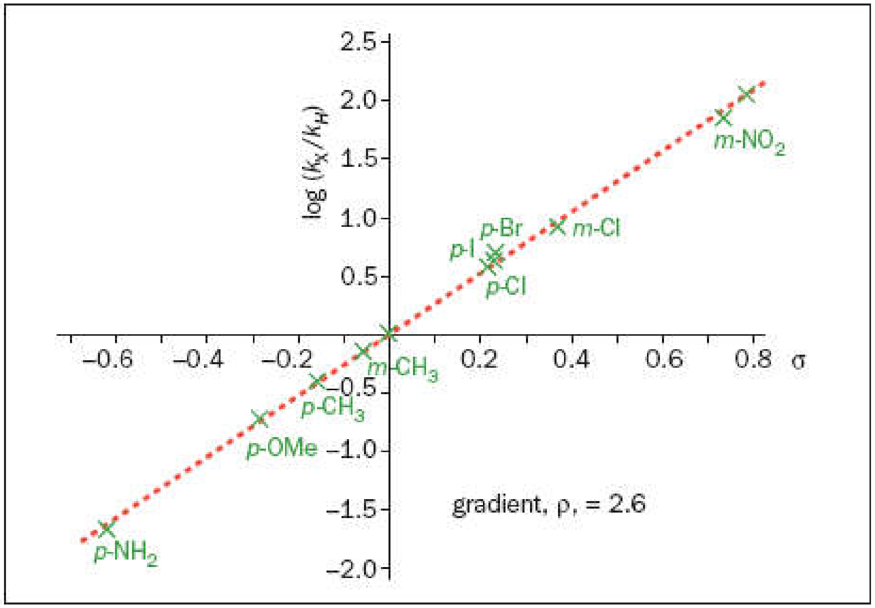
Substituent, X	pK_a of p-XC_6H_4COOH	pK_a of m-XC_6H_4COOH
NH_2	4.82	4.20
OCH_3	4.49	4.09
CH_3	4.37	4.26
H	4.20	4.20
F	4.15	3.86
I	3.97	3.85
Cl	3.98	3.83
Br	3.97	3.80
CO_2CH_3	3.75	3.87
$COCH_3$	3.71	3.83
CN	3.53	3.58
NO_2	3.43	3.47

Substituent, X	σ_p	σ_m	Comments
NH_2	–0.62	0.00	groups that donate electrons have negative σ
OCH_3	–0.29	0.11	
CH_3	–0.17	–0.06	
H	0.00	0.00	there are no values for *ortho* substituents
F	0.05	0.34	
I	0.23	0.35	
Cl	0.22	0.37	$\sigma_p < \sigma_m$ for inductive withdrawal
Br	0.23	0.40	
CO_2CH_3	0.45	0.33	
$COCH_3$	0.49	0.37	$\sigma_p > \sigma_m$ for conjugating substituents
CN	0.67	0.62	
NO_2	0.77	0.73	groups that withdraw electrons have positive σ

$$\sigma_X = \log\left(\frac{K_a(X\text{–}C_6H_4COOH)}{K_a(C_6H_5COOH)}\right) = pK_a(C_6H_5COOH) - pK_a(X\text{–}C_6H_4COOH)$$

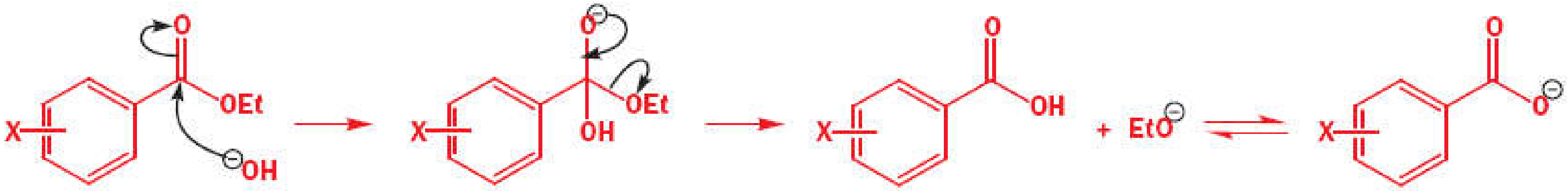
MEANING OF σ : When **+ve**= **e$^-$** withdrawing ; When **–ve** = **e$^-$** donating

The Hammett reaction constant ρ



There is a good correlation between how fast the reaction goes and the value of σ; in other words, the points lie more or less on a straight line. The gradient of this best fit line, given the symbol ρ (rho), tells us how sensitive the reaction is to substituent effects in comparison with the ionization of benzoic acids. The gradient is ρ = +2.6.

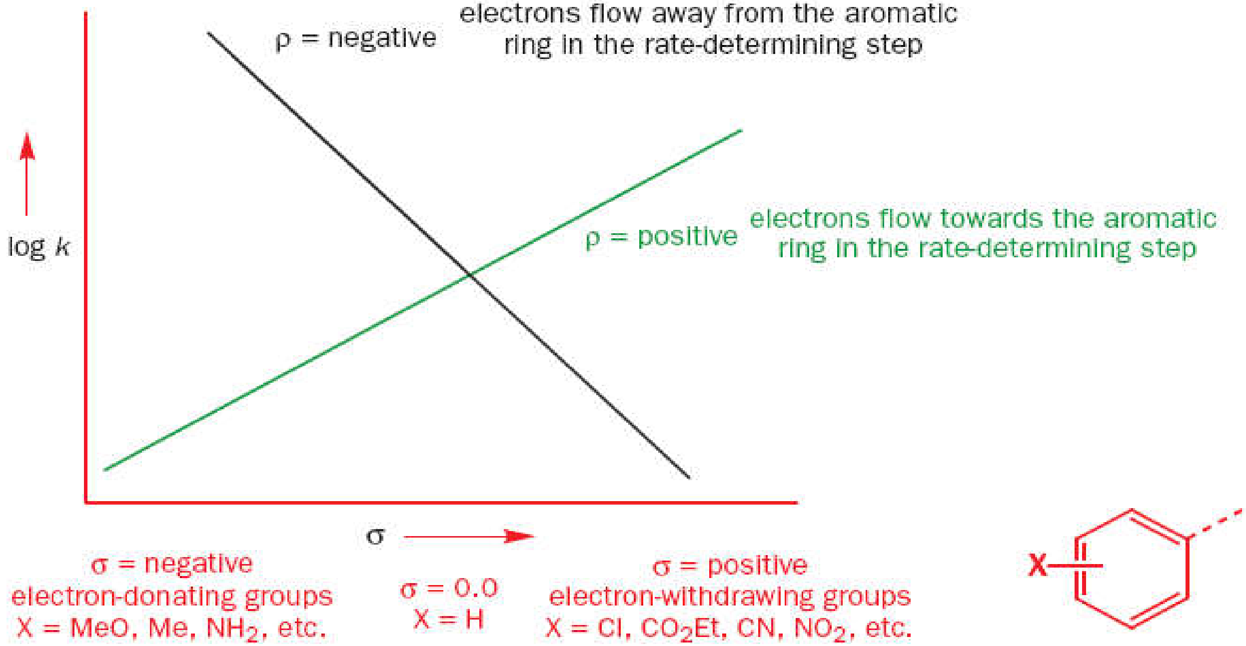
This tells us that the reaction responds to substituent effects in the same way (because it is +) as the ionization of benzoic acids but by much more (101.6 times more) because it is 2.6 instead of 1.0.



ρ measures the sensitivity of the reaction to electronic effects.

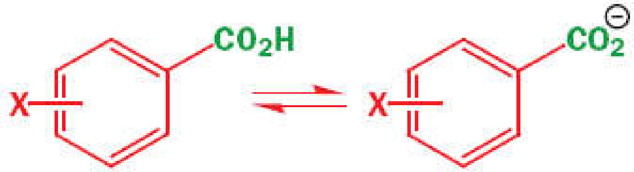
- **A *positive* ρ value means *more* electrons in the transition state than in the starting material**
- **A *negative* ρ value means *fewer* electrons in the transition state than in the starting material**

TYPICAL HAMMETT PLOTS

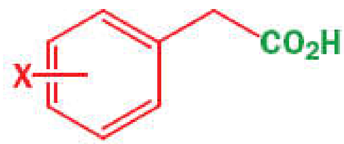


Equilibria with positive Hammett ρ values

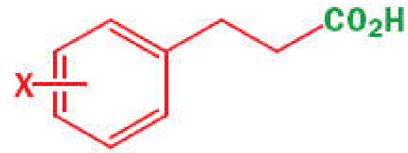
Positive ρ values mean electrons flowing towards the ring in the *rds*.



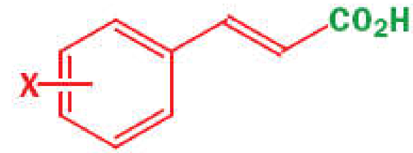
$\rho = 1.0$ (by definition)



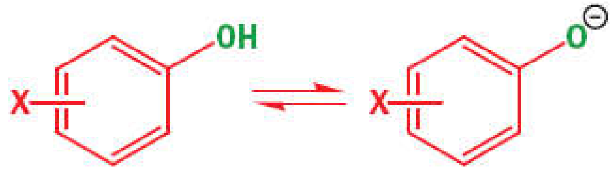
$\rho = 0.5$



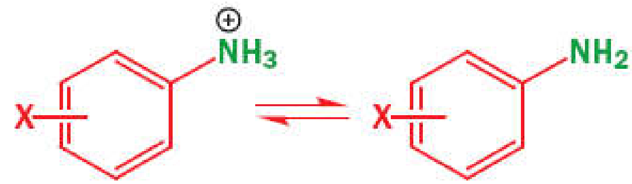
$\rho = 0.2$



$\rho = 0.5$

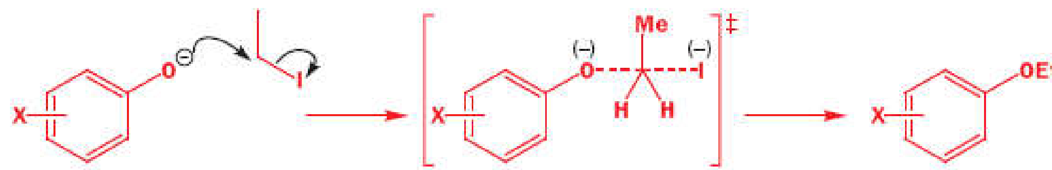


$\rho = +2.3$



$\rho = +3.2$

Reactions with negative Hammett ρ values

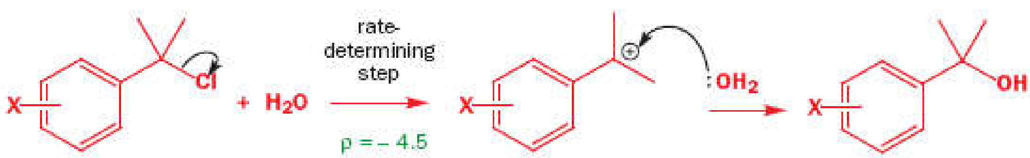


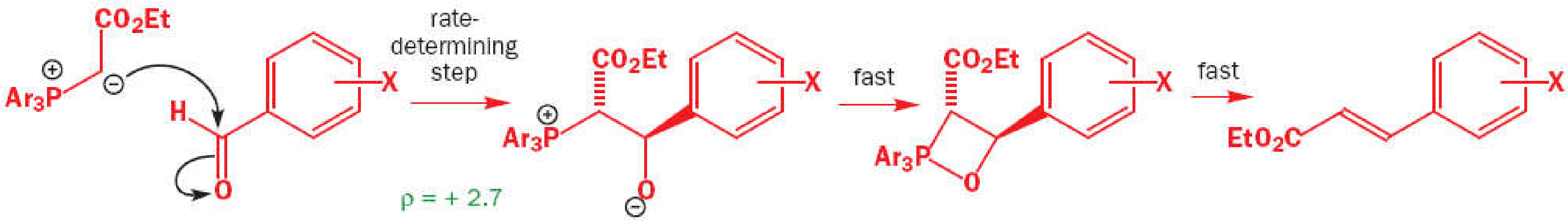
full negative charge delocalized round ring

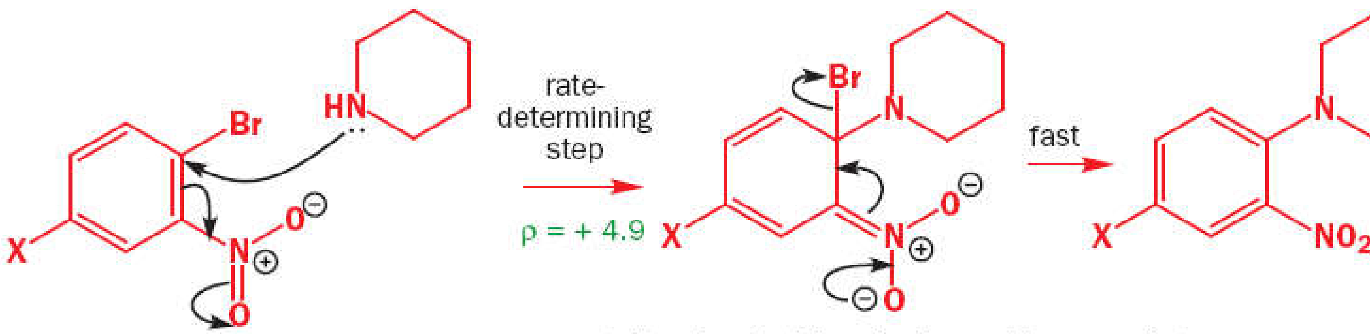
partial negative charge delocalized round ring

$\rho = -1$

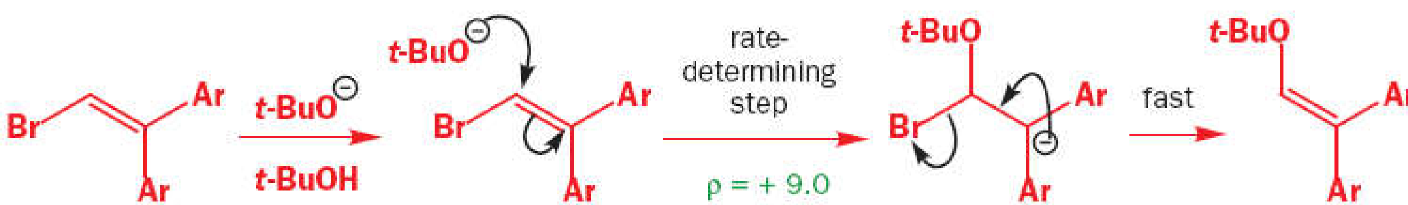
Negative ρ values mean electrons flowing away from the ring in the *rds*.

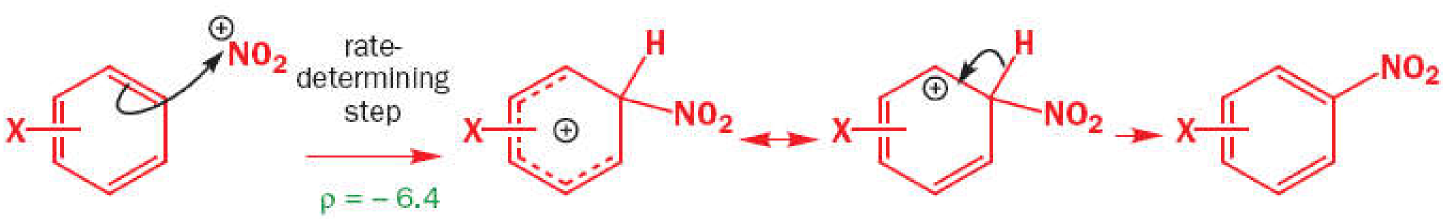




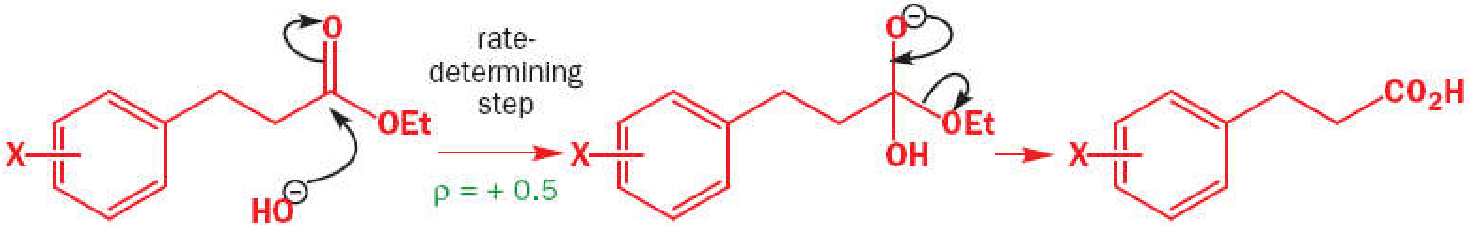


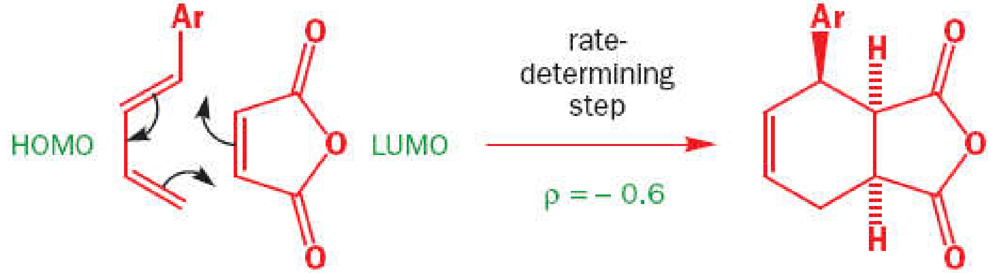
negative charge delocalized round benzene ring

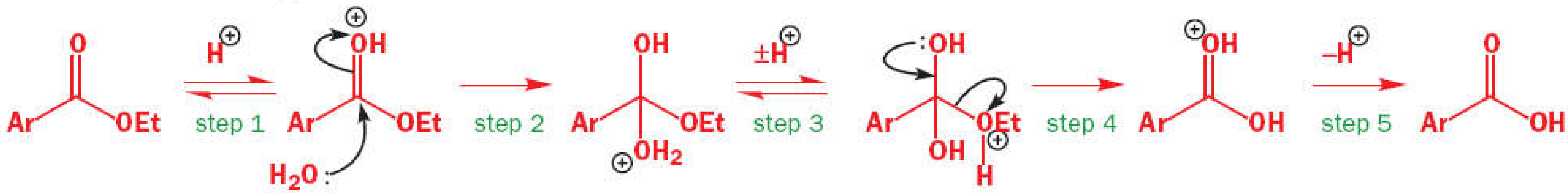




Small ρ values:





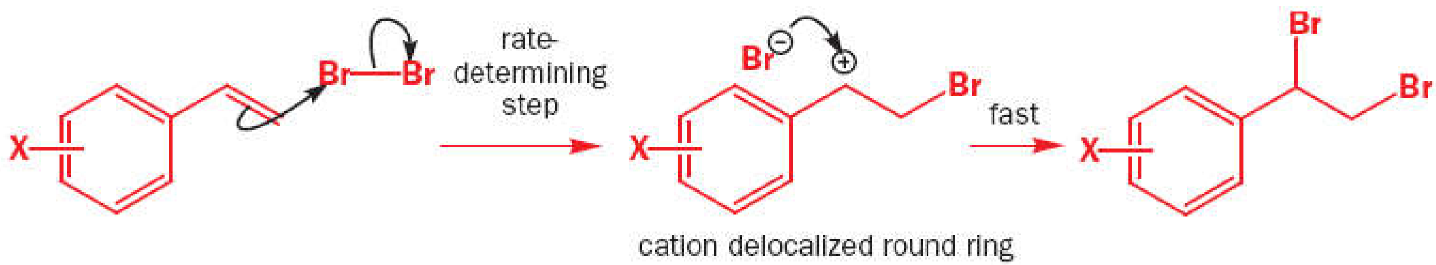


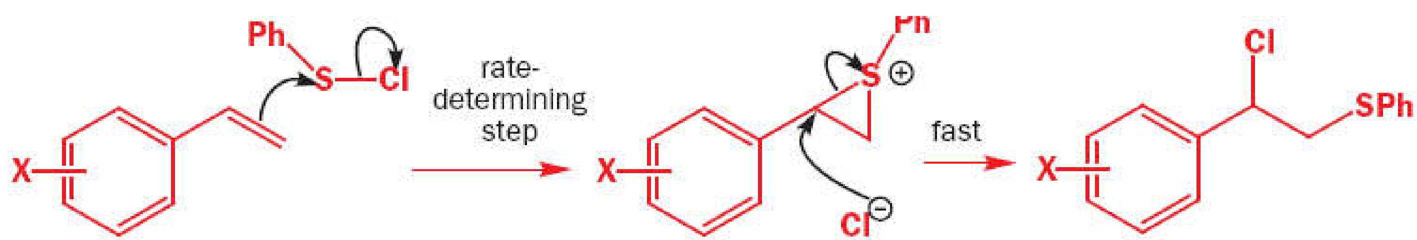
Alkaline hydrolysis ρ value of +2.6 vs. acid hydrolysis ρ value of +0.1

● The meaning of Hammett ρ values

This then is the full picture. You should not, of course, learn these numbers but you need an idea of roughly what each group of values means. You should see now why it is unimportant whether the Hammett correlation gives a good straight line or not. We just want to know whether **ρ** is + or – and whether it is, say, 3 or 6. It is meaningless to debate the significance of a r value of 3.4 as distinct from one of 3.8.

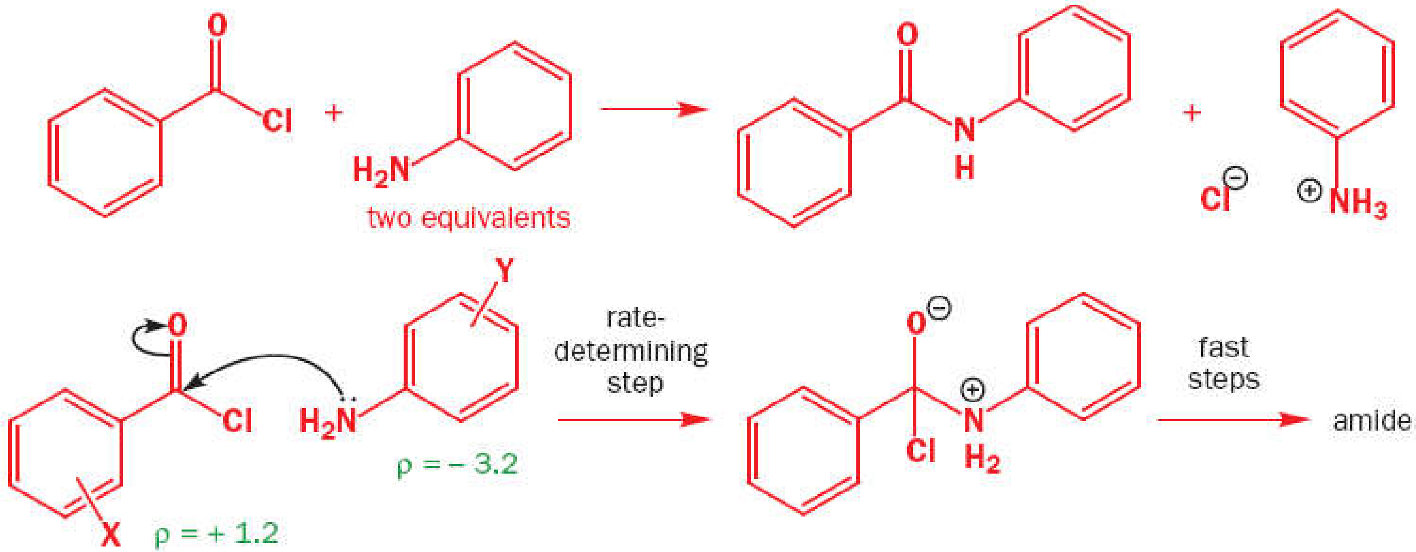
–6 –5	–4 –3 –2	–1 0 +1	+2 +3 +4	+5 +6
large negative ρ-values	moderate negative ρ-values	small ρ-values	moderate positive ρ-values	large positive ρ-values
positive charge on ring or delocalized round benzene ring	electrons flow out of TS positive charge near ring loss of conjugation	1. Ar too far away 2. No electron change 3. Two ρ-values cancel each other out	electrons flow into TS negative charge near ring loss of conjugation	negative charge on ring or delocalized round benzene ring





Picture of the transition state from Hammett plots

Reaction between a nucleophile and an electrophile



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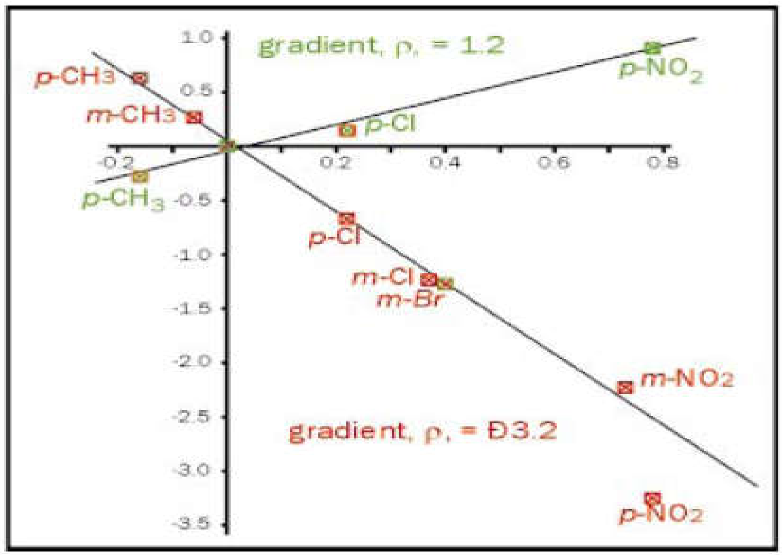
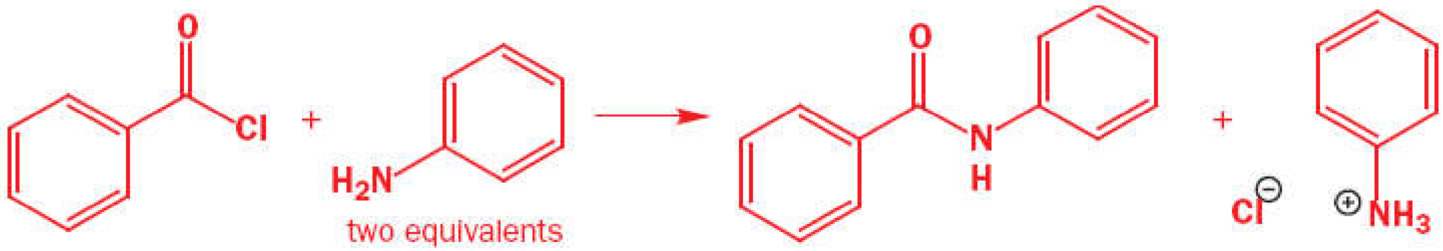
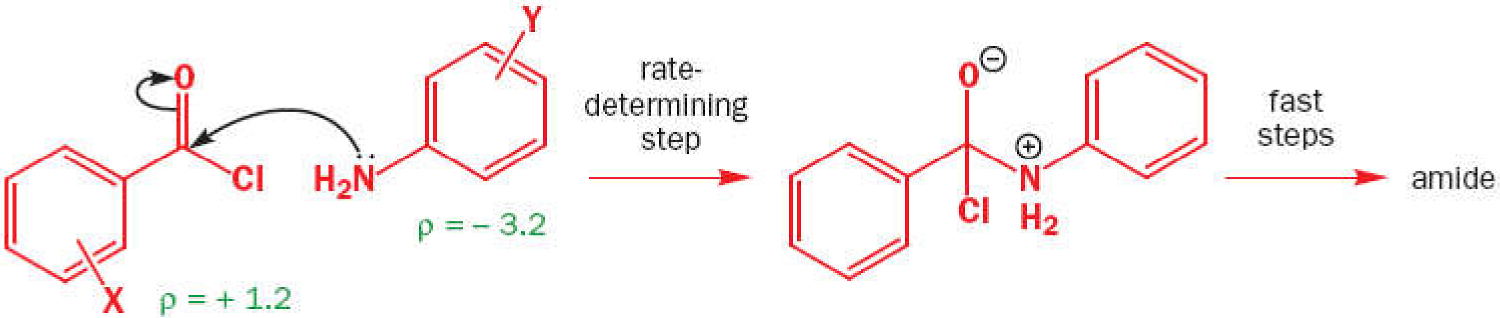


Rate depends on the nucleophilicity of the amine 100 times more than on the electrophilicity of the acid chloride

Picture of the transition state from Hammett plots

Reaction between a nucleophile and an electrophile





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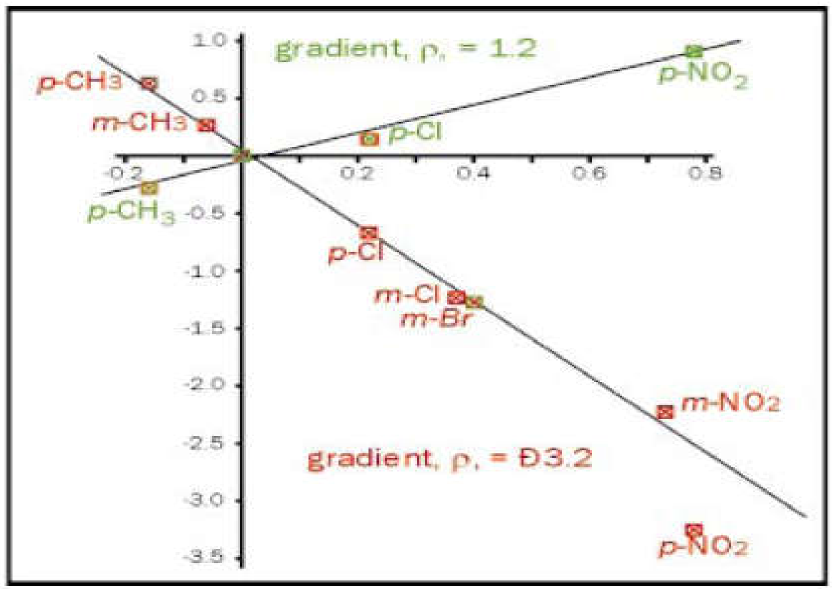
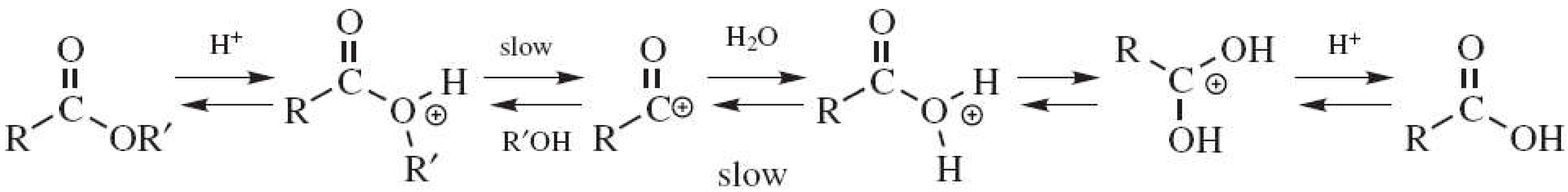


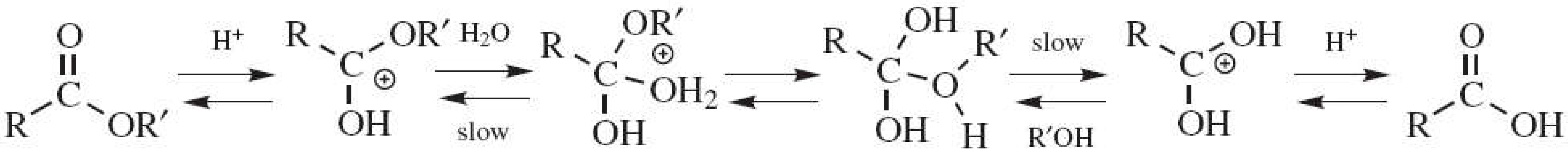
Rate depends on the nucleophilicity of the amine 100 times more than on the electrophilicity of the acid chloride

Ester Hydrolysis Mechanisms

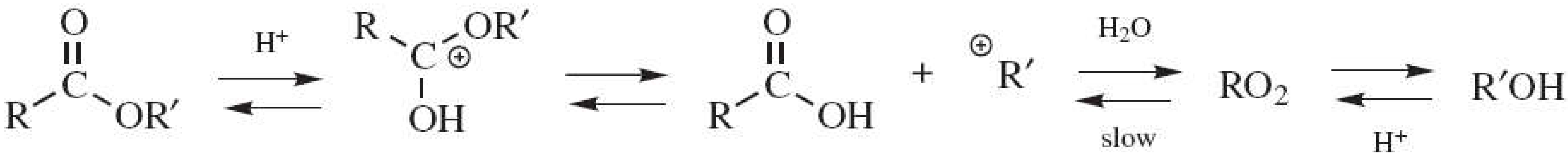
$A_{AC}1$ [S_N1]



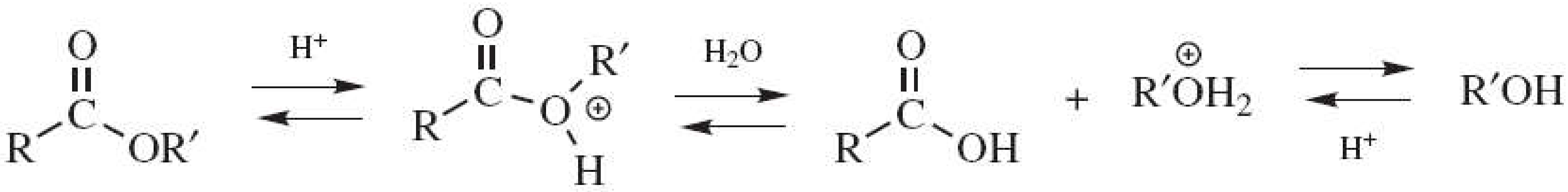
$A_{AC}2$ [Tetrahedral Intermediate ; Most Common]



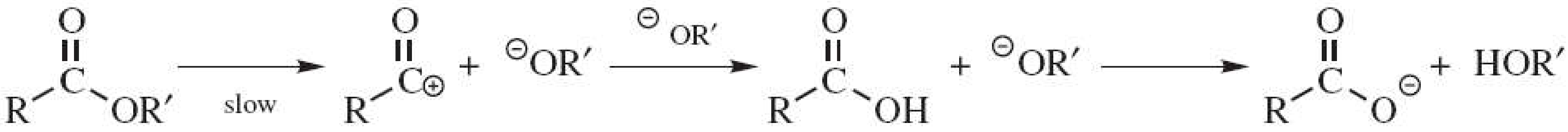
$A_{AL}1$ [S_N1]



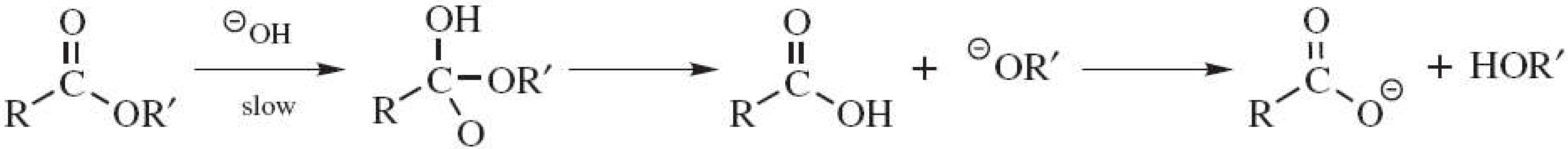
$A_{AL}2$ [S_N2]



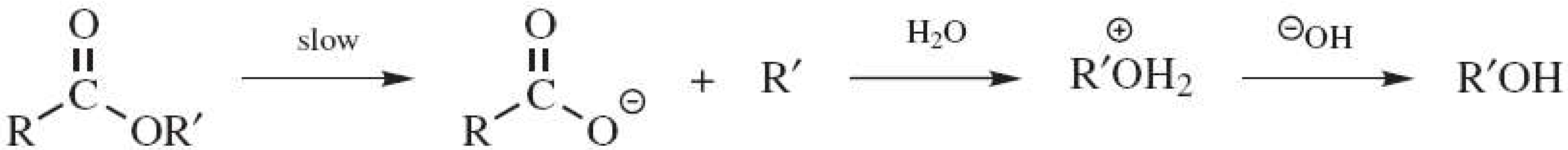
$B_{AC}1$ [Not observed experimentally]



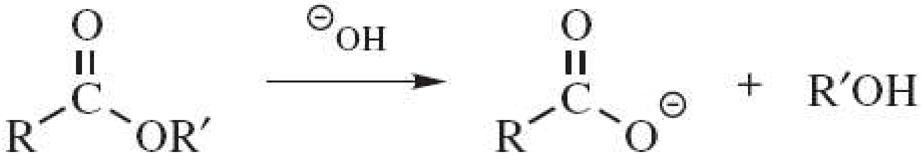
$B_{AC}2$ [Tetrahedral Intermediate ; Most Common]



$B_{AL}1$ [S_N1]



$B_{AL}2$ [S_N2]

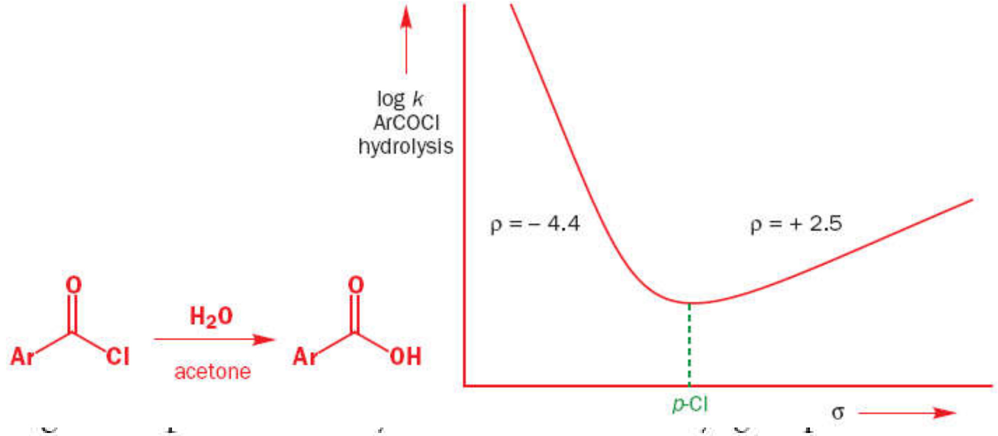


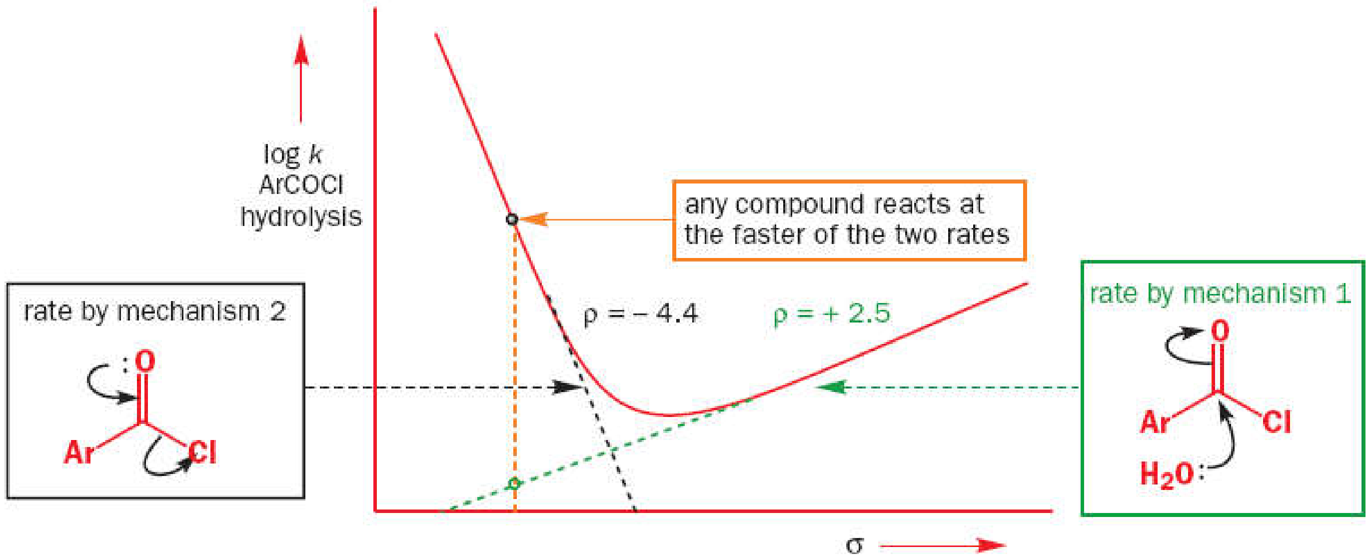
The evidence for most common mechanism involving a Tetrahedral intermediate is:

1. Hydrolysis with $H_2{}^{18}O$ results in the ^{18}O appearing in the acid and not in the alcohol
2. Esters with chiral R′ groups give alcohols with retention of configuration
3. Allylic R′ gives no allylic rearrangement;
4. Neopentyl R′ gives no rearrangement;1555 all these facts indicate that the O-R′ bond is not broken.

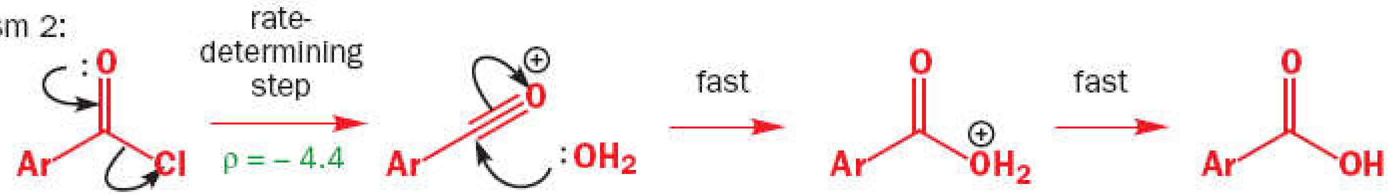
Nonlinear Hammett Plots

MINIMA IN THE PLOT

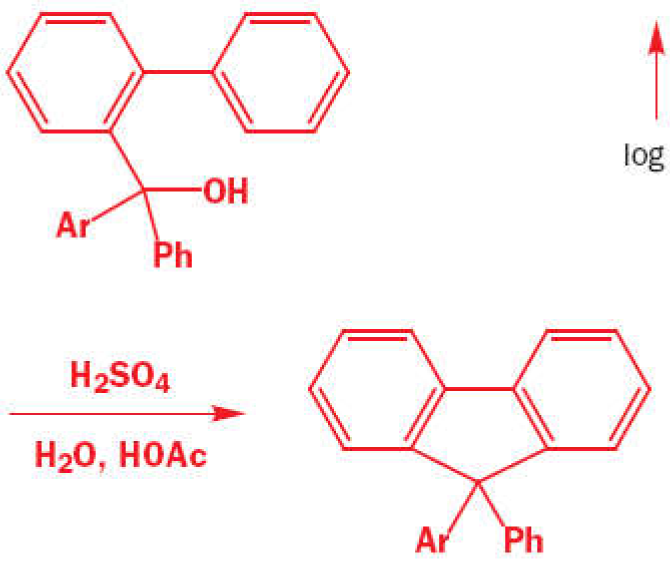


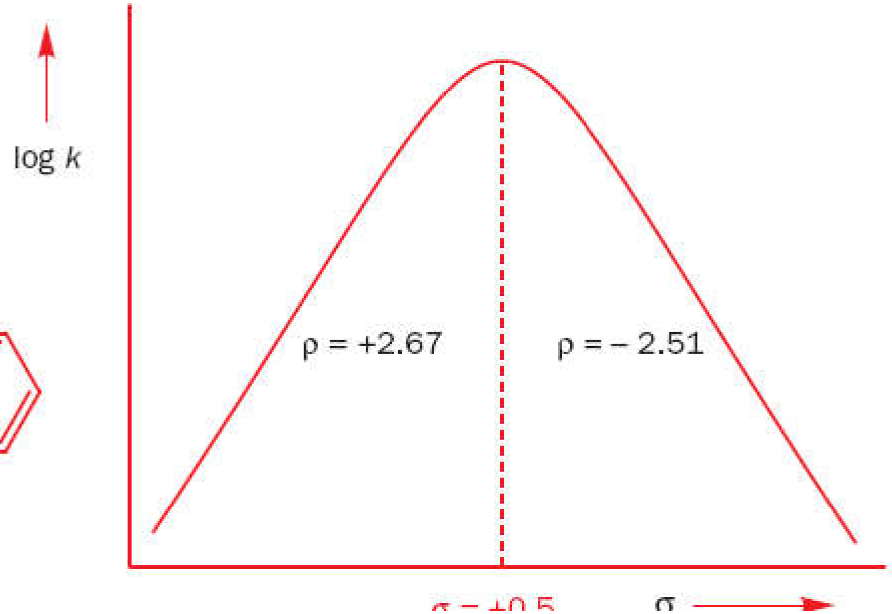


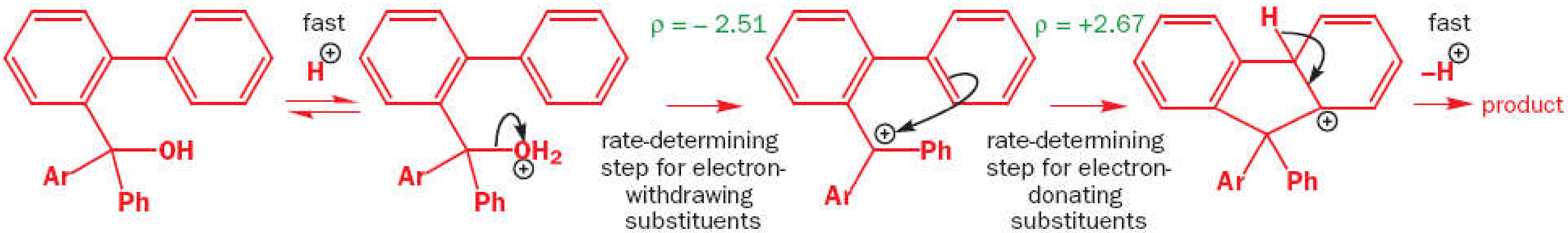
mechanism 2:



MAXIMA IN THE PLOT



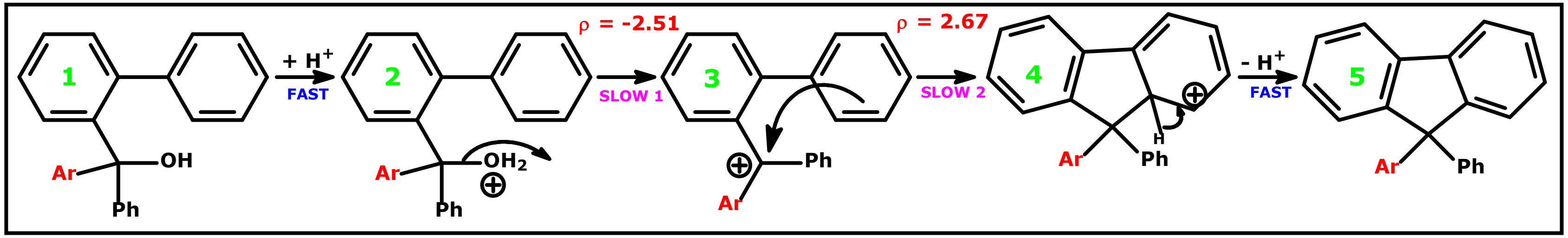




The carbocation intermediate in the Friedel–Crafts reaction is rather stable, being ***tertiary*** and ***benzylic***, and the formation of the cation, normally the rate-determining step, with inevitably a negative ρ value [*because charge developing in the TS++ is +ve*], goes faster and faster [**i.e. E_a for the rate determining step DECREASES, moving from right to left on the RHS part of the curve**] as the electron-donating power of the substituents increases ***until it is faster than the cyclization which becomes the rate-determining step***.

The cyclization puts electrons back into the carbocation and has a positive ρ value. As the two steps have more or less the reverse electron flow to and from the same carbon atom, it is reasonable for the size of ρ to be about the same but of opposite sign.

This explanation for the graph with the MAXIMA holds ONLY when the reporter group X is on either the Ar or Ph designated rings in the mechanism shown above

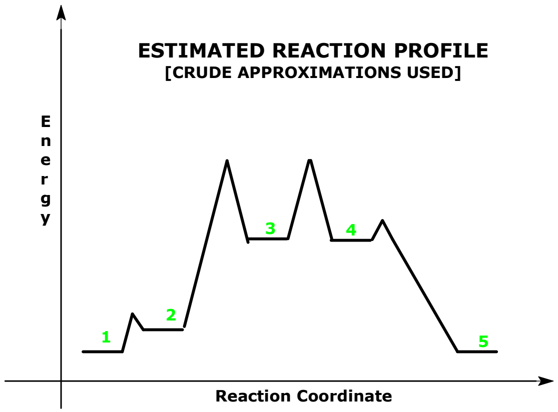


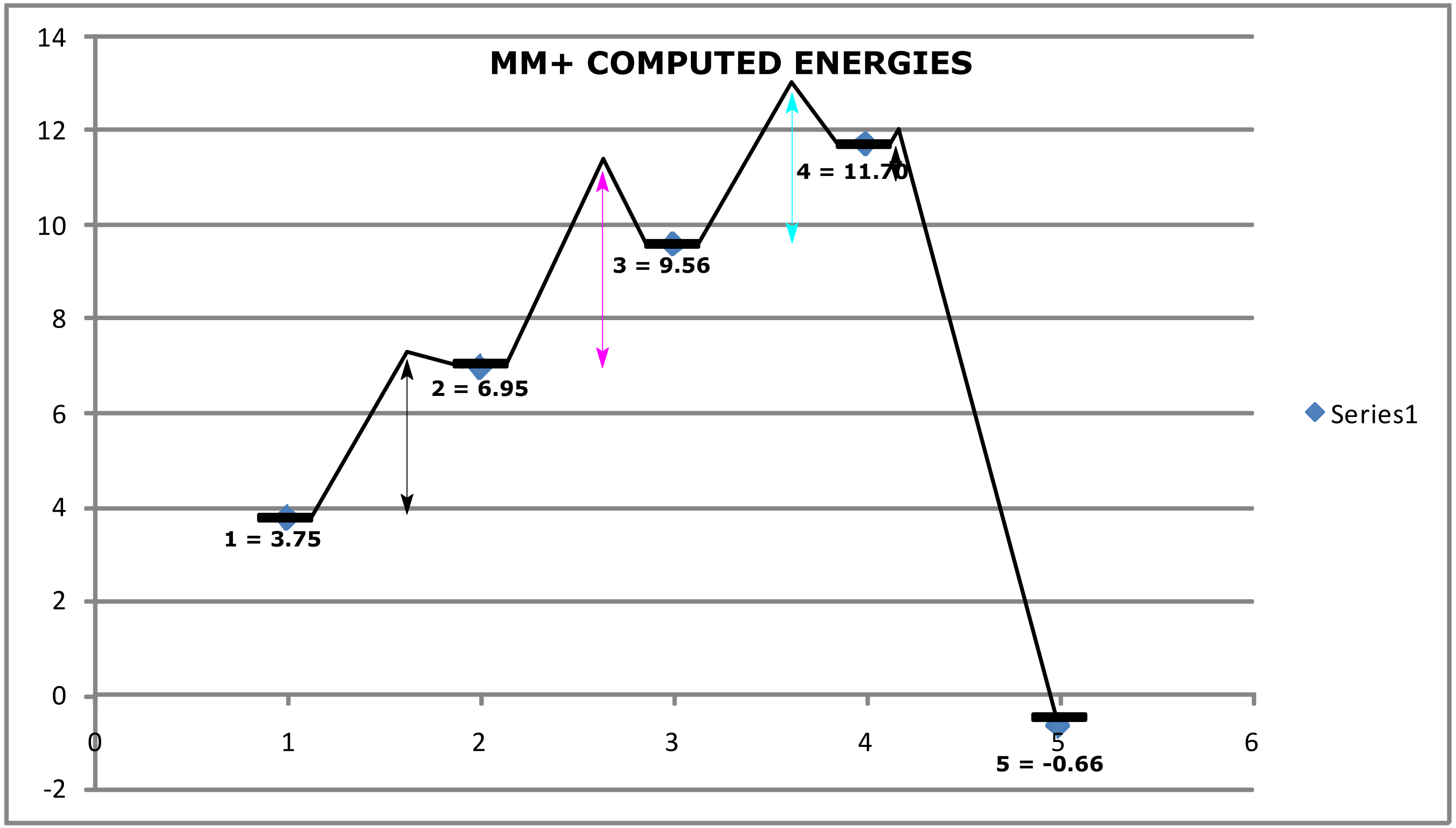
SLOW 1
Carbocation *formation*

+ve charge INCREASES in the TS++
(i.e. LESS e- density in TS)
e- WITHDRAWING groups BAD
E_a higher = Rate slower

SLOW 2
Carbocation *quenching*

-ve charge INCREASES in the TS++
(i.e. MORE e- density in TS ++)
e- WITHDRAWING groups GOOD
E_a lower = Rate faster





TAKE HOME LESSON

A reaction occurs by:

The *faster of two possible mechanisms* [MINIMA IN THE HAMMETT PLOT]

BUT

By the *slower of two possible rate-determining steps* [MAXIMA IN THE HAMMETT PLOT].

WHAT DOES THIS HAVE TO DO WITH CATALYSIS?

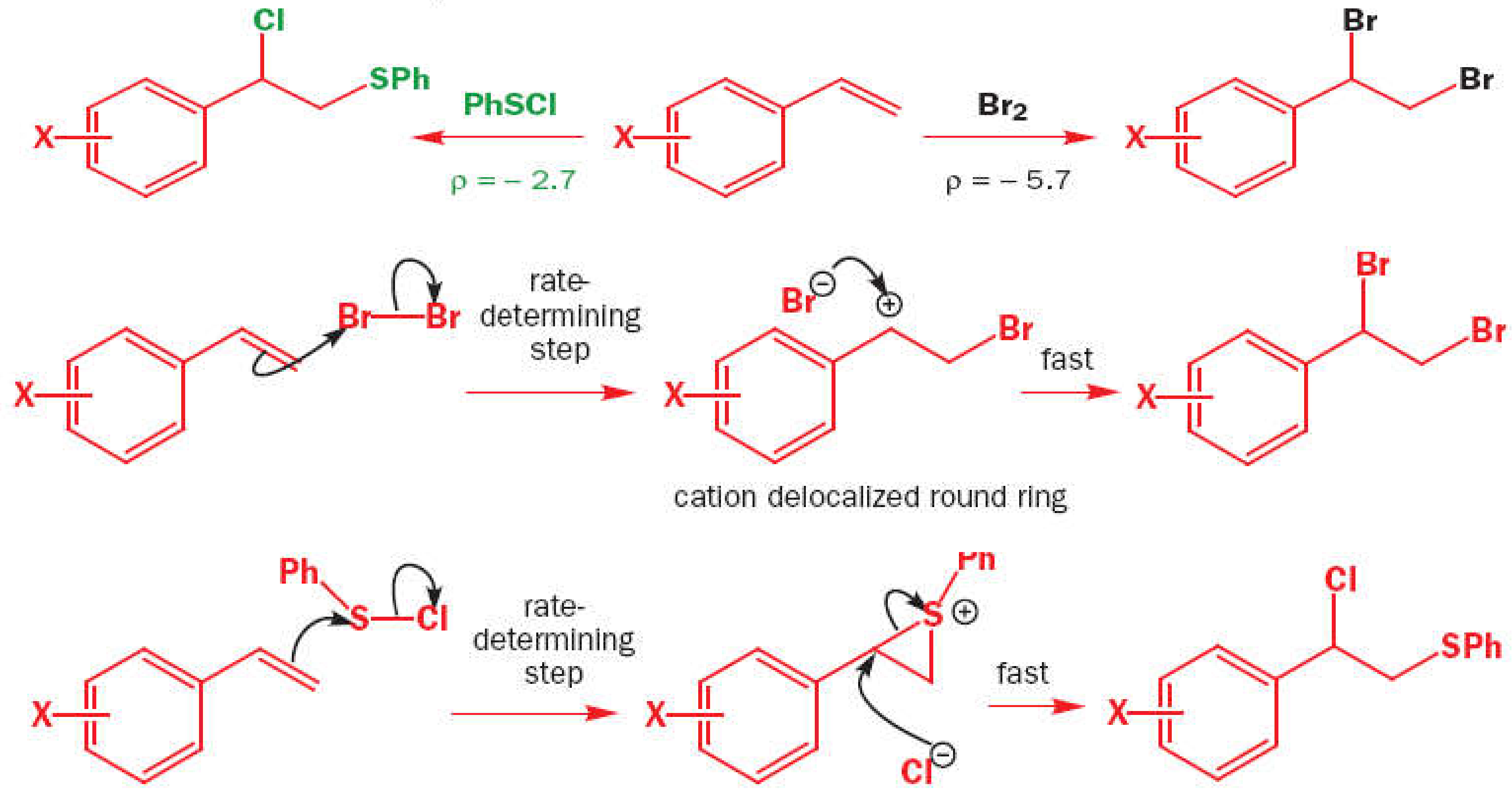
ONCE CHEMICALS ARE MIXED, *MULTIPLE OUTCOMES ARE POSSIBLE*.

IT IS POSSIBLE, *BY MODULATING THE TS^{++}*, TO FAVOUR A POSSIBLE PATHWAY.

The meaning of Hammett ρ values

This then is the full picture. You should not, of course, learn these numbers but you need an idea of roughly what each group of values means. You should see now why it is unimportant whether the Hammett correlation gives a good straight line or not. We just want to know whether **ρ** is + or – and whether it is, say, 3 or 6. It is meaningless to debate the significance of a r value of 3.4 as distinct from one of 3.8.

–6 –5	–4 –3 –2	–1 0 +1	+2 +3 +4	+5 +6
large negative ρ-values	moderate negative ρ-values	small ρ-values	moderate positive ρ-values	large positive ρ-values
positive charge on ring or delocalized round benzene ring	electrons flow out of TS positive charge near ring loss of conjugation	1. Ar too far away 2. No electron change 3. Two ρ-values cancel each other out	electrons flow into TS negative charge near ring loss of conjugation	negative charge on ring or delocalized round benzene ring



Catalysis: Transition State Stabilisation

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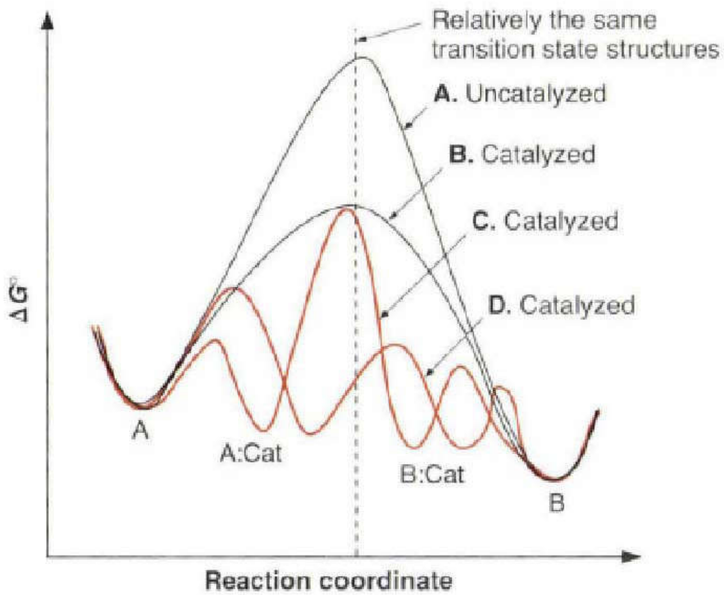
Inherent in catalysis is the idea that the activation energies for any catalysed reaction must be lower than the activation energies for the uncatalysed reaction.



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C: A more realistic view of how catalysis can be achieved without a complete mechanism change.

Binding of the reactant is REQUIRED first.

D: A totally new mechanism with a completely new reaction coordinate can also give catalysis.

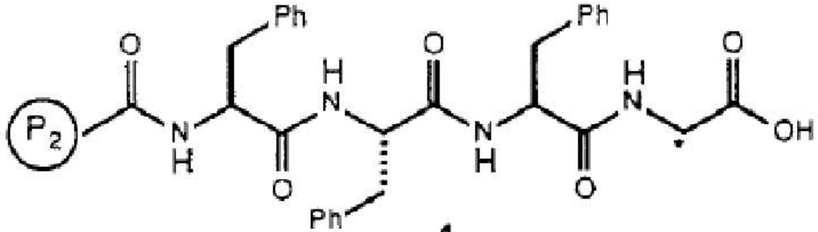
GOAL: 'Binding' the Transition State *Better* than the Ground State

Introduction to catalysis:

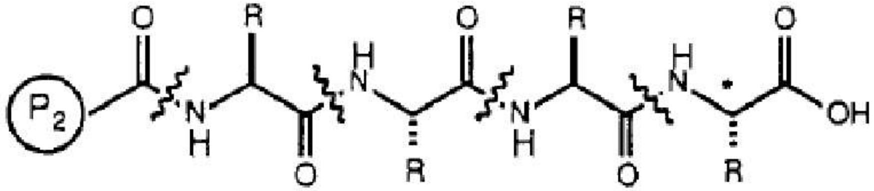
How slow is slow?

Hydrolysis of a Peptide Bond in Neutral Water

[At neutral pH & room temperature = 3 X 10^{-9} s^{-1}(?)~$t_{1/2}$=7 years.]



4



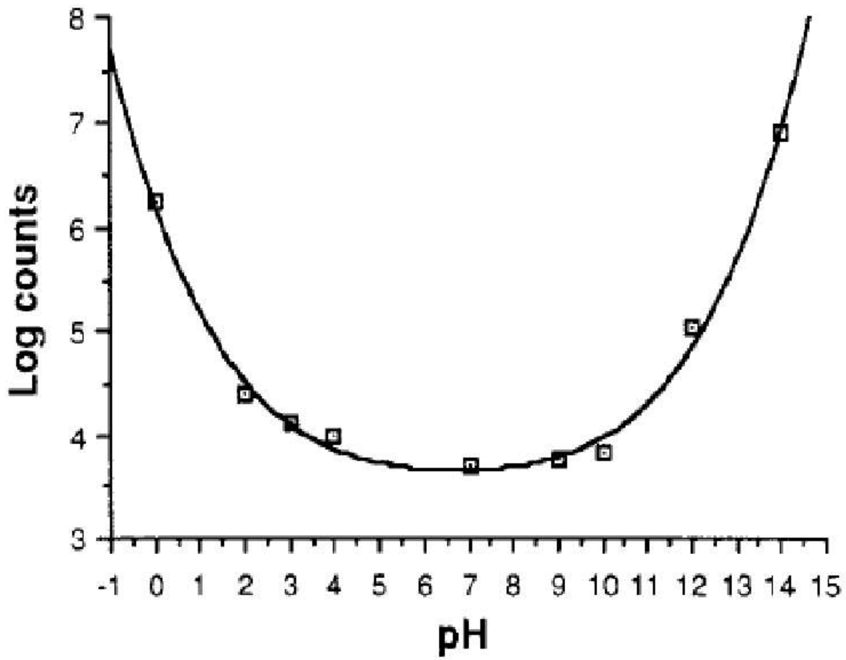


Figure 2. Release of radiolabel from resin **4** vs pH.

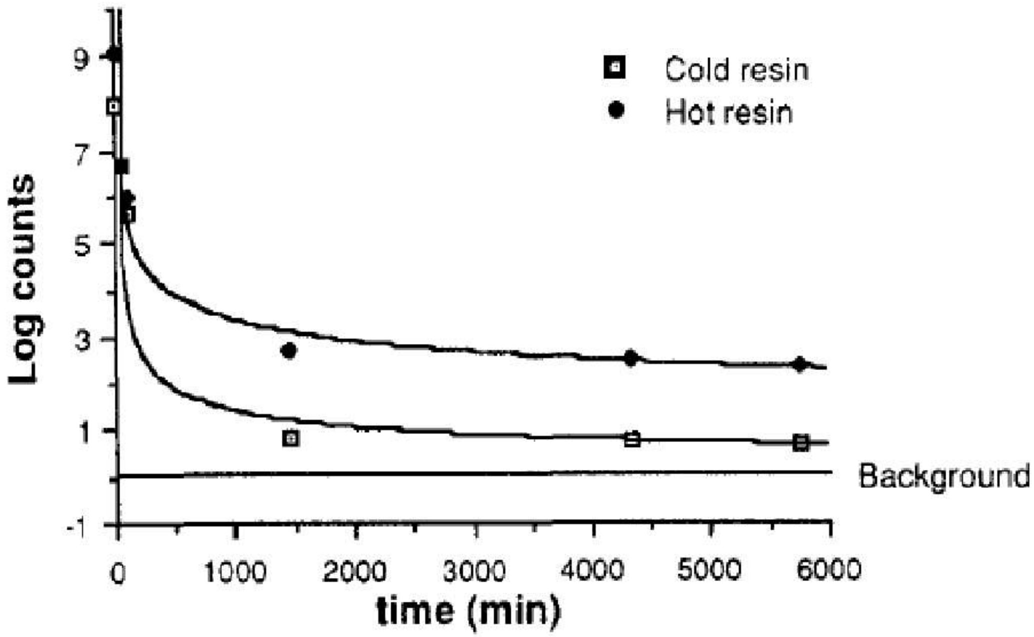
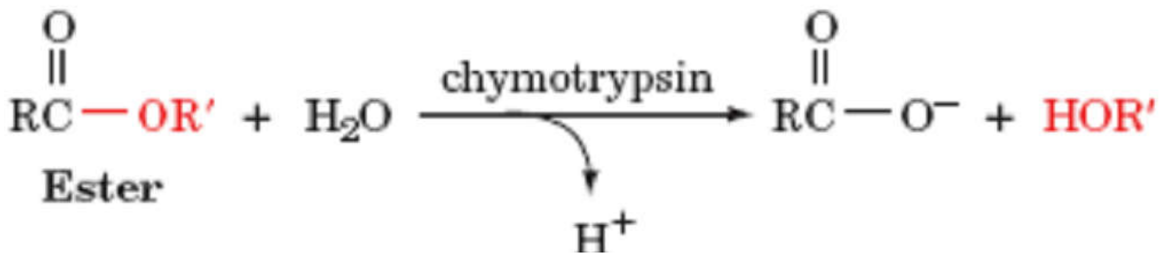
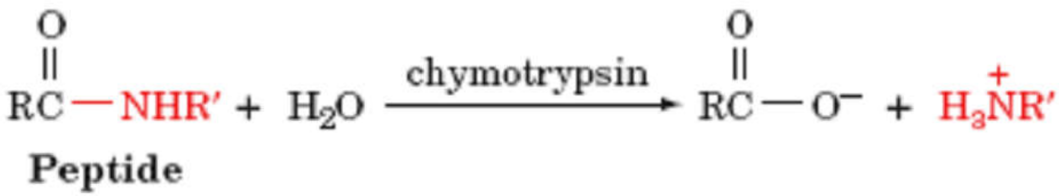


Figure 3. Release of radiolabel vs washing of polymer **4**.



[Done by you CMP100]



[Happens inside you! (digestion of food)]

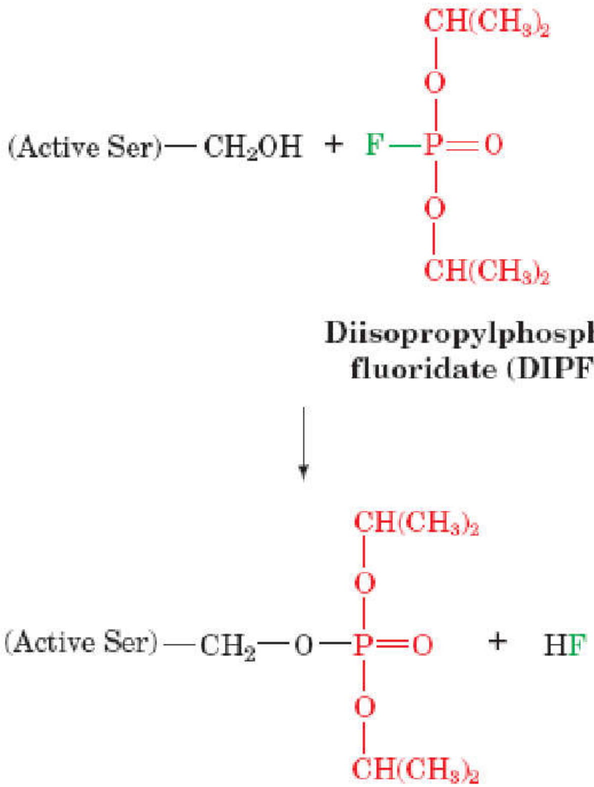
So just WHAT is CHYMOTRYPSIN?

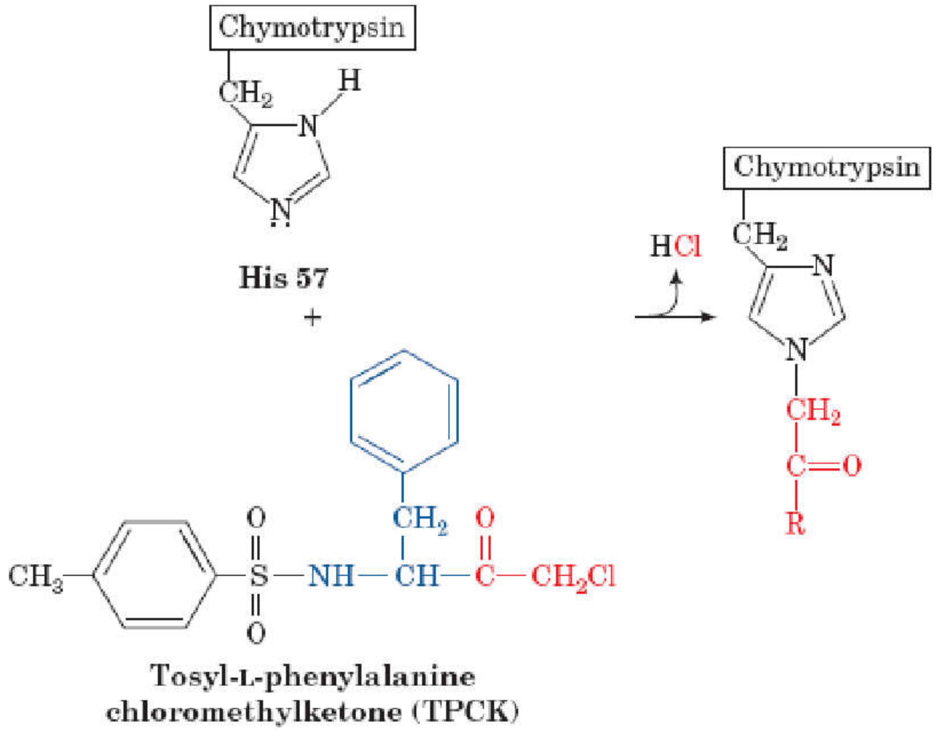
It is a BIOLOGICAL catalyst.

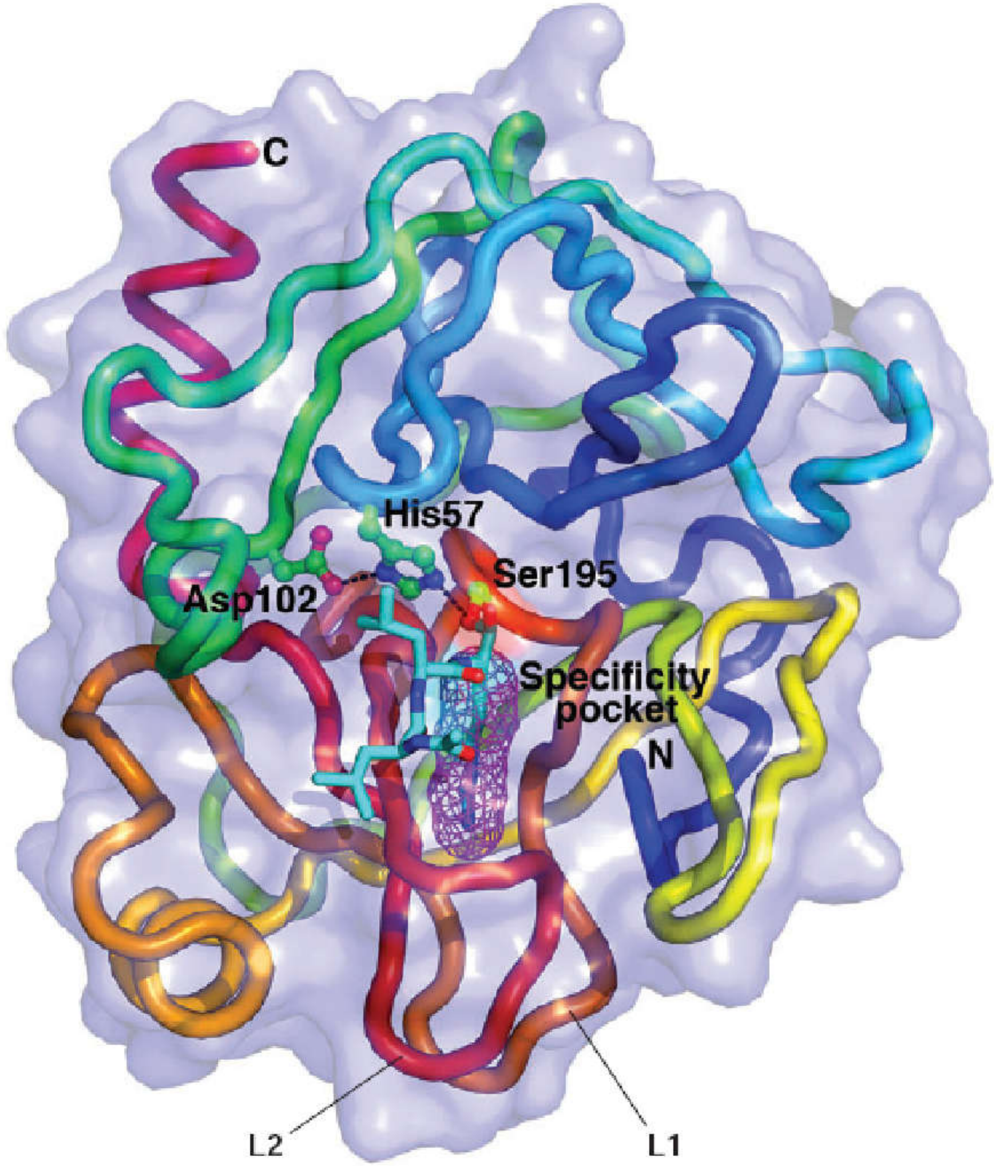
A protein molecule capable of highly specialised CATALYSIS.

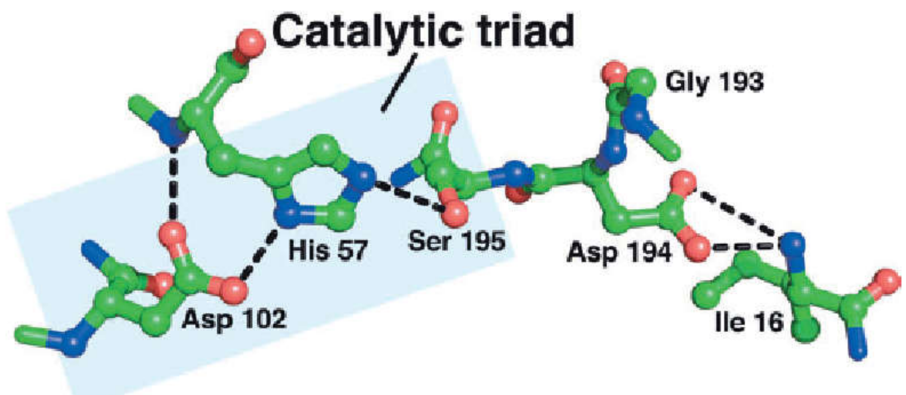
How does it work?

Preliminary Data

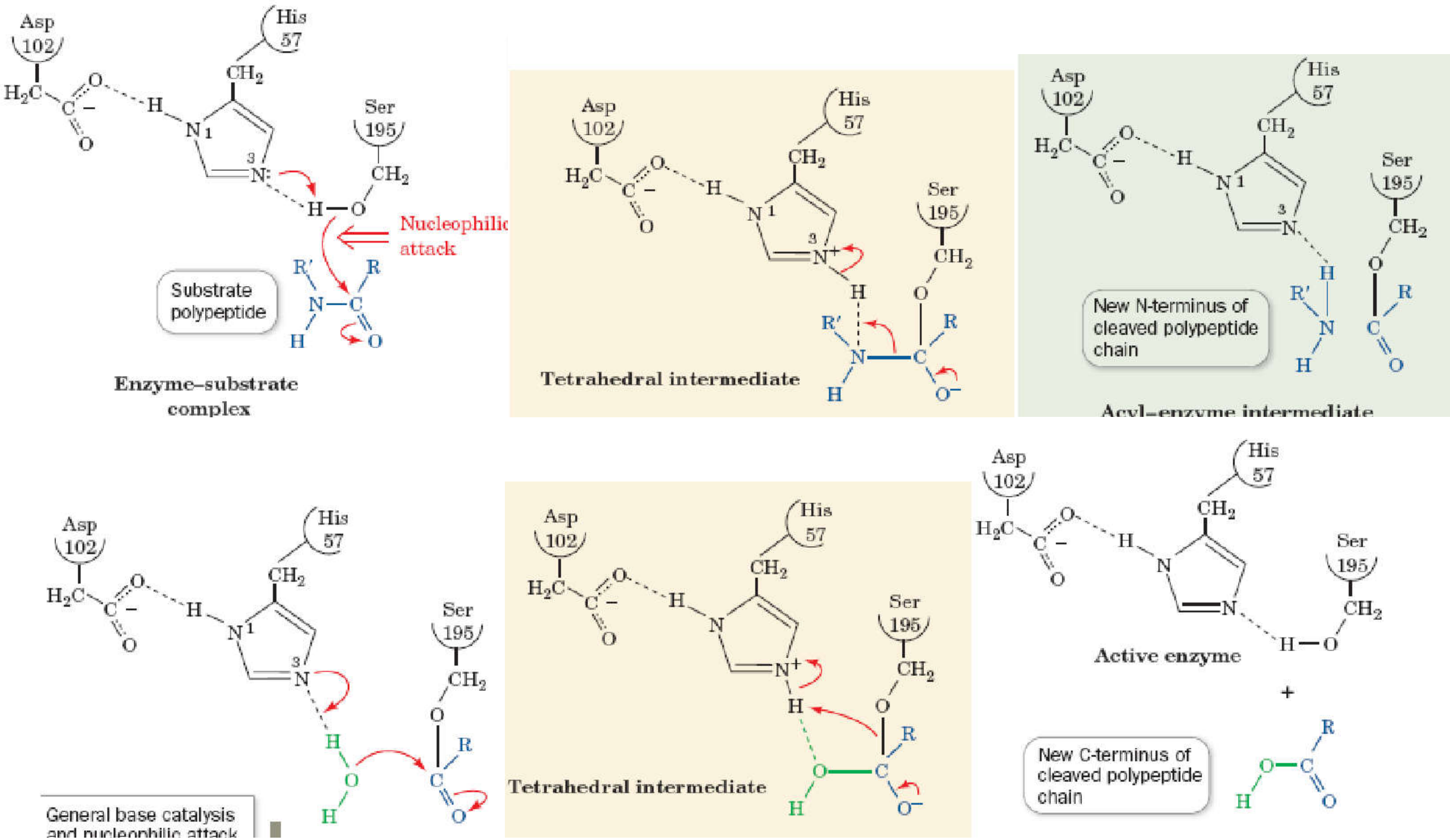





C
His57
Ser195
Asp102
Specificity pocket
N
L2
L1


Catalytic triad
Gly 193
His 57
Ser 195
Asp 194
Asp 102
Ile 16

Stepwise Mechanism



Lets ‘*remove*’ of the “Catalytic Triad”.

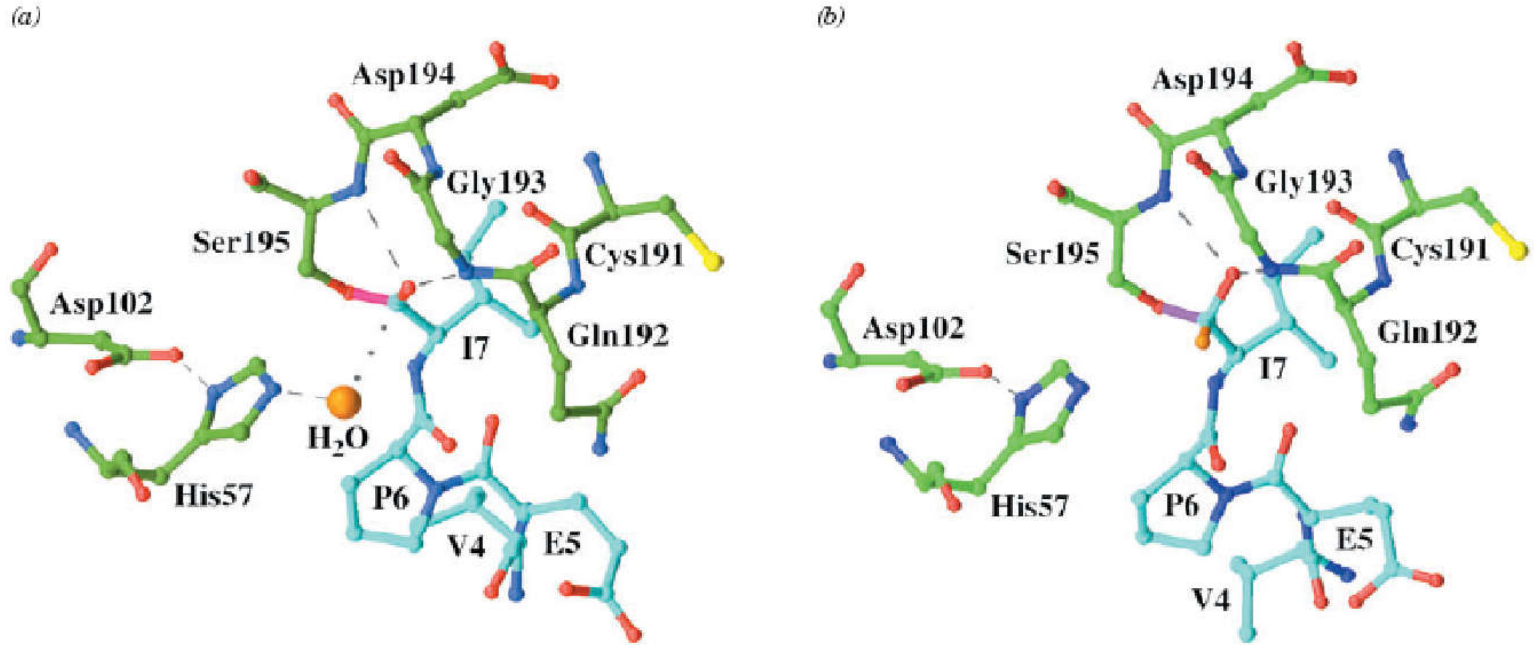
Does catalysis still happen?

YES!

[Not as efficiently as before but still faster than the UNCATALYSED reaction!]

10^{0}[Uncatalysed] vs 10^{+10}[Catalysed] vs 10^{+4}[Without Triad]

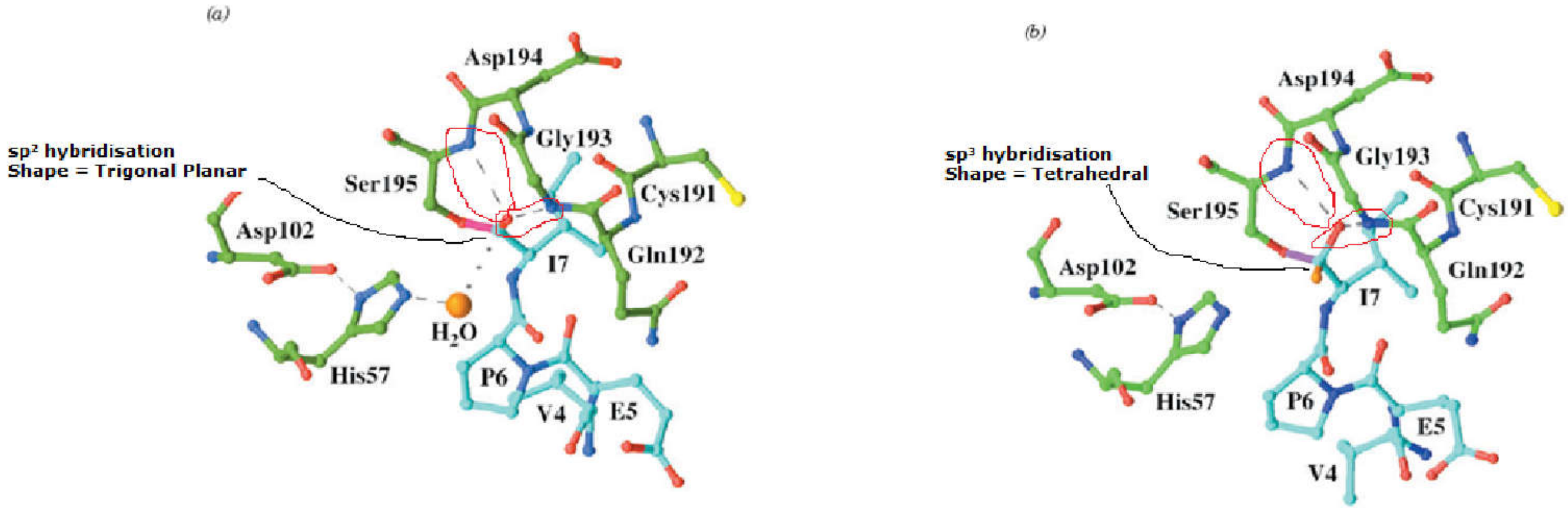
WHY?



(*a*) At pH 5.0, a covalent bond (*magenta*) links the Ser 195 O atom to the C-terminal (I7) C atom of the substrate.
A water molecule (*orange sphere*) appears poised to nucleophilically attack the acyl–enzyme's carbonyl C atom.
The dashed lines represent catalytically important hydrogen bonds, and the dotted line indicates the trajectory that the water molecule presumably follows in nucleophilically attacking the acyl group's carbonyl C atom.

(*b*) When the complex is brought to pH 9.0 and then rapidly frozen, the water molecule becomes a hydroxyl substituent (*orange*) to the carbonyl C atom, thereby yielding the tetrahedral intermediate, which resembles the transition state.

Transition State Stabilisation



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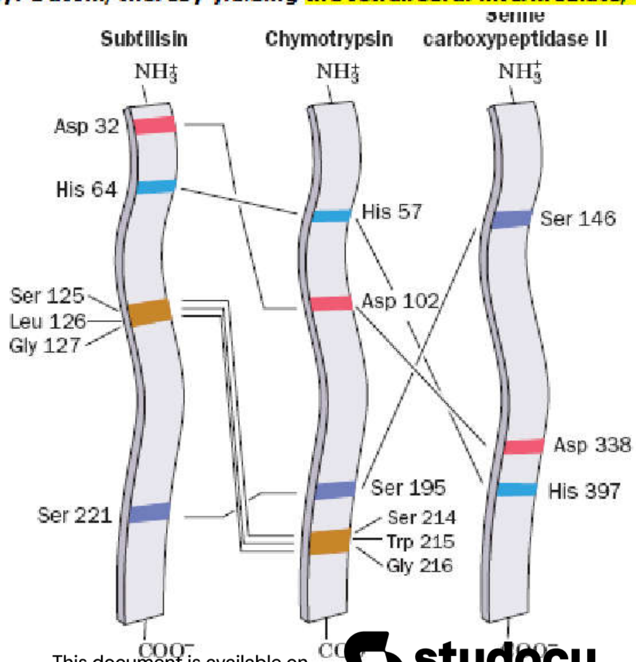

Explanation of 'TRANSITION STATE STABILISATION' as the basis for catalysis

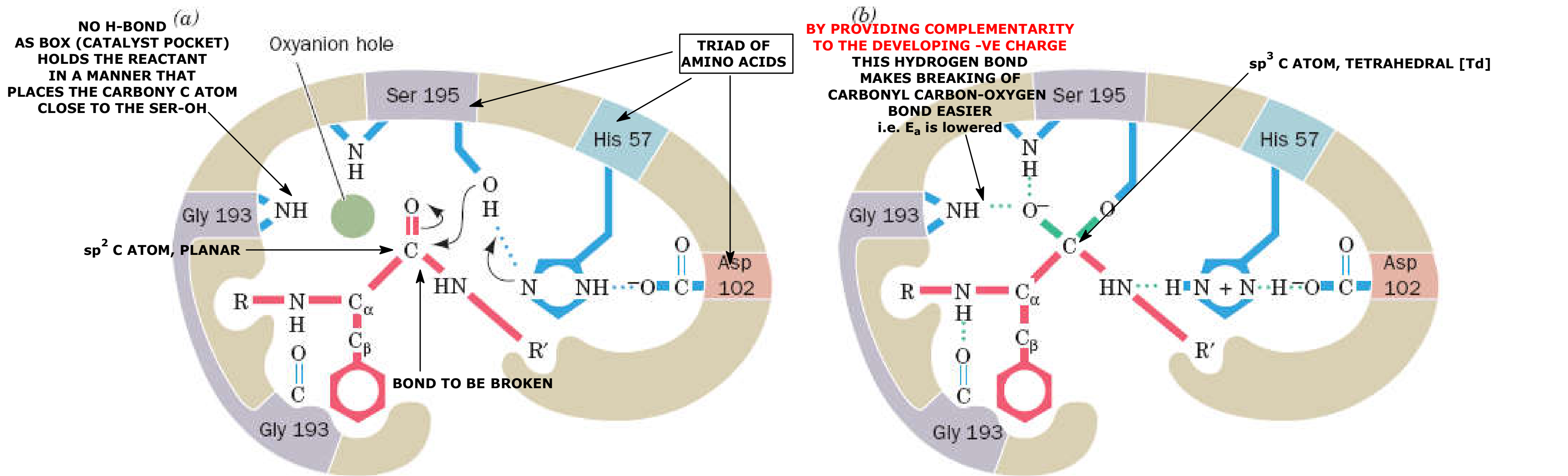

FIG. 11-30 Transition state stabilization in the serine proteases. (*a*) When the substrate binds to the enzyme, the trigonal carbonyl carbon of the scissile peptide is conformationally constrained from binding in the oxyanion hole (*upper left*). (*b*) In the tetrahedral intermediate, the now charged carbonyl oxygen of the scissile peptide (the oxyanion) enters the oxyanion hole and hydrogen bonds to the backbone NH groups of Gly 193 and Ser 195. The consequent conformational change permits the NH group of the residue preceding the scissile peptide bond to form an otherwise unsatisfied hydrogen bond to Gly 193. Serine proteases therefore preferentially bind the tetrahedral intermediate. [After Robertus, J.D., Kraut, J., Alden, R.A., and Birktoft, J.J., *Biochemistry* **11**, 4302 (1972).] **See Kinemage Exercise 10-3.**

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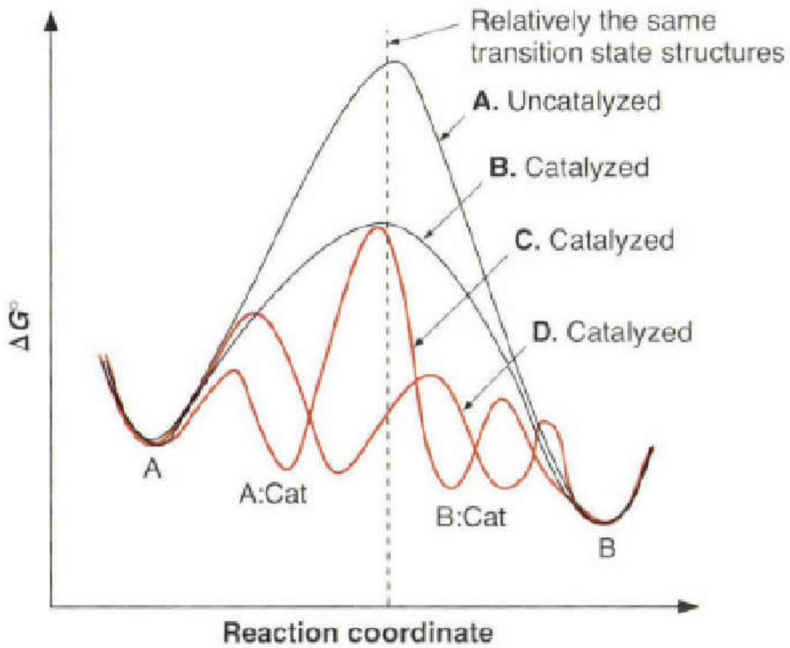

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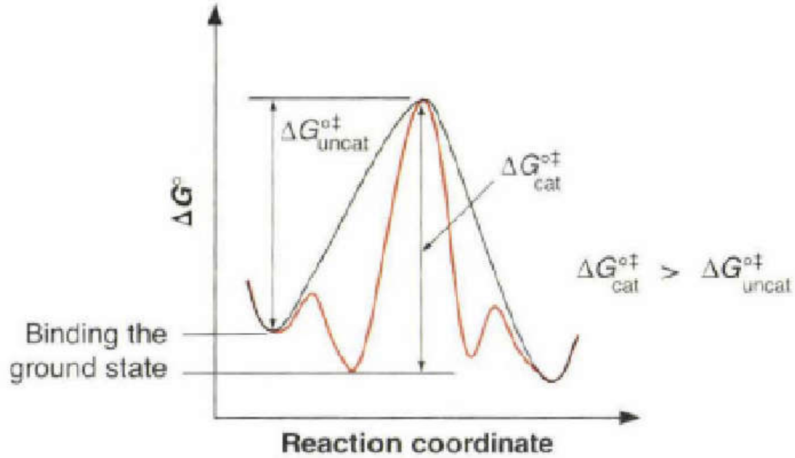
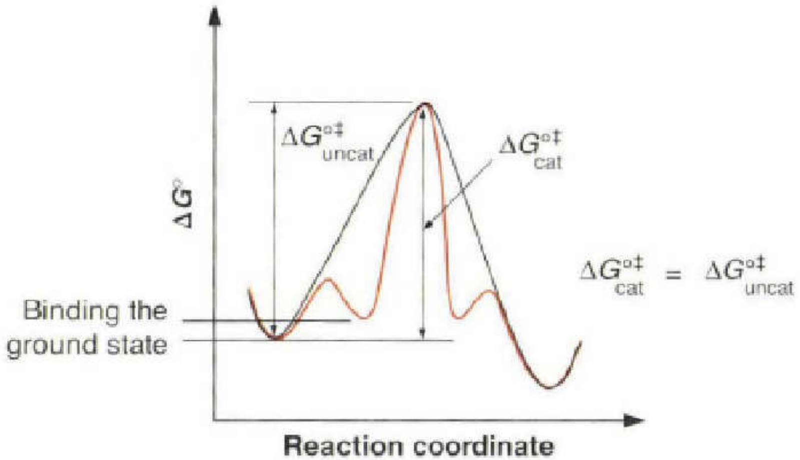
Reaction coordinate diagrams for various types of catalysts interacting with ground states and transition states.

A.



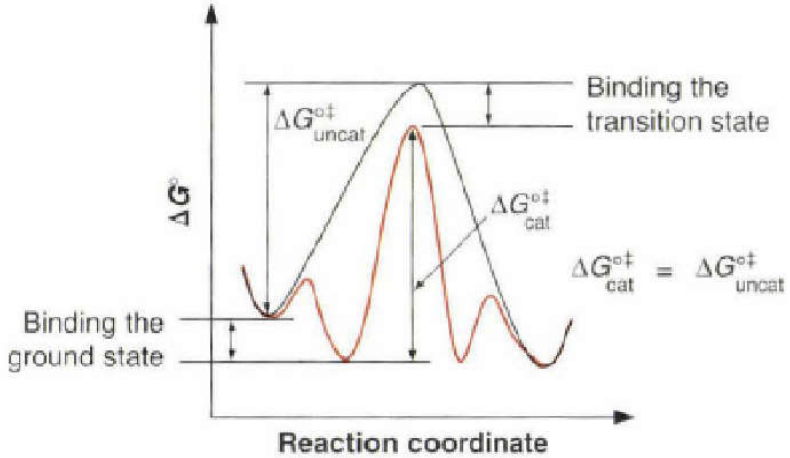
A. The substrate is bound by the catalyst, but there is no stabilisation of the transition state and no catalysis occurs. The rate in the presence of the catalyst is ***actually slower*** than the background rate.



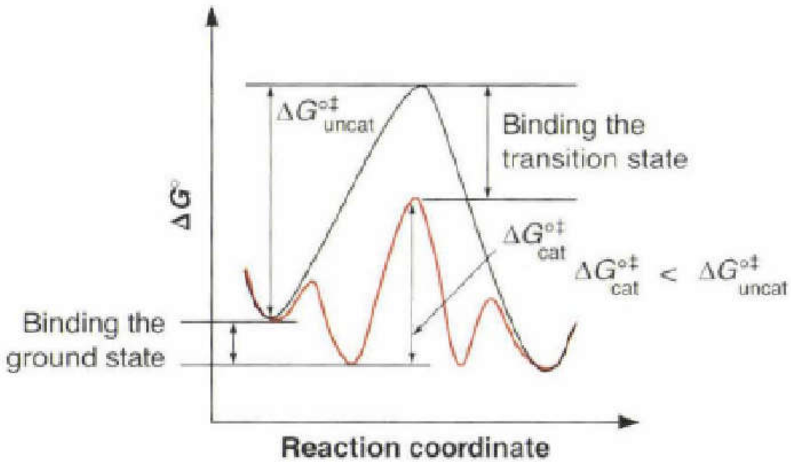
B.

B. The substrate is bound weakly, but there is no stabilisation of the transition state. The rate is the same as the background.

C.



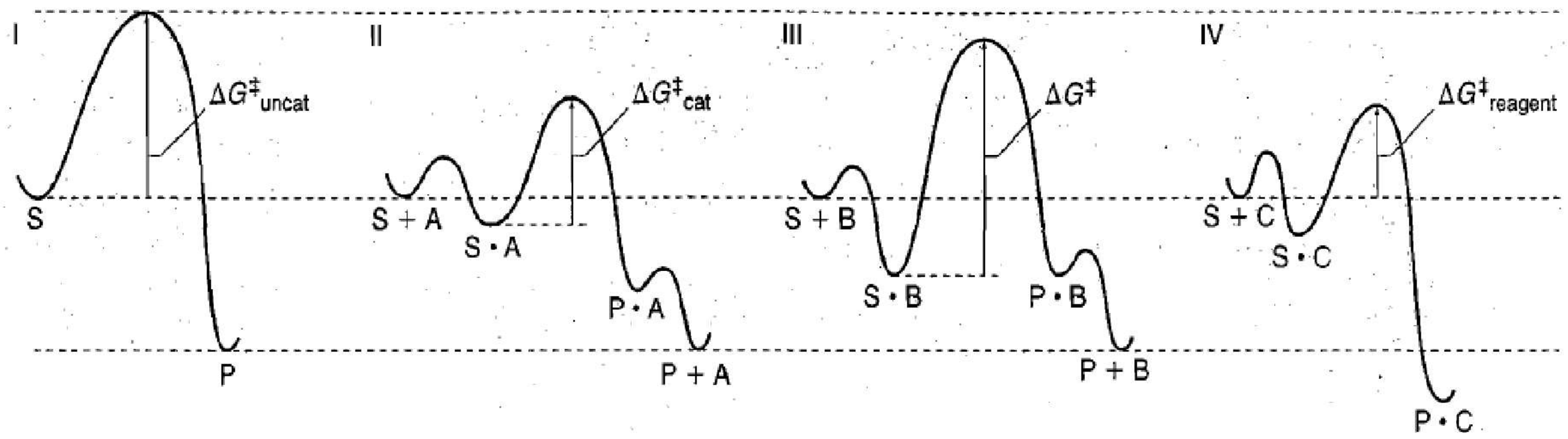
C. The substrate and transition state are bound to the same extent, and the uncatalysed rate is the same as the catalyzed rate.



D.

D. The transition state is bound better than the substrate, and **catalysis occurs**.

Uncatalyzed reaction | **Reaction catalysed by A** | **Reaction promoted by an additive B** | **Reaction involving reagent C.**



Uncatalyzed reaction

Catalyzed reaction via substrate–catalyst complexes

Greater stabilization of ground state than transition state–no catalysis

Reaction coordinate for a reaction "mediated" by C, where C is a reagent not a catalyst

Scenario I: UNCATALYSED process. Thermal, reaction of substrate (S) to generate product (P).

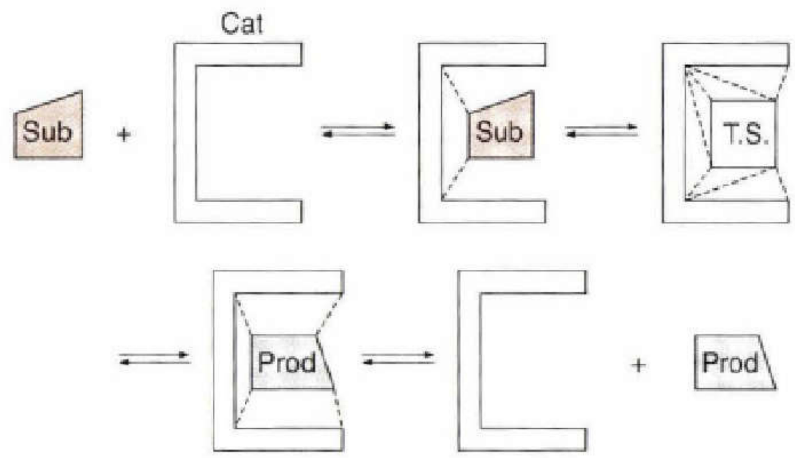
Scenario II: A CATALYTIC PROCESS
Species A stabilizes the transition state more than it stabilizes the ground state S (by forming S*A).
In addition, species A releases the free product in this reaction coordinate in a manner that reforms A.
'A' IS THE CATALYST.

Scenario III: NO CATALYSIS
The complex S*B formed between additive B and substrate S also forms product and regenerates B. *However, the overall energy landscape in this reaction coordinate does not lead to catalysis.* Here, additive B stabilizes S (by formation of S*B) more than it stabilizes the transition state. Hence, the activation energy in this scenario is higher than that for the reaction without additive B. S would then convert to P *by the lower energy uncatalyzed pathway in the presence of substoichiometric amounts of B* (i.e. S*B->S->P because the E_a is lower for this route).

Scenario IV: Species C forms an adduct with BOTH the substrate & the product of the reaction.
Reaction ends with P*C, the complex of the product with C. A second process (not shown) is needed to be conducted on P*C by some other reagent (such as water, acid, or an oxidant) to release the product P.

Reaction in the presence of C will occur faster than in the absence of C and could affect selectivity, *compound C is changed by the reaction*. Thus, compound C is a reagent in this scenario, instead of a catalyst, and must be used in stoichiometric quantities.

GENERIC SCHEME FOR CATALYSIS



Sub = Substrate, Cat = Catalyst, T.S. = Transition State, and Prod = Product.

Substrate first binds to the catalyst, this is followed by interactions with the catalyst that stabilize the transition state, and finally the catalyst releases the product.

Here, the catalyst is shown as having a pocket, ***but a surface, or even a single bond (such as to a proton), can act in a similar manner***.

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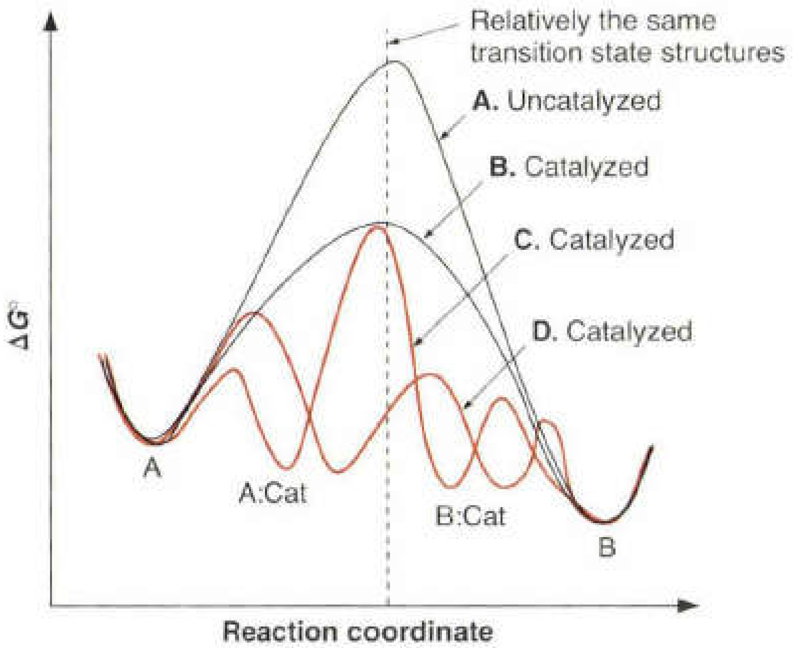

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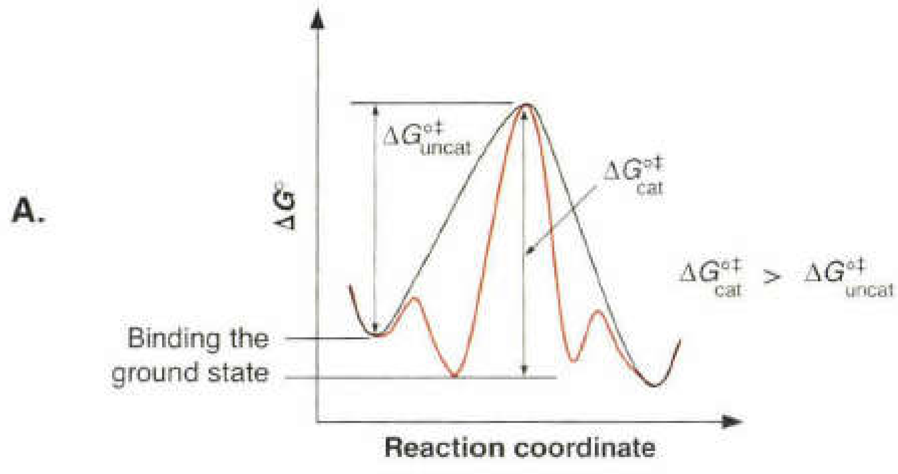

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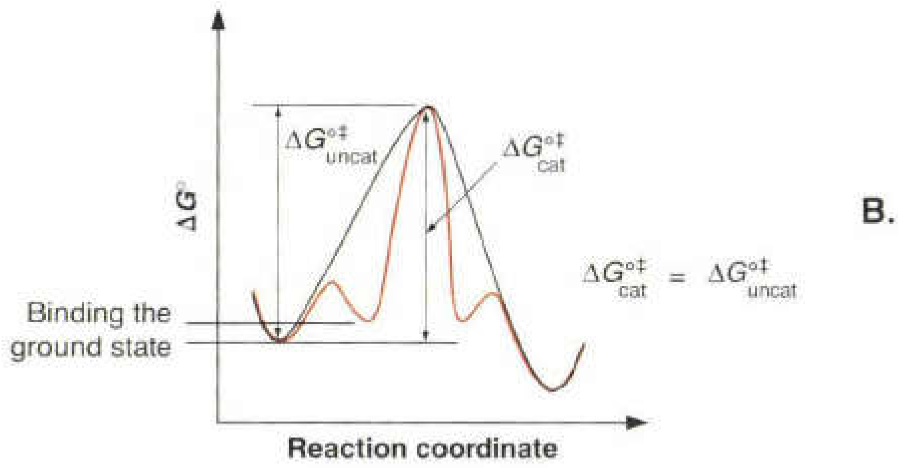

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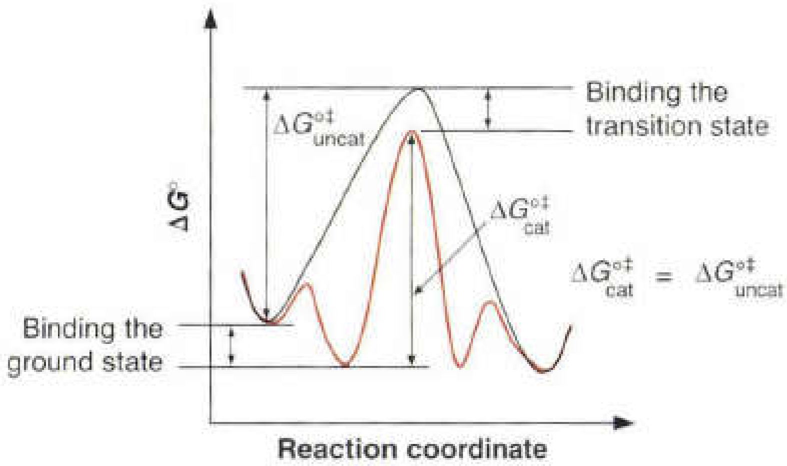

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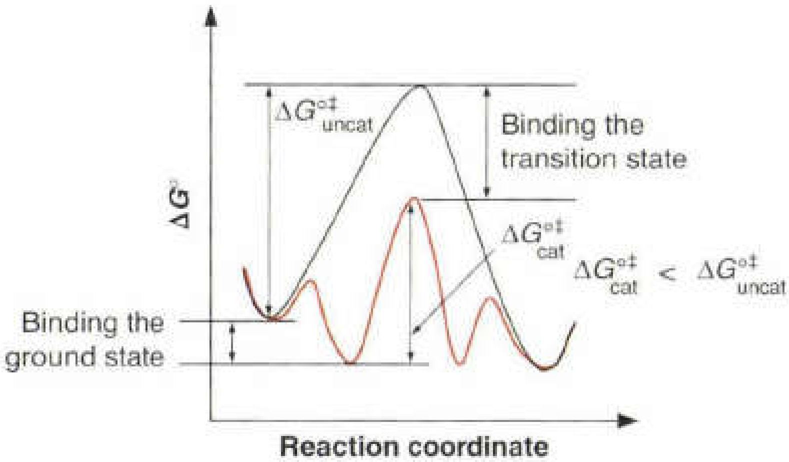

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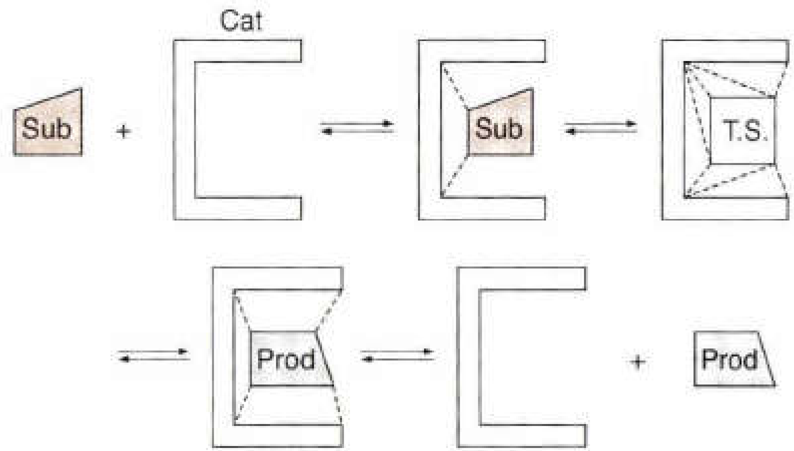
GENERIC SCHEME FOR CATALYSIS



Sub = Substrate, Cat = Catalyst, T.S. = Transition State, and Prod = Product.

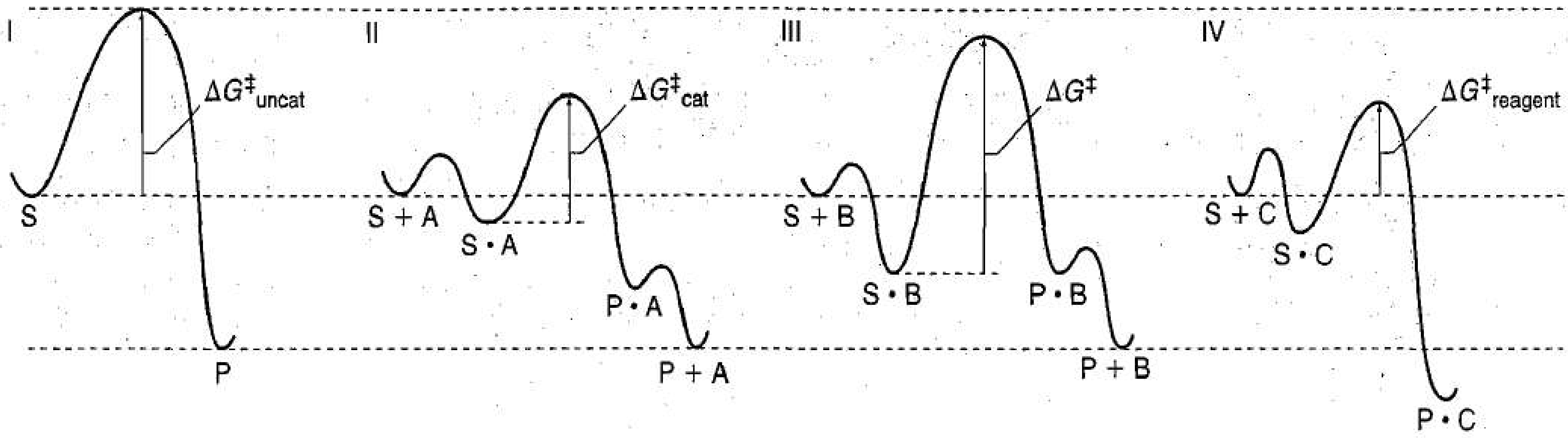
Substrate first binds to the catalyst, this is followed by interactions with the catalyst that stabilize the transition state, and finally the catalyst releases the product.

Here, the catalyst is shown as having a pocket, ***but a surface, or even a single bond (such as to a proton), can act in a similar manner***.

In short:

Basically, in a TS++, there is charge polarisation / separation relative to the reactants.
This can result in excess of either + or - charge relative to the reactant.
The *job* of the catalyst is to *provide a complementary charge* which makes this polarisation easier.
i.e. LOWERING the E_a

Uncatalyzed reaction **Reaction catalysed by A** **Reaction promoted by an additive B** **Reaction involving reagent C.**



Uncatalyzed reaction

Catalyzed reaction via substrate–catalyst complexes

Greater stabilization of ground state than transition state–no catalysis

Reaction coordinate for a reaction "mediated" by C, where C is a reagent not a catalyst

In I, the graph represents the thermal, uncatalyzed reaction of substrate (S) to generate product (P).

In II, species A stabilizes the transition state more than it stabilizes the ground state S (by forming S*A). In addition, species A releases the free product in this reaction coordinate in a manner that reforms A. This scenario constitutes a catalytic process, and A would be the catalyst.

In III, the complex S*B formed between additive B and substrate S also forms product and regenerates B. *However, the overall energy landscape in this reaction coordinate does not lead to catalysis.* Here, additive B stabilizes S (by formation of S*B) more than it stabilizes the transition state. Hence, the activation energy in this scenario is higher than that for the reaction without additive B. S would then convert to P *by the lower energy uncatalyzed pathway in the presence of substoichiometric amounts of B*.

In IV, a species C forms an adduct with the substrate and the product of the reaction. Here, the reaction coordinate ends with P*C, the complex of the product with C. A second process is then conducted on P*C by addition of another reagent (such as water, acid, or an oxidant) to release the organic product (not shown)**. Although reaction in the presence of C will occur faster than in the absence of C and could affect selectivity, *compound C is changed by the reaction*. Thus, compound C is a reagent in this scenario, instead of a catalyst, and must be used in stoichiometric quantities.**

Forms Of Catalysis

Spatial Temporal Approach:

The rate of reaction between functionalities A and B is proportional to the *time* that A and B reside *within a critical distance*.

TIME & DISTANCE CRITICAL

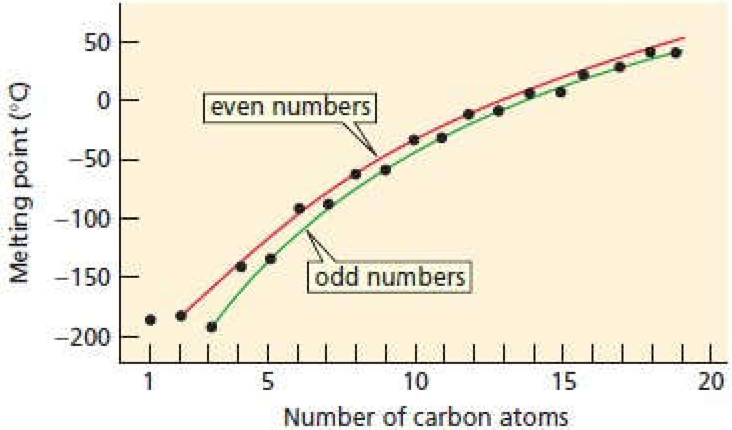
***Binding* is the key element in the *most widely accepted theory of how catalysis is achieved* (without completely changing the mechanism),**

"Binding" is *similar* to Solvation

?HOW?

(Kinds of SOLVENTS: Non-polar & Polar ; Polar can be Protic or Aprotic)

Figure 2.2 ▶
Melting points of straight-chain alkanes. Alkanes with even numbers of carbon atoms fall on a melting-point curve that is higher than the melting-point curve for alkanes with odd numbers of carbon atoms.



Even # C atoms have higher mp's because the can BIND better to each other

"Molecular Recognition" for the TS^{++} is greater than that for the ground state

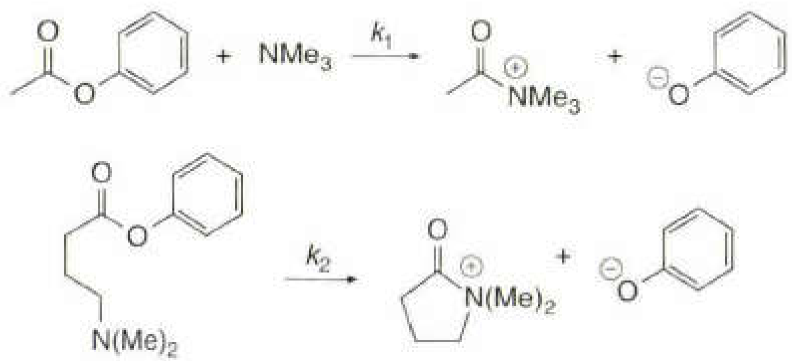
In other words:

The catalyst <u>HAS</u> to BIND the ground state (NECESSARY)

BUT

It HAS to BIND the TS^{++} BETTER! (SUFFICIENT)

Proximity as a Binding Phenomenon



Rate inter-molecular (k_1)= 1.3 x 10^{-4} $M^{-1}s^{-1}$ [SECOND ORDER]

Rate intra-molecular (k_2)= 0.17 s^{-1} = 1.7 x 10^{-1} s^{-1} [FIRST ORDER]

Rate enhancement for the intramolecular reaction ? ~1000 fold!

k_2 ~ k_1

Effective Molarity (E.M.):

Ratio of the first order(s^{-1}) to second order rate constants (M^{-1} s^{-1})

(*for the analogous reactions*)

For the previous case:

EFFECTIVE MOLARITY: k_2 / k_1 ; 0.17 s^{-1}/0.00013 M^{-1} s^{-1}~ 1308 M!!!!!!!!!!!!!

WHAT DOES THIS NUMBER EVEN MEAN

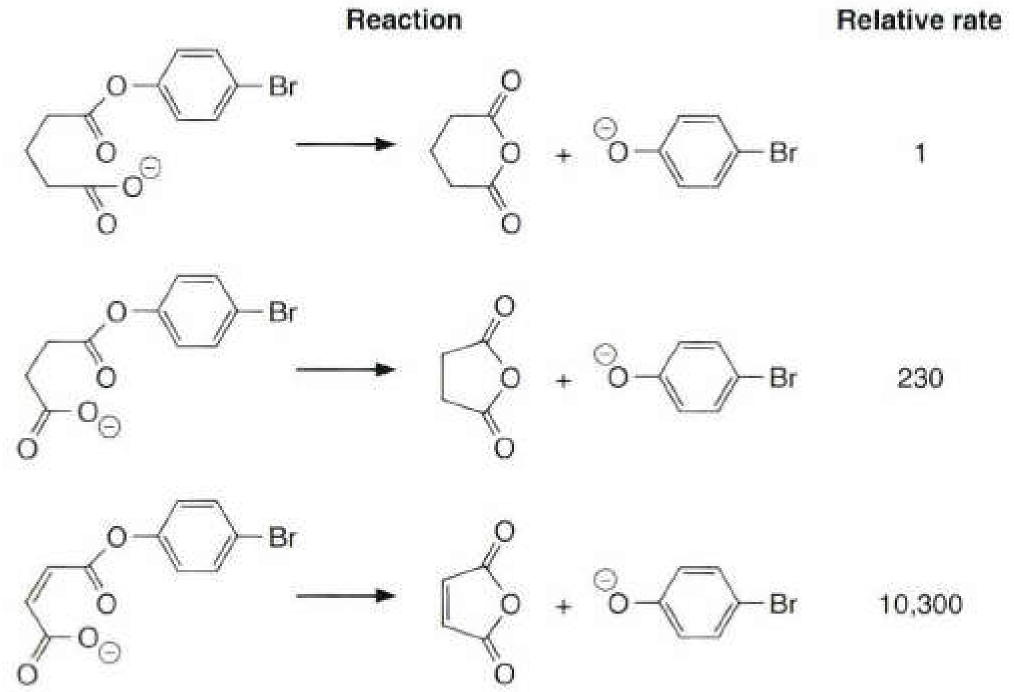
Recall:

Molarity of H_2O in H_2O? (PURE WATER) **55.56 M (1000 g L^{-1} / 18 g M^{-1})**

Also:

Rate $\propto$ [Concentration]

By making an inter-molecular reaction, intra-molecular [BINDING], observed RATE INCREASES CAN BE EXPLAINED AS BEING DUE TO A MASSIVE increase in the concentrations of the reactants.



Comparison of intra-molecular cases (Rotamer Freezing)

Case 1 *vs*. Case 2:

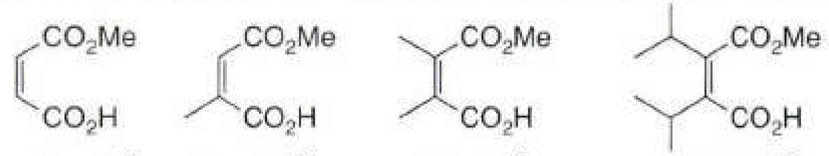
# of SINGLE BONDS between the two reacting centers DECREASES→Rate↑: INCREASE~ 10^2

Case 2 *vs*. Case 3:

As above

Effective Molarities (M) for a Variety of Cyclization Reactions*

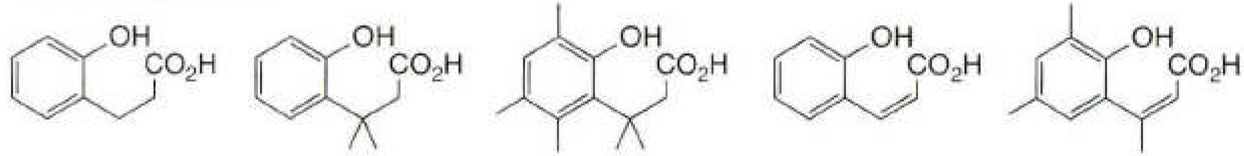
A.



3×10^{9} | 6×10^{10} | 4×10^{12} | 5×10^{12}

Anhydride formation

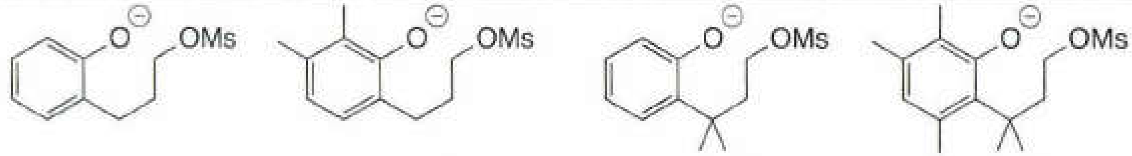
B.



4×10^{4} | 7×10^{6} | 2.3×10^{7} | 3.7×10^{11} | 2×10^{13}

Lactone formation

C.



6.6×10^{3} | 2.3×10^{4} | 2.0×10^{7} | 5.7×10^{9}

Ether formation

Acid & Base Catalysis:

Acids & Bases provide the *best known ways of speeding up reactions*.

[Make an ester—add some acid || Break an ester—add some base]

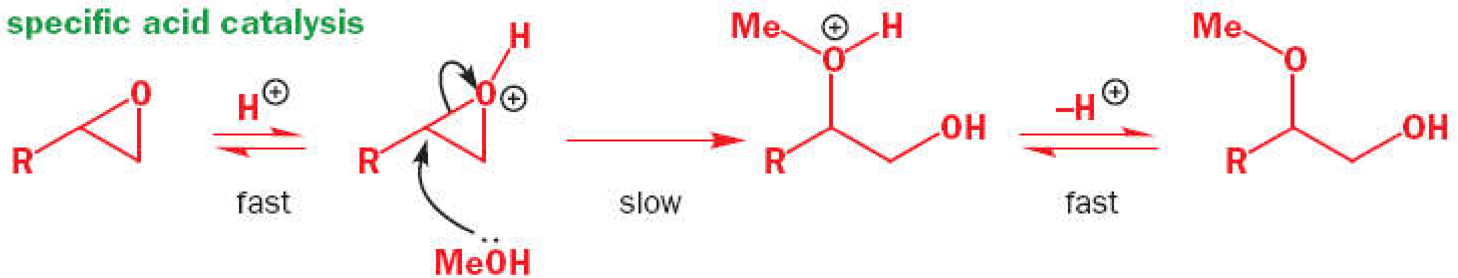
[Review ACID-BASE theories: Arrhenius, Bronsted, Lewis & Hard-Soft]

Actually TWO KINDS of acid catalysis (and two kinds of base catalysis)!

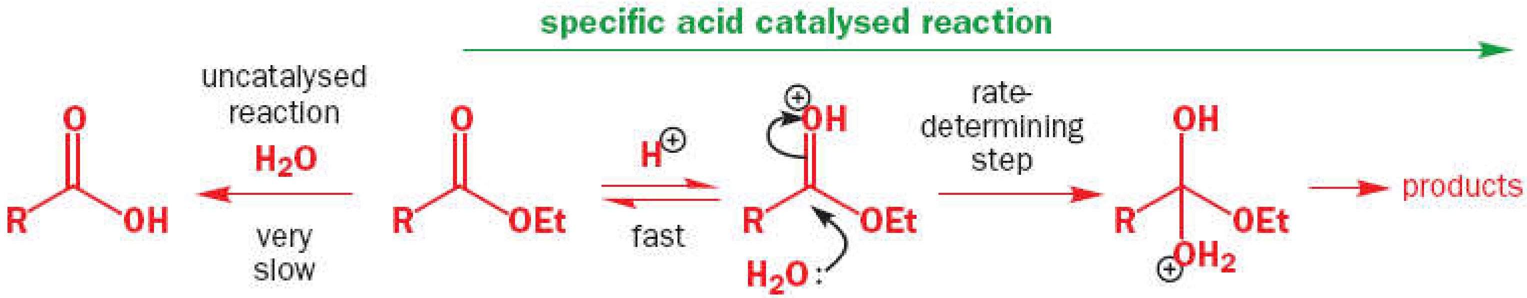
When we talk about acid (base) catalysis we normally mean:

Specific Acid Catalysis [SAC]

(and also Specific Base Catalysis [SBC].



Here, methanol attacks the neutral epoxide very slowly.
However, it attacks the protonated epoxide much more quickly.



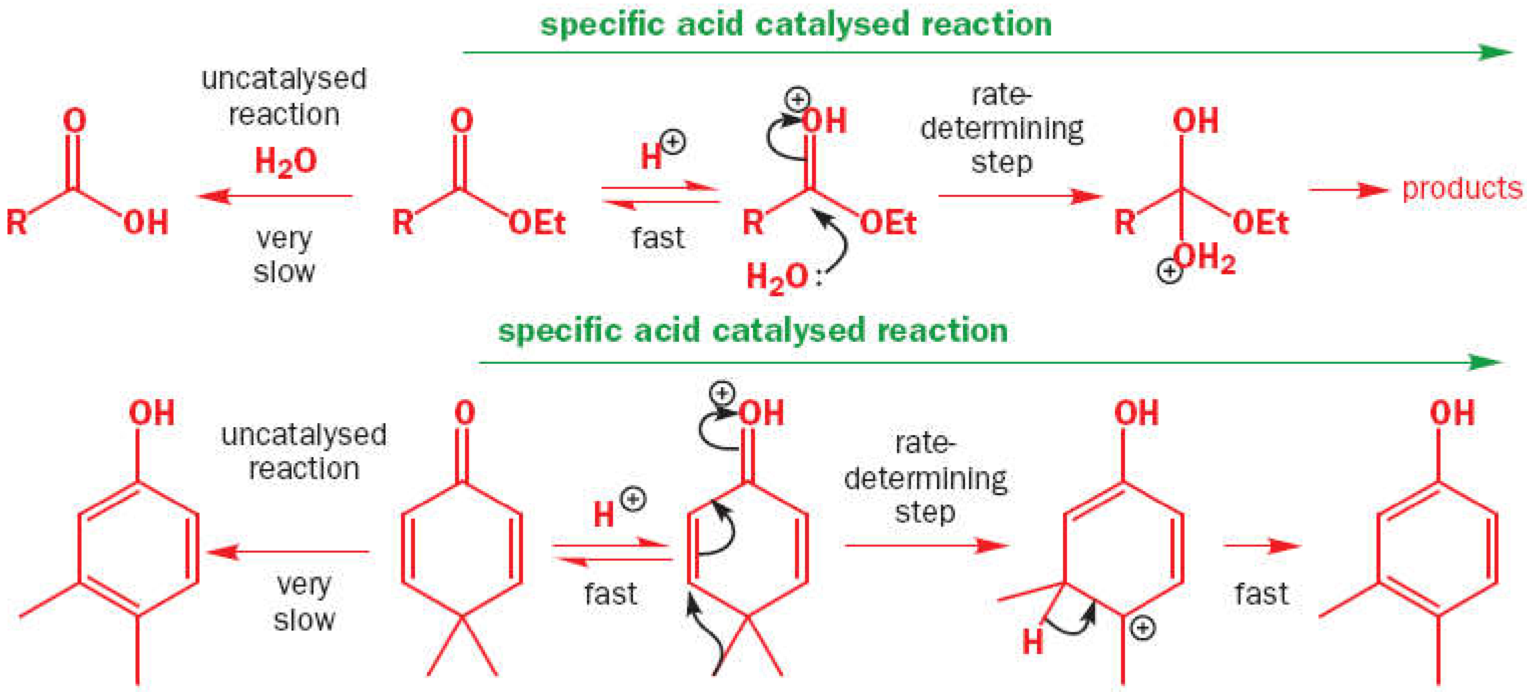
WHAT'S GOING ON?

Specific acid catalysis (SAC)

Involves a *rapid protonation* of the compound *followed by* the *slow step*
Now the slow step is accelerated in comparison with the uncatalysed reaction *because of the greater reactivity of the protonated compound*.

This catalysis depends *only* on the protonating power of the *solution*.

? SO, HOW HAS THE TS^{++} BEEN STABILISED HERE ?



More Generally:

Consider the acid-catalysed reaction in water of compound S to form product P, as shown Here, k_1 and k_{-1} are fast relative to k_2, and H^+ represents protonated solvent.

$$\text{(fast)} \quad S + H^+ \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} SH^+$$

$$\text{(rate-limiting)} \quad SH^+ \xrightarrow{k_2} P$$

$$\text{Rate} = \frac{d[P]}{dt} = k_2[SH^+]$$

$$K = \frac{[SH^+]}{[S][H^+]}$$

$$\frac{d[P]}{dt} = k_2K[S][H^+] = k_{SH^+}[S][H^+]$$

The concentrations of any acids that might ionize to produce H^+ do not appear in Rate equation because the proton has been fully transferred to the substrate *before the rate-limiting step of the reaction*.

The reaction rate depends only on the concentrations of S and H^+, so the reaction is said to be subject to specific acid catalysis.

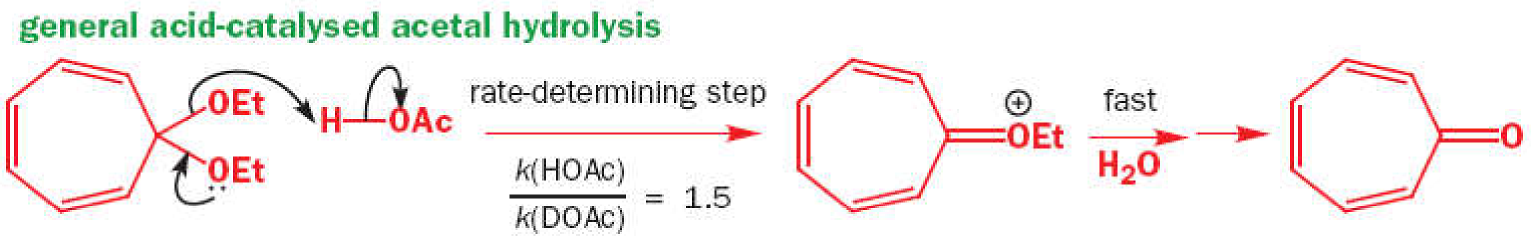
***Specific Acid Catalysis* means that:**
rate depends only on the concentration of protons in the solvent [pH].

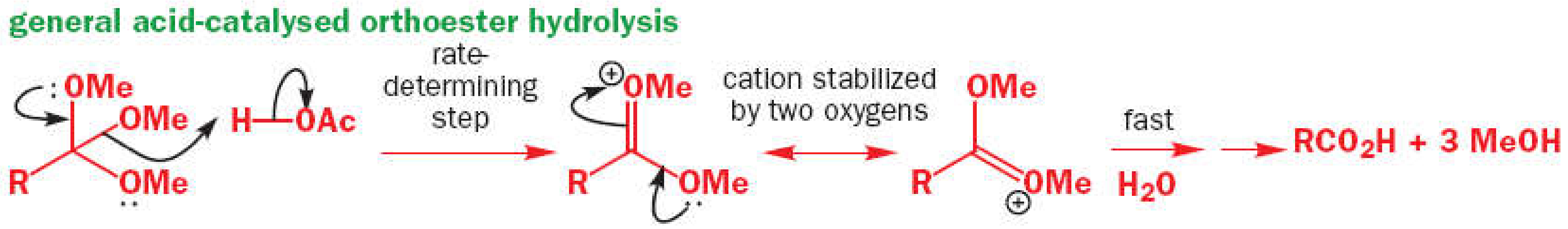
General Acid Catalysis [GAC]

Second kind of acid (base) catalysis is called 'general' rather than 'specific', abbreviated GAC (GBC).

As the name implies this kind of *catalysis depends not only on pH* but *also on the concentration of undissociated acids (bases)*.

The *proton transfer is not complete before the rate-determining step but occurs during it*. If partial / complete proton transfer occurs *during* the rate-limiting step of a reaction, then the reaction is subject to general acid catalysis.





(This is a *milder kind of catalysis* and *occurs in living things*.)

More Generally:

Two processes can give rise to general acid catalysis:
In the first, a proton is transferred from an acid, AH, to a substrate, S, *in the rate-limiting step*.

This transfer may occur as the rate-limiting step in a two-step reaction or the proton may be transferred during a one-step reaction leading directly to protonated product, PH^+.

$$\text{(rate-limiting)} \quad S + AH \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} SH^+ + A^-$$

$$\text{(fast)} \quad SH^+ \xrightarrow{k_2} P$$

$$\text{(rate-limiting)} \quad S + AH \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} PH^+ + A^-$$

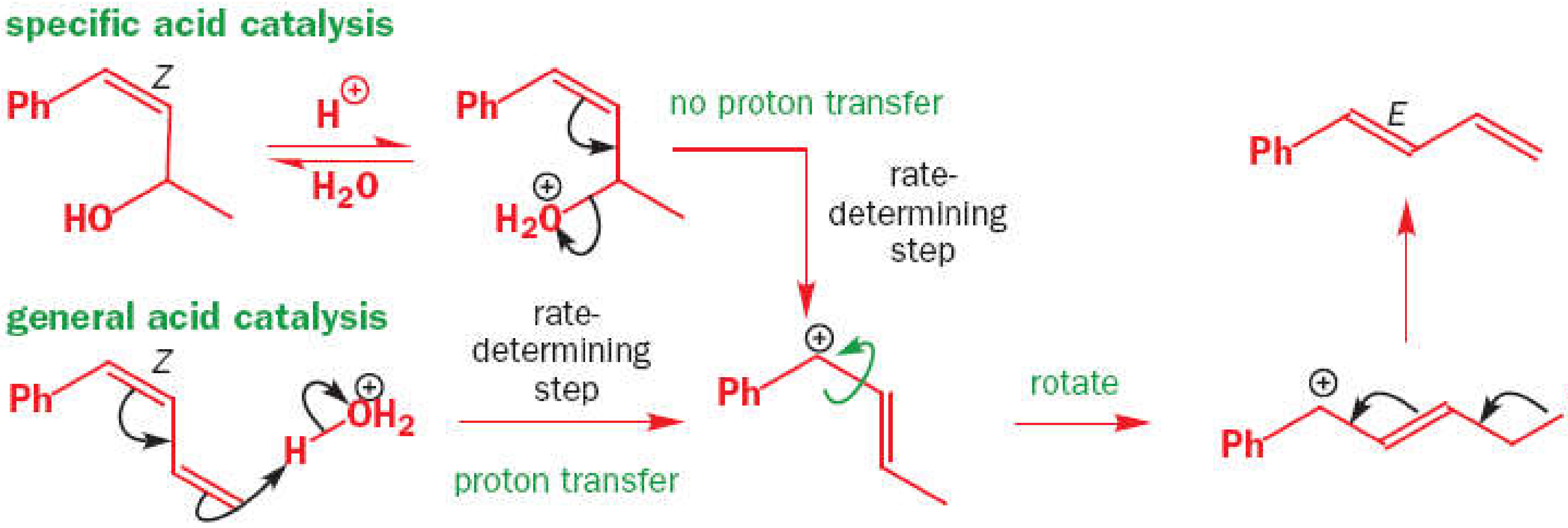
IN EITHER CASE, THE RATE LAW IS:

$$\frac{d[P]}{dt} = k_1[S][AH]$$

The reaction is said to be subject to *general* acid catalysis because *acids in general, not just H^+, catalyse the reaction*. If more than one acid is available to transfer protons, then the rate is a summation of the individual rates from the acid catalysis of all of the acids present

$$\frac{d[P]}{dt} = \sum_i k_1[S][A_iH]$$

SAC vs. GAC



Isomerization of the allylic alcohol is SAC—NO protonation in the slow step.

Isomerization of the diene is GAC—protonation at carbon is the slow step

RATE LAW SAC

$$\frac{d[\mathrm{P}]}{dt} = k_2\mathrm{K}[\mathrm{S}][\mathrm{H}^+] = k_{\mathrm{SH}^+}[\mathrm{S}][\mathrm{H}^+]$$

RATE LAW GAC

$$\frac{d[\mathrm{P}]}{dt} = k_1[\mathrm{S}][\mathrm{AH}]$$

(Strong electrolytes *vs.* Weak electrolytes)

Summary of features of specific acid catalysis

1. Only H_3O^+ is an effective catalyst; pH alone matters
2. Usually means rate-determining reaction of protonated species
3. Effective only at pHs near or below the pK_{aH} of the substrate
4. Proton transfer is not involved in the rate-determining step
5. Only simple unimolecular and bimolecular steps—moderate + or $-\Delta S^{\ddagger}$

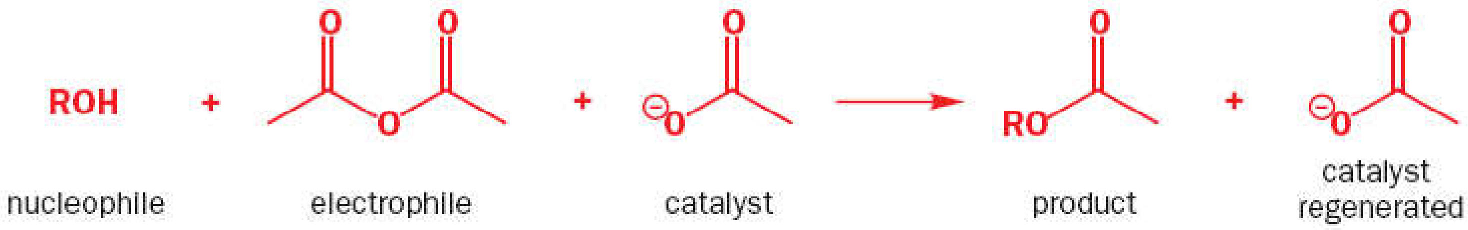
Summary of features of general acid catalysis

1. Any acid is an effective catalyst; pH also matters
2. Proton transfer is involved in the rate-determining step
3. Effective at neutral pHs even if above the pK_{aH} of the substrate
4. Catalyst often much too weak an acid to protonate reagent
5. Catalyst adds proton to a site that is becoming more basic in the rate-determining step
6. Some other bond-making or bond-breaking also involved unless proton is on carbon
7. Often termolecular rate-determining step: large $-\Delta S^{\ddagger}$

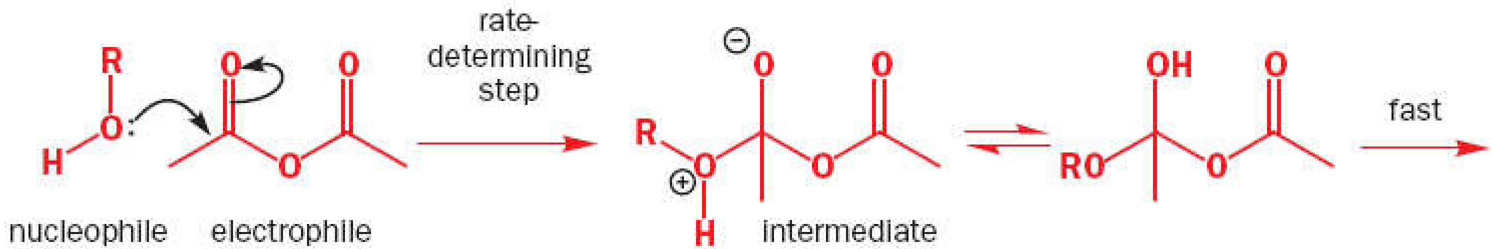
Specific Base Catalysis [SBC]

Removal of a proton from heteroatoms by heteroatom bases (B:)is always fast, but removal of a proton from carbon is slower, hence, rate-determining.

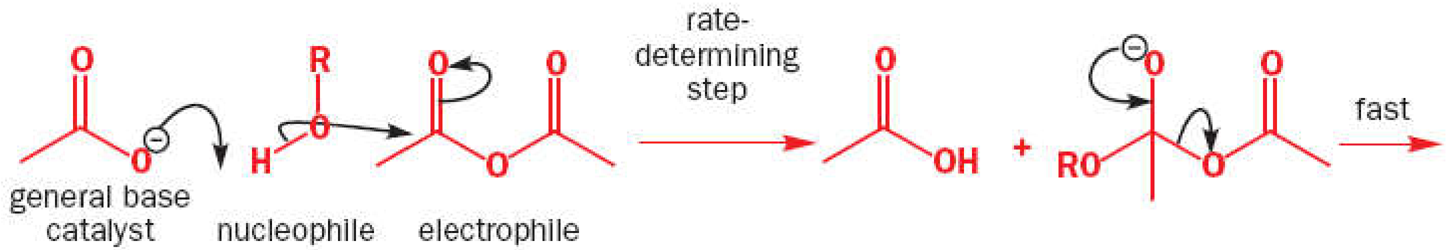
General Base Catalysis [GBC]



MECHANISM: UNCATALYSED



MECHANISM: CATALYSED [Note *rds* termolecular!]



Normally effective only if one of the three molecules is present in large excess [WHY?]
For example, this reaction might be done in ROH as a solvent, so that ROH is always present

(GBC is a *milder kind of catalysis* the kind that *occurs in living things*.)

Summary of features of specific base catalysis

1. Only HO^- is an effective catalyst; pH alone matters
2. Usually means rate-determining reaction of deprotonated species
3. Effective only at pHs near or above the pK_a of the substrate
4. Proton transfer is not involved in the rate-determining step, unless C–H bonds are involved
5. Only simple unimolecular and bimolecular steps—moderate + or $-\Delta S^\ddagger$

Summary of features of general base catalysis

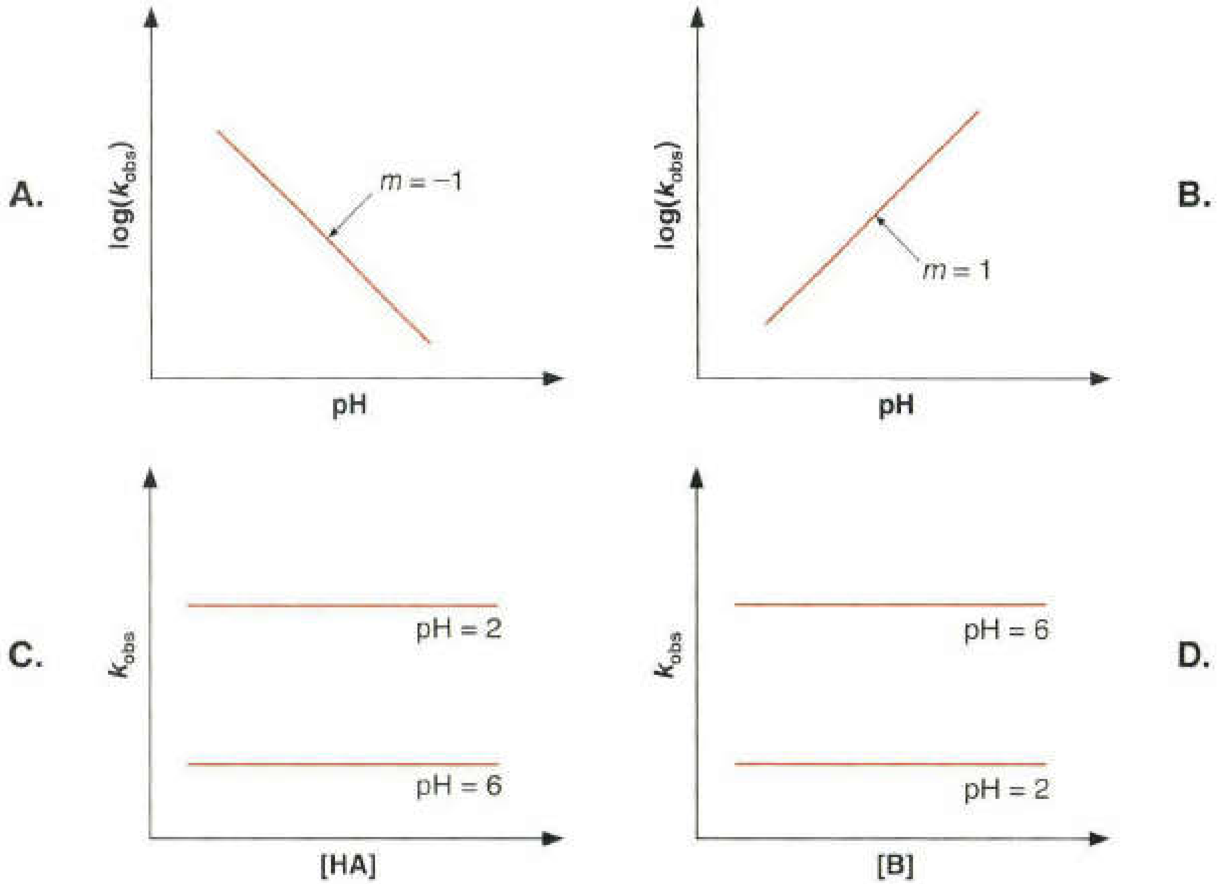
1. Any base is an effective catalyst; pH also matters
2. Proton transfer is involved in the rate-determining step
3. Effective at neutral pHs even if below the pK_a of the substrate
4. Catalyst often much too weak a base to deprotonate reagent
5. Catalyst removes proton, which is becoming more acidic in the rate-determining step
6. Some other bond-making or bond-breaking also involved unless proton is on carbon
7. Often termolecular rate-determining step: large $-\Delta S^\ddagger$

Catalytic rate expression for the reaction of iodine with acetone in buffer solutions was determined to be

$$k = k_{H_2O}[H_2O] + k_H[H^+] + k_{OH}[HO^-] + k_A[A^-] + k_{AH}[AH]$$

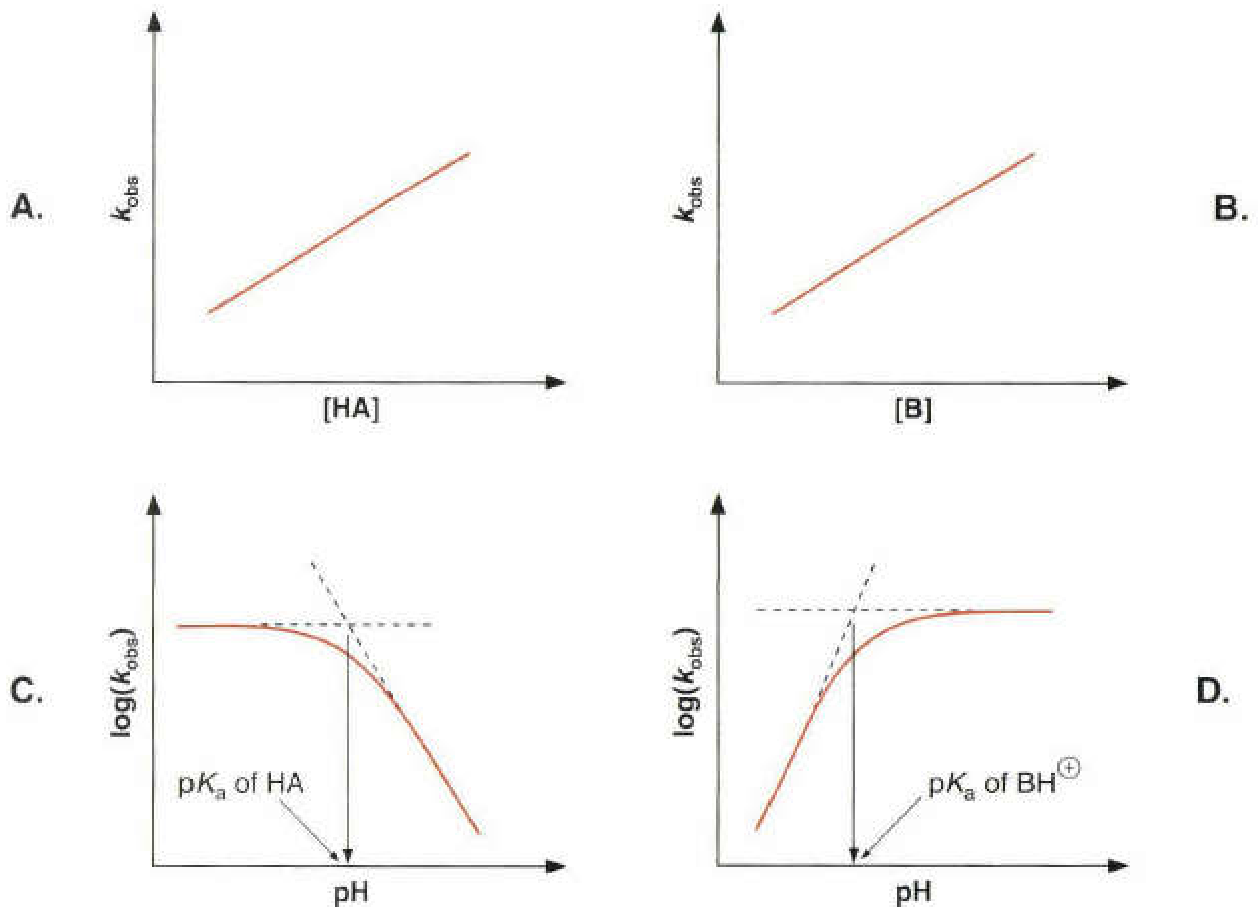
Where kH_2O is the rate constant for the uncatalysed or water-catalysed reaction, and [AH] and $[A^-]$ are the concentrations of the protonated and deprotonated form of the acid catalyst

Distinctive kinetic plots for SAC & SBC



A. The pH dependence of $\log(k_{obs})$ for a SAC reaction.
B. The pH dependence of $\log(k_{obs})$ for a SBC reaction.
C. The dependence of k_{obs} for a SAC reaction on the concentration of added acid HA at constant pH.
D. The dependence of k_{obs} for a SBC reaction on the concentration of added base B at constant pH.
(pH values 2 & 6 are just chosen examples and not indicative of any particular scenario)

Distinctive kinetic plots for GAC & GBC



A. Dependence of k_{obs}, for a GAC reaction on the concentration of an added acid HA at constant pH.
B. Dependence of k_{obs}, for a GBC reaction on the concentration of an added base B at constant pH.
(pH values 4 and 7 are chosen simply for example purposes)
C. The pH dependence of $\log(k_{obs})$ for a GAC reaction.
D. The pH dependence of $\log(k_{obs})$ for a GBC reaction.

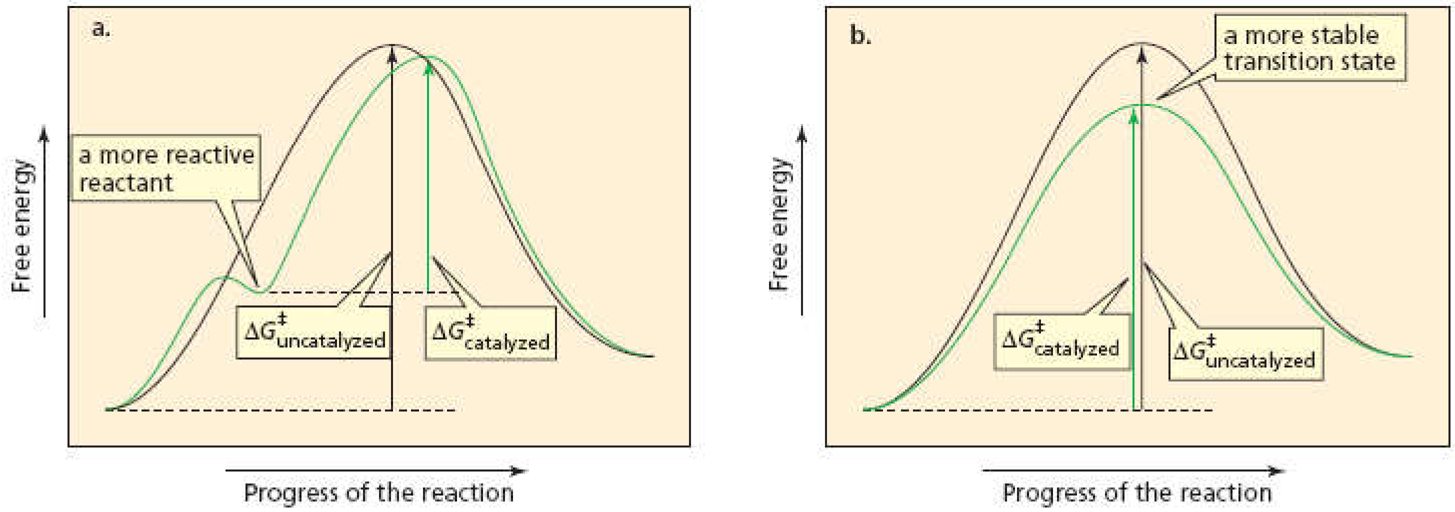


Figure 23.1

Reaction coordinate diagrams for an uncatalyzed reaction (black) and for a catalyzed reaction (green).

(a) The catalyst converts the reactant to a more reactive species.

(b) The catalyst stabilizes the transition state.

the *rate* at which the product is formed.

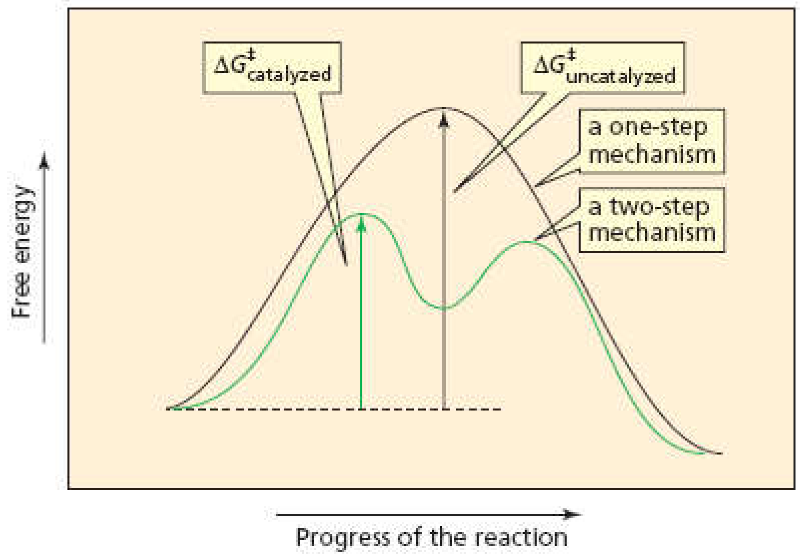
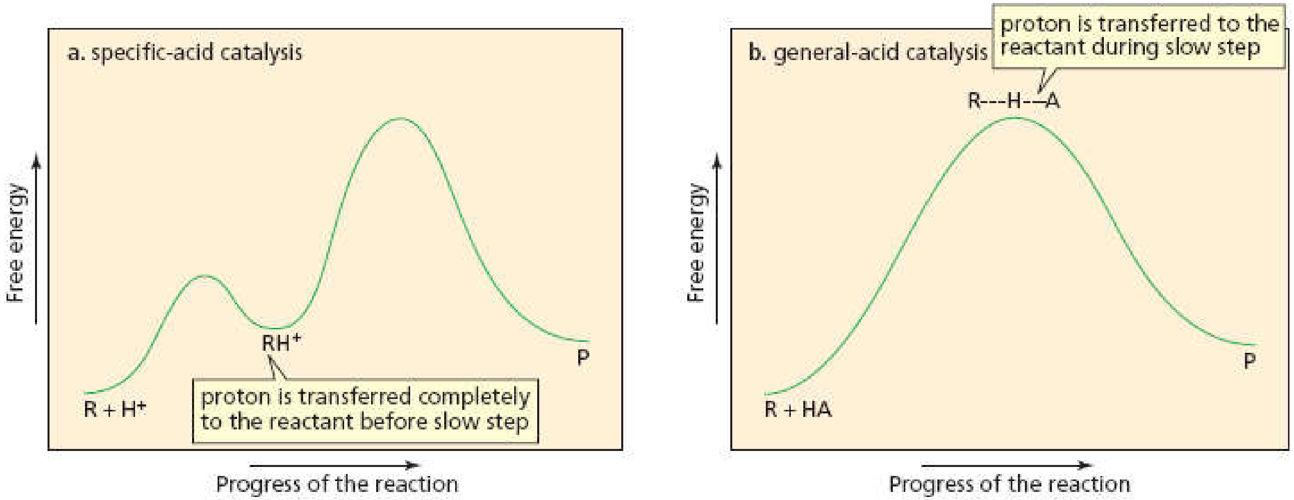


Figure 23.2

Reaction coordinate diagrams for an uncatalyzed reaction (black) and for a catalyzed reaction (green). The catalyzed reaction takes place by an alternative and energetically more favorable pathway.

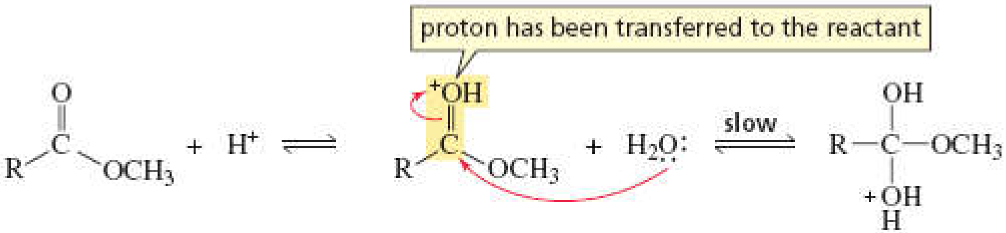


▲ **Figure 23.3**

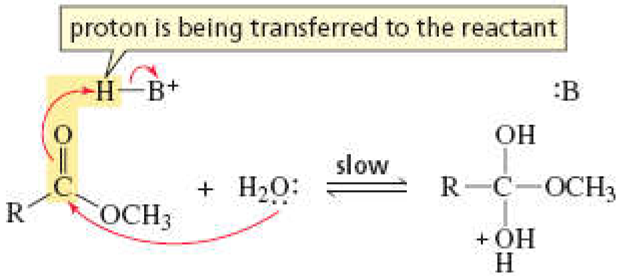
(a) Reaction coordinate diagram for a specific-acid-catalyzed reaction. The proton is transferred completely to the reactant before the slow step of the reaction begins (R = reactant; P = product).

(b) Reaction coordinate diagram for a general-acid-catalyzed reaction. The proton is transferred to the reactant during the slow step of the reaction.

specific-acid-catalyzed addition of water



general-acid-catalyzed addition of water



The proton is donated to the reactant *before* the slow step in a specific-acid-catalyzed reaction, and *during* the slow step in a general-acid-catalyzed reaction.

Electrophilic Catalysis

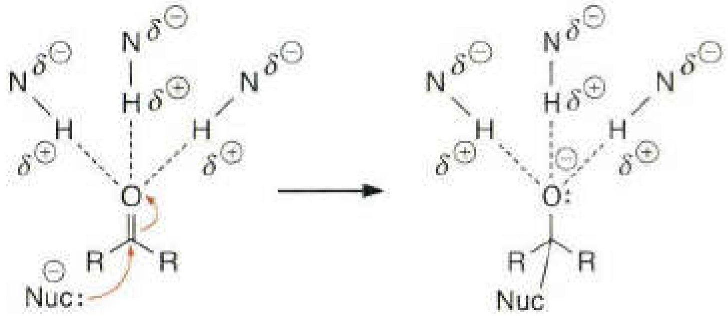
Catalysis due to "Proximal Charges" that can create electric fields which catalyse reactions.

Charges can be:
Full formal charges (Metal cations) OR Dipoles/ quadrupoles OR Hydrogen Bond donors / acceptors

This means that:

CATALYSIS OCCURS BECAUSE THE CATALYST ENSURES

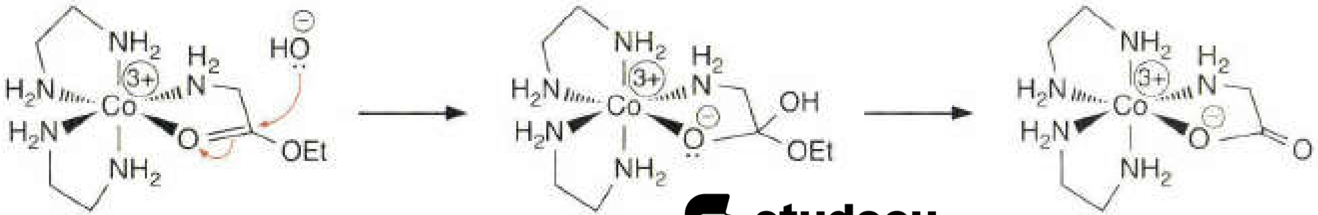
Electrostatic *stabilization of a charge that is developing along a reaction coordinate*
By dipoles / hydrogen bonds
Oriented in such a manner that best complements (solvates) the developing charge



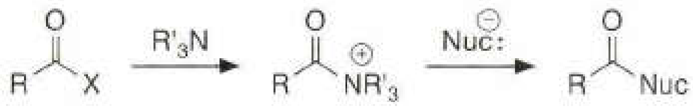
Such electrostatic interactions are much more important in organic solvents with lower dielectric constants than in water

Metal Ion Catalysis

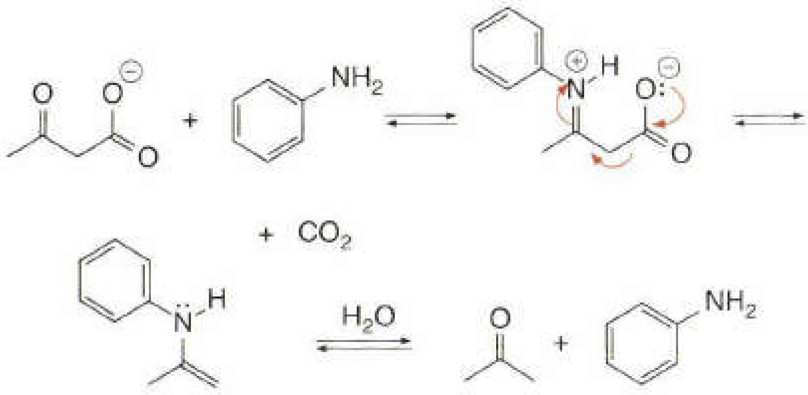
Metal coordination can polarize bonds, thereby enhancing their inherent reactivity.



Nucleophilic Catalysis:



Covalent Catalysis:



Some definitions

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Homogeneous catalysis
Reagents and catalyst are all in the same phase (typically all are in solution).

Heterogeneous catalysis ('surface catalysis')
Reagents are in a different phase from the catalyst - usually the reagents are gases (or liquids) and are passed over a solid catalyst (e.g. catalytic convertors in car exhausts).

Biocatalysis
Using enzymes to catalyse a reaction (see lecture 6).

Heterogeneous versus Homogeneous

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General features:

Heterogeneous	Homogeneous
Readily separated ✓	Difficult to separate ✗
Readily recycled / regenerated ✓	Difficult to recover ✗
Long-lived ✓	Short service life ✗
Cheap ✓	Expensive ✗
Lower rates (diffusion limited) ✗	Very high rates ✓
Sensitive to poisons ✗	Robust to poisons ✓
Lower selectivity ✗	Highly selective ✓
High energy process ✗	Mild conditions ✓
Poor mechanistic understanding ✗	Mechanisms often known ✓

Ultimate goal: to combine the fast rates and high selectivities of homogeneous catalysts with the ease of recovery /recycle of heterogeneous catalysts

Heterogeneous Catalysis

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Used in refining / bulk chemical syntheses much more than in fine chemicals and pharmaceuticals (which tend to use homogeneous catalysis).

Seven stages of surface catalysis:

1. **Diffusion** of the substrate(s) towards the surface.
2. **Physisorption** - i.e. physical absorption via weak interactions, e.g. van der Waals, adhering the substrate(s) to the surface.
3. **Chemisorption** - formation of chemical bonds between the surface and the substrate(s).
4. **Migration** of the bound substrate(s) to the active catalytic site - also known as **surface diffusion**.
5. **Reaction**.
6. **Desorption** of product(s) from the surface.
7. **Diffusion** away from the surface.

Heterogeneous Catalysts

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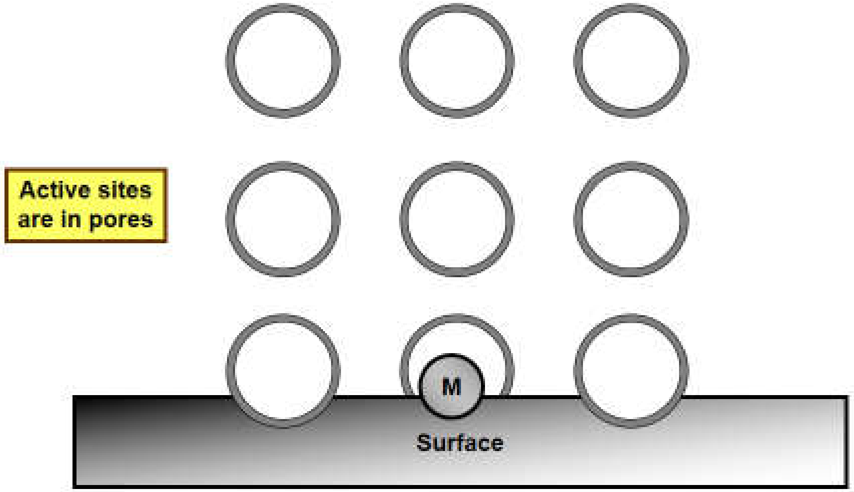

Heterogeneous Catalysts

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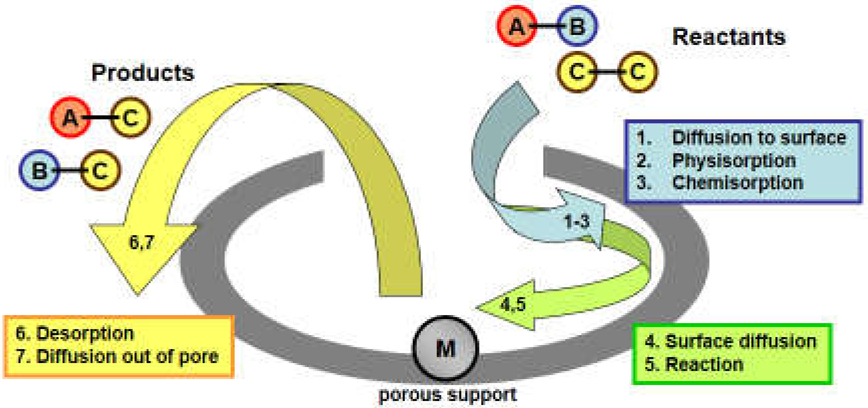
Typical features:

Metal or metal oxide impregnated onto a support (typically silica and / or alumina).

Three dimensional highly porous structure with very high surface area



Heterogeneous acid-base catalysis

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ca. 130 industrial process use solid acid-base catalysts

- Mainly found in bulk/ petrochemicals production e.g. dehydration, condensation, alkylation, esterification etc.
- Most are acid-catalysed processes.

ca. 180 different catalysts employed

- 74 of these are zeolites, ZSM-5 is the largest group.
- Second largest group are oxides of Al , Si , Ti , Zr.

ZEOLITES

***Zeolite*, from the Greek ζέω (*zéō*), meaning "to boil" and λίθος (*líthos*), meaning "stone"**

Microporous, aluminosilicate minerals commonly used as commercial adsorbents and catalysts

Occur naturally but are also produced industrially on a large scale

Aluminosilicate members of the family of microporous solids known as "molecular sieves."
Molecular sieve refers to a particular property of these materials:
The *ability to selectively sort molecules based primarily on a size exclusion process*.
This is due to a very *regular pore structure of molecular dime*nsions.
The maximum size of the molecular or ionic species that can enter the pores of a zeolite is controlled by the dimensions of the channels.

Industrially important zeolites are produced synthetically. Typical procedures entail heating aqueous solutions of alumina and silica with sodium hydroxide.

Synthetic zeolites hold some key advantages over their natural analogs. The synthetic materials are manufactured in a uniform, phase-pure state. It is also possible to produce zeolite structures that do not appear in nature. Zeolite A is a well-known example. Since the principal raw materials used to manufacture zeolites are silica and alumina, which are among the most abundant mineral components on earth, the potential to supply zeolites is virtually unlimited.

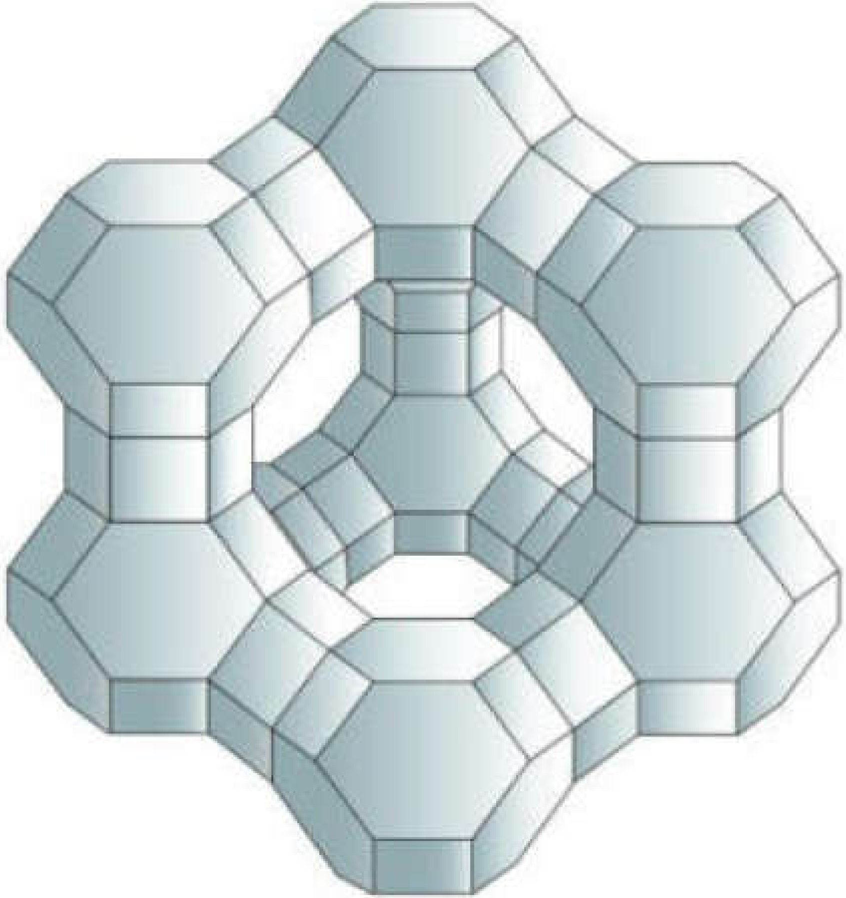
World's annual production of natural zeolite is about 3 million tonnes

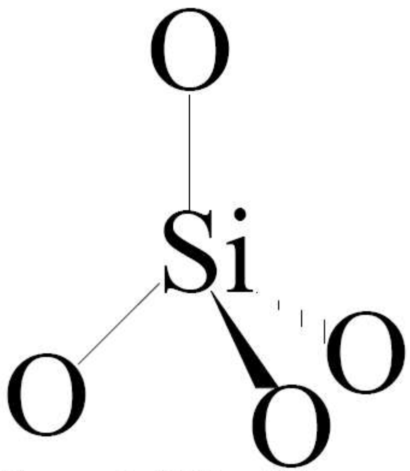
Used as ion-exchange beds in domestic and commercial water purification, softening, and other applications.

Used to separate molecules (only molecules of certain sizes and shapes can pass through), and as traps for molecules so they can be analyzed.

Widely used as catalysts and sorbents. Their well-defined pore structure and adjustable acidity make them highly active in a large variety of reactions.[9]

Synthetic zeolites are widely used as catalysts in the petrochemical industry, for instance in fluid catalytic cracking and hydrocracking. *Zeolites confine molecules in small spaces*, which causes changes in their structure and reactivity. The hydrogen form of zeolites (prepared by ion-exchange) are powerful solid-state acids, and can facilitate a host of acid-catalyzed reactions, such as isomerisation, alkylation, and cracking.





Tetrahedra made up of
EITHER

SiO_4

OR

AlO_4^- units

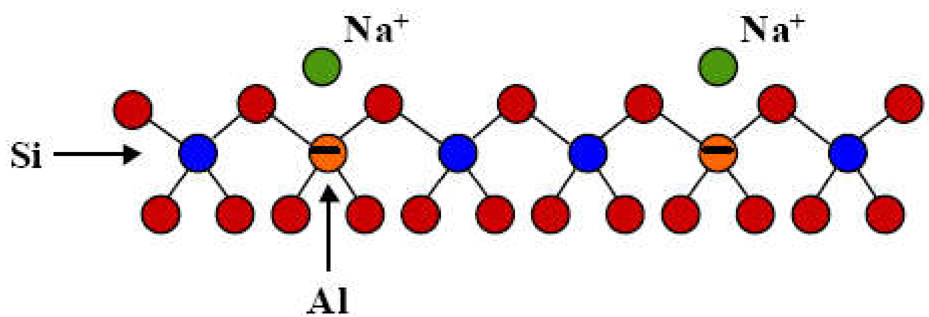
Every unit of AlO_4^- will have an associated cation in order to maintain charge balance, H^+, Li^+, Na^+, K^+. NH_4^+ etc.

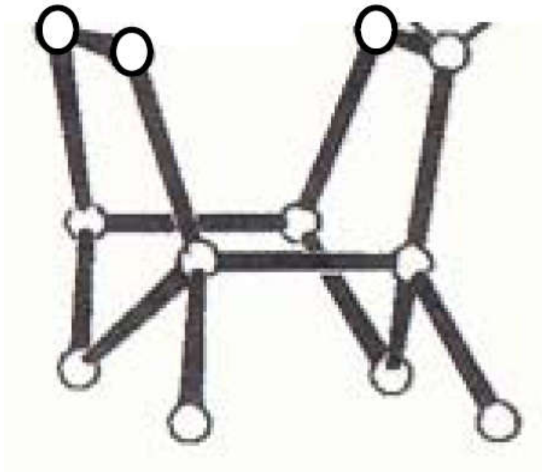
- General formula for the composition of a zeolite is

$$M_{x/n}[(AlO_2)_x(SiO_2)_y] \cdot mH_2O$$

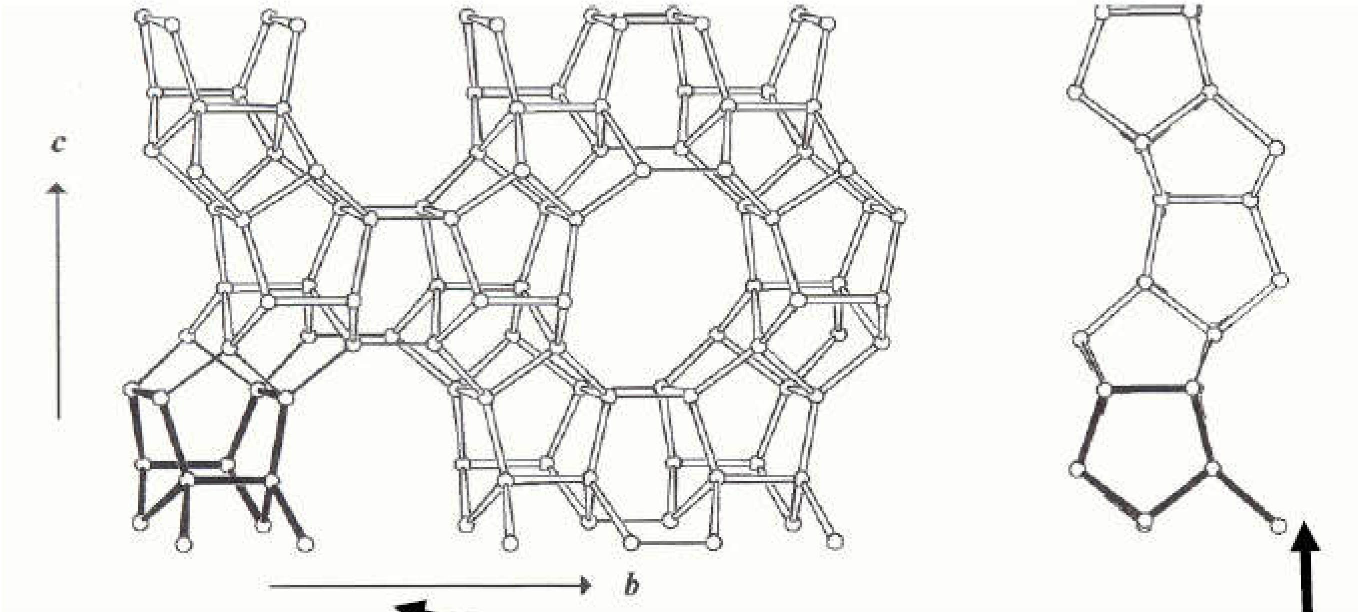
where cations M of valence n neutralize the negatively charged zeolite framework.

- SiO_2 tetrahedra are electrically neutral (*e.g.*, quartz) Substitution of Si(IV) by Al(III) creates an electrical imbalance and neutrality is provided by an exchangeable cation



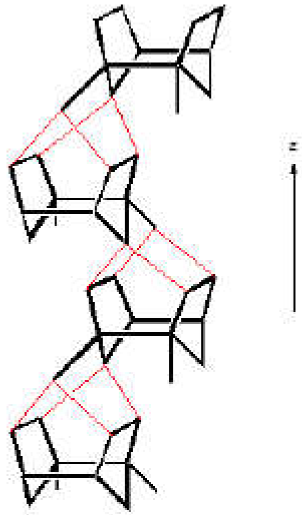


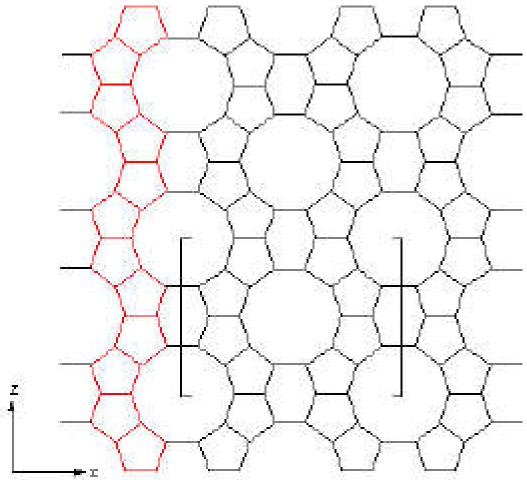
SBU of ZSM-5 zeolite

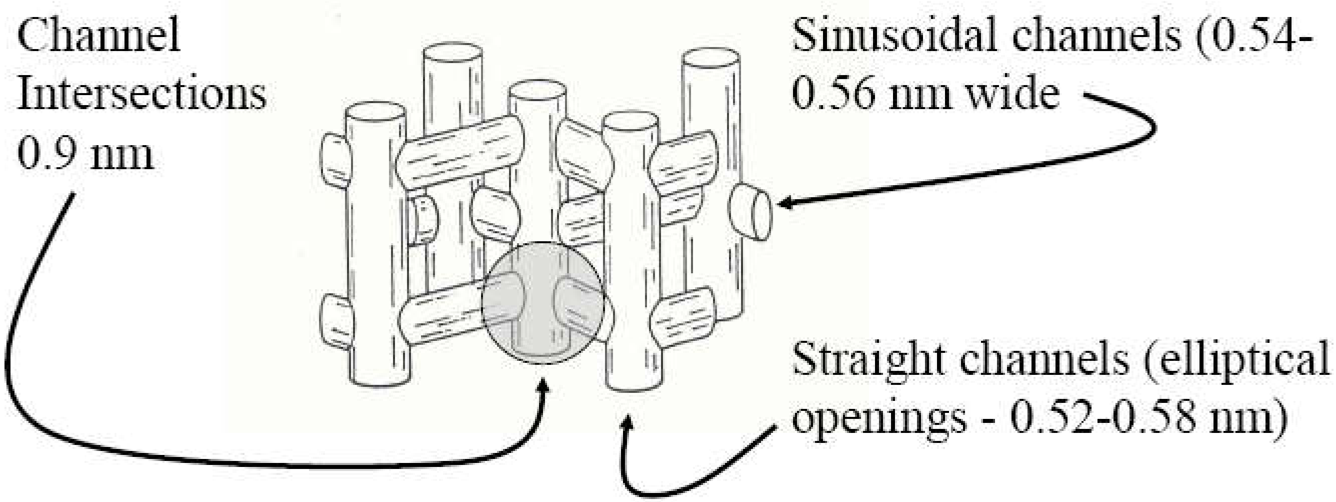


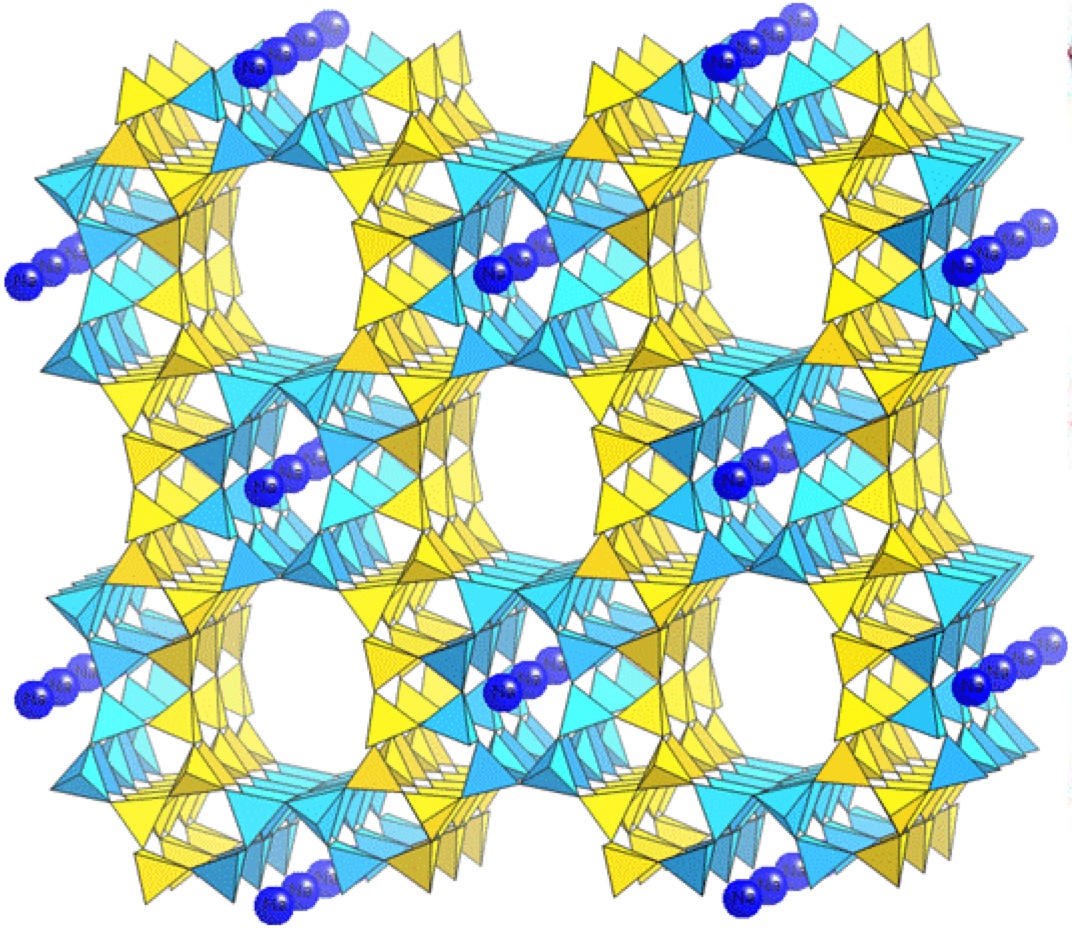
Combination of ZSM-5 SBUs shown along the a axis and

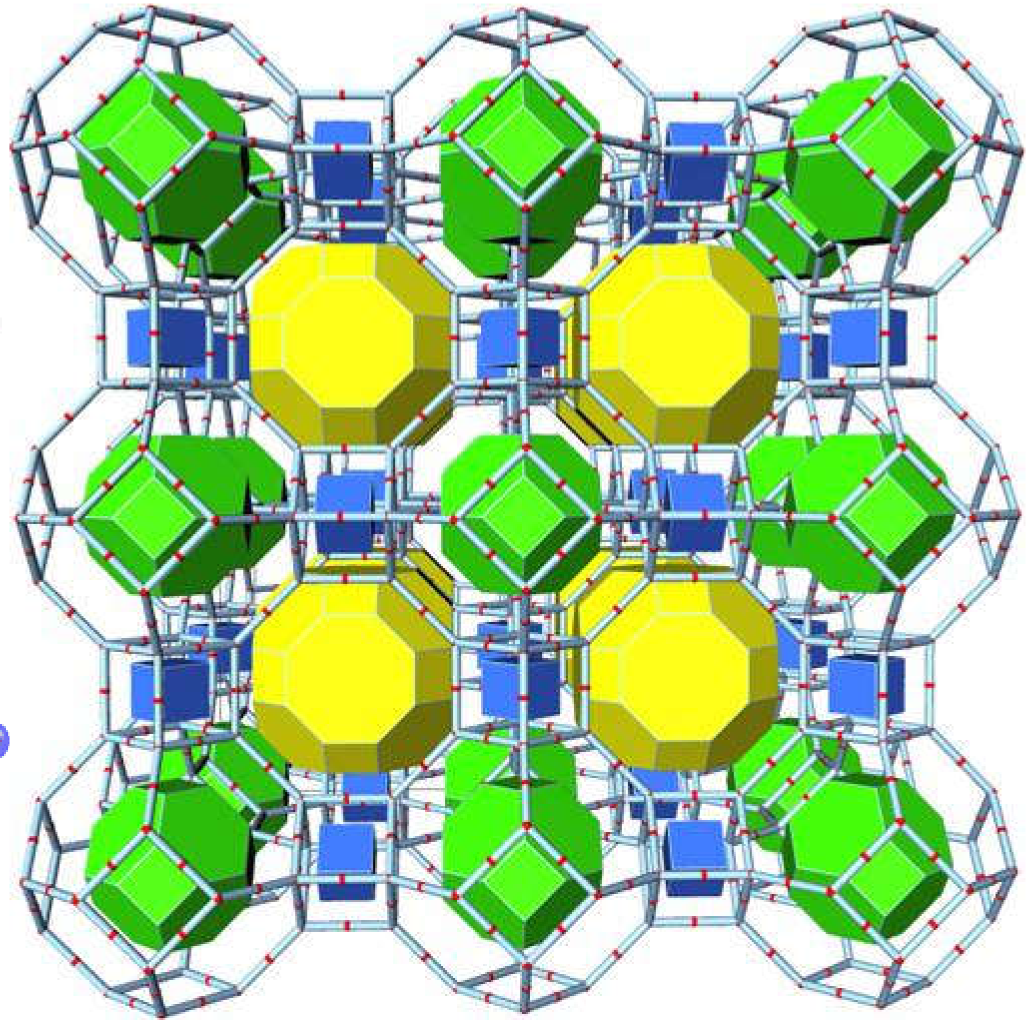
as a parallel projection along the b axis

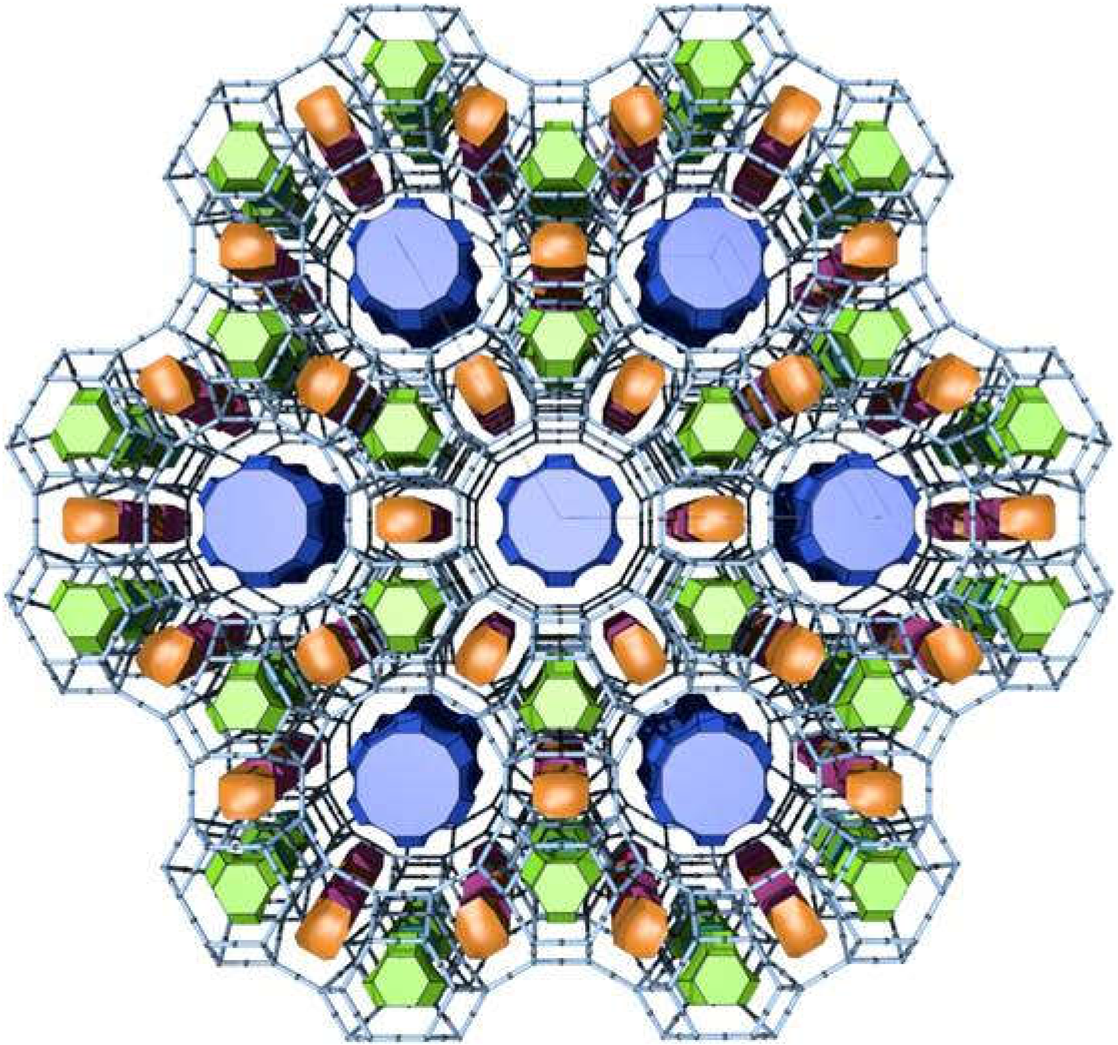


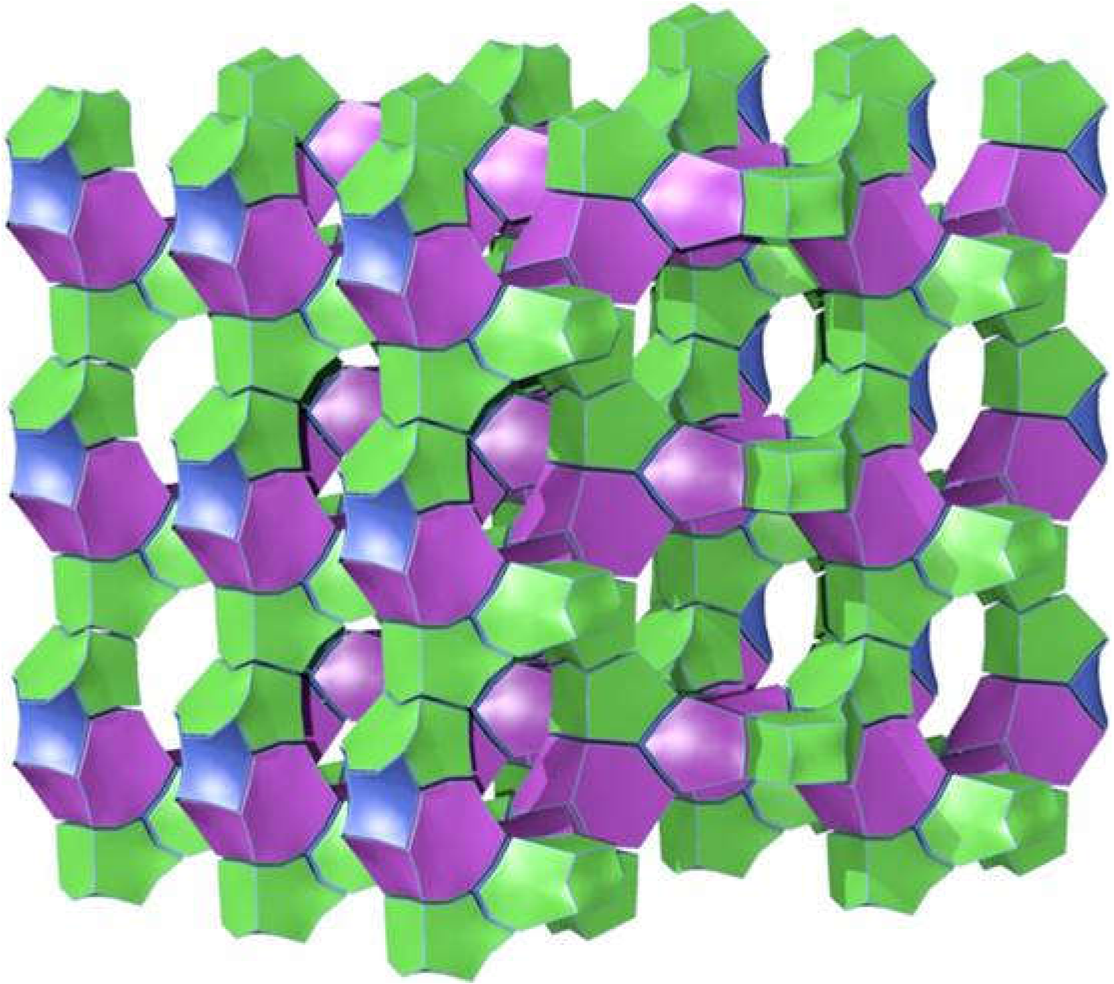


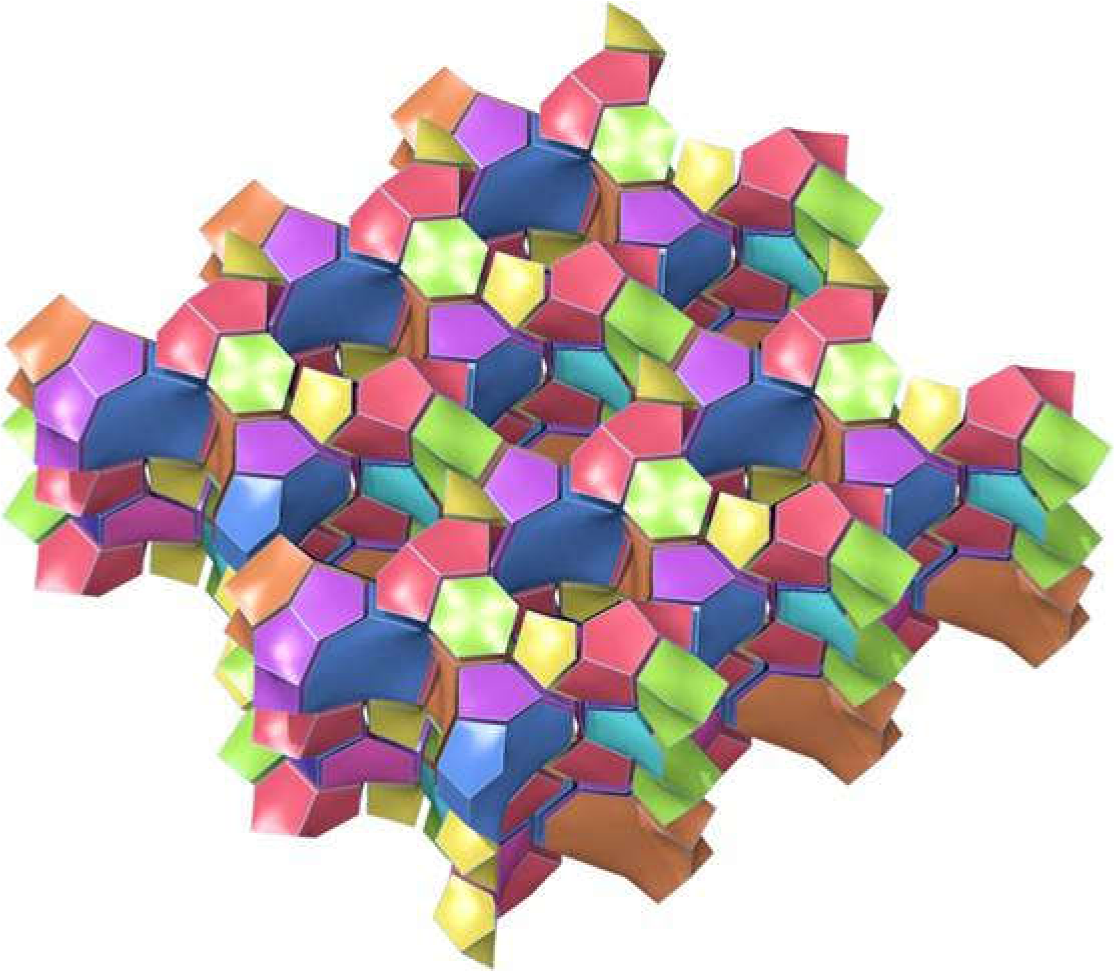

Channel
Intersections
0.9 nm
Sinusoidal channels (0.54-
0.56 nm wide
Straight channels (elliptical
openings - 0.52-0.58 nm)


Na



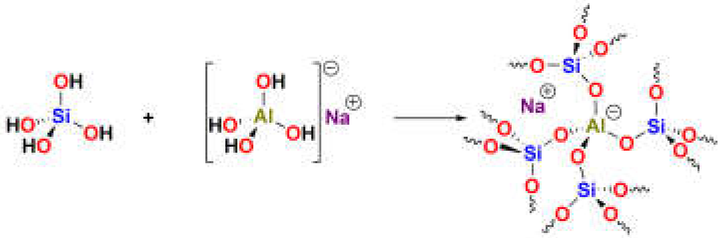






Zeolites - crystalline, hydrated aluminosilicates

Crystalline inorganic polymer comprising SiO_4 and AlO_4^- tetrahedra (formally derived from $Si(OH)_4$ and $Al(OH)_4^-$ with metal ions balancing the negative charge).

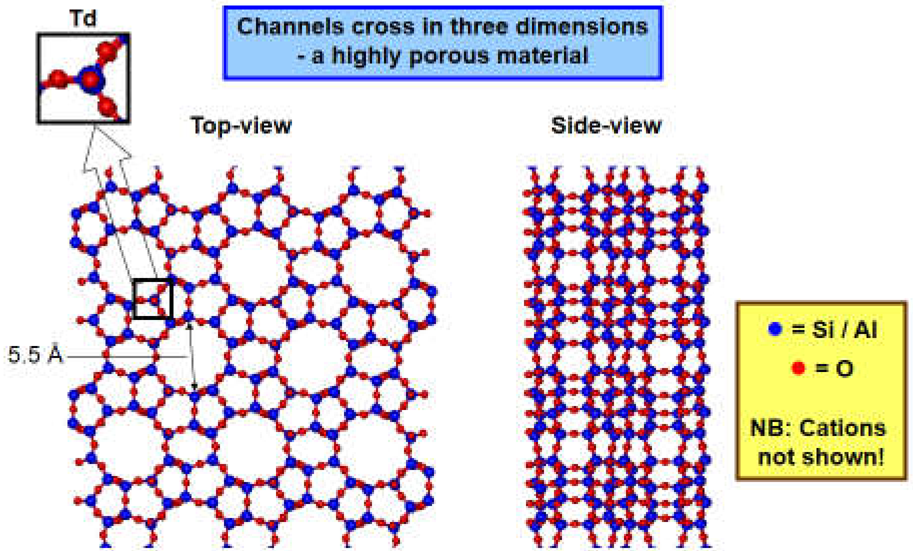


Lattice consists of interconnected cage-like structures featuring a mixture of pore (channel) sizes depending upon the Al : Si ratio, the counter-cation employed, the level of hydration, the synthetic conditions etc.

Hydrated nature of zeolites allows them to behave as Brønsted acids

e.g. ZSM-5

Channels cross in three dimensions - a highly porous material

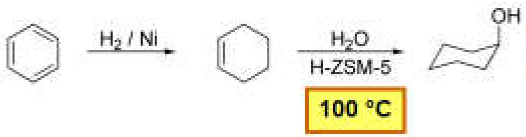


Zeolites - Asahi Cyclohexanol process

Traditional synthesis

For selectivity reasons, the reaction is run at low conversions (approx 6% per tank) and the hot cyclohexane stream is continuously recycled.

Zeolite catalysed process:

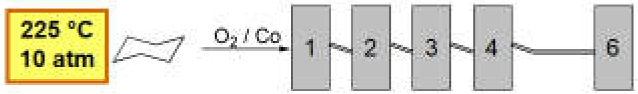


98 % selectivity

Why is the Asahi process important?

Imperial College London

Flixborough 1974 - 28 deaths



Tank 5 removed for repairs

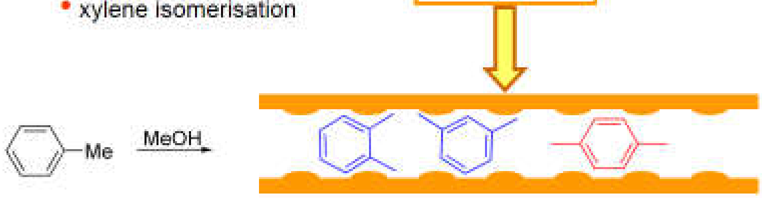
Tanks 1, 2 and 3

Temporary pipework between tanks 4 and 6 ruptured and cyclohexane cloud exploded

Zeolites - shape selective alkylation of toluene

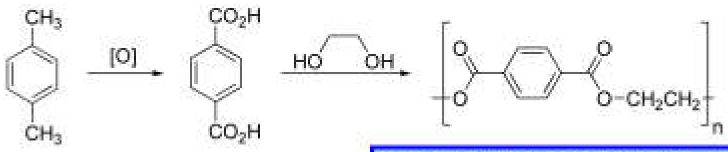
H-ZSM-5 catalyses:
- toluene alkylation
- xylene isomerisation

H-ZSM-5 (acidic ZSM-5)



Channel size only allows para-xylene to emerge

Only para-xylene is required for PET synthesis:



poly(ethylene terephthlate) - PET

Influence of Si/Al Ratio

Zeolites with a low [Al] are hydrophobic (and *vice versa*)

Lowensteins' rule, **Al-O-Al** linkages forbidden (Si/Al must be > or = 1)

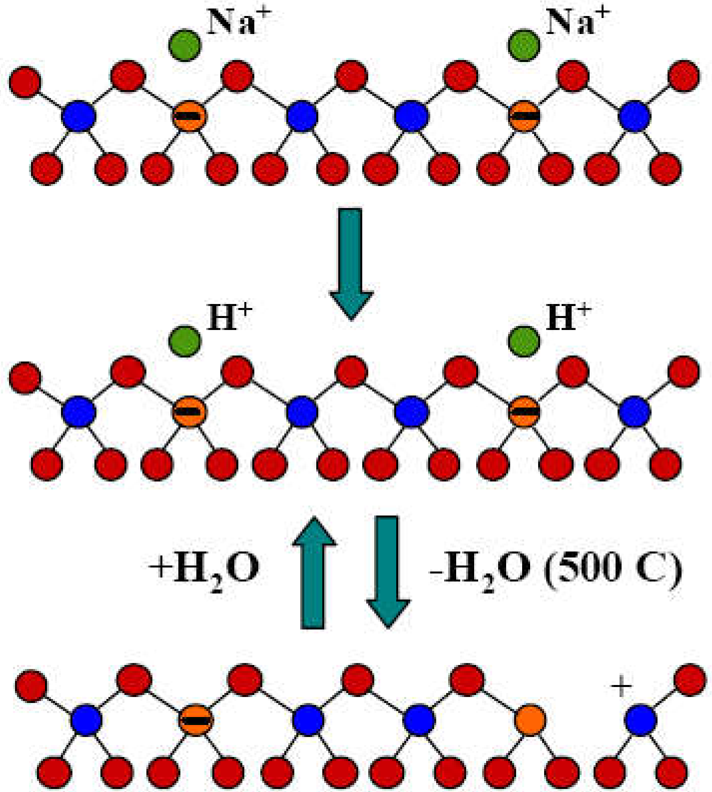
If the counter ion is a proton then this is hydrogen bonded to the lone pairs of the neighbouring Oxygen bridging atom generating Bronstead Acidity

High temperature treatment can de-hydroxylate the zeolite and generate a Lewis acid site (i.e. lone pair acceptor) on Al atoms

High concentrations of protons (from a low Si/Al) give a high acidity but lower concentrations of protons yield **STRONG** acid sites

Acid Sites

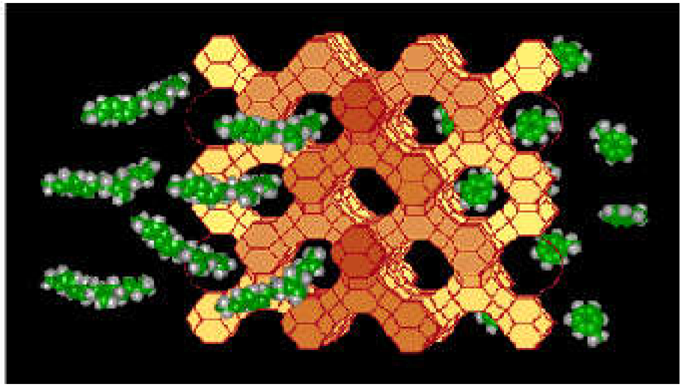
- Zeolite as synthesized
- Bronsted acid form
- Lewis acid form



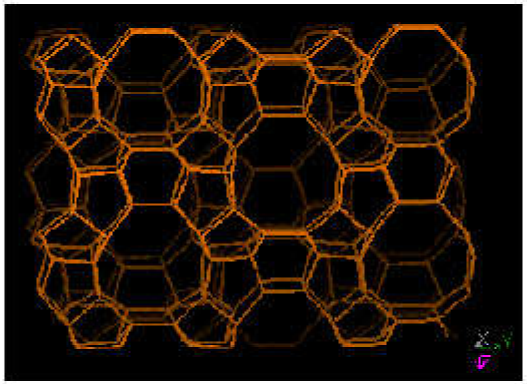
Properties that increase catalytic activity of ZEOLITES.

- molecular sieving (for shape selective catalysis)
- well defined active sites
- cationic exchange capacity,
- high surface area,
- variable acidity and controllable electrostatic fields (M^{2+} and M^{3+}),
- relatively good chemical and thermal stability.
- sites for occluded species – generate "internal" metal particles

Schematic illustration of straight-chain hydrocarbons diffusing with the channel system of a faujasite zeolite.

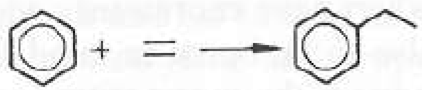
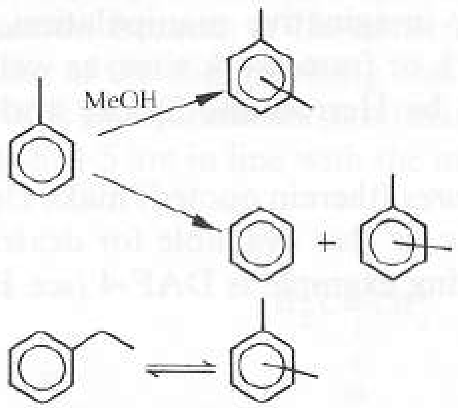
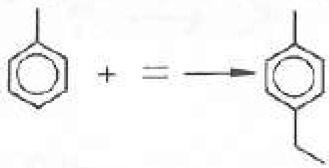


Zeolite ZSM-5, a member of the MFI structural family. This schematic shows the 10-membered ring straight channels normal to the page. The acid form of this catalyst is useful for methanol conversion, olefin oligomerization, and toluene alkylation.



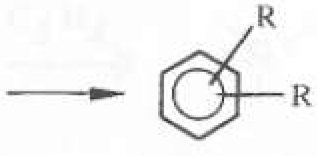
Examples of zeolites acting as selective catalysts in **ACID CATALYSED** reactions

Table 8.4. Acid-catalysed reactions giving hydrocarbon products.

Reaction	Zeolite catal
1 Ethylbenzene from benzene and ethylene 	ZSM-5
2 Xylenes production, isomerization (including ethylbenzene) and disproportionation 	Variously modified ZSM-5
3 *p*-Ethyltoluene from toluene plus ethylene 	ZSM-5

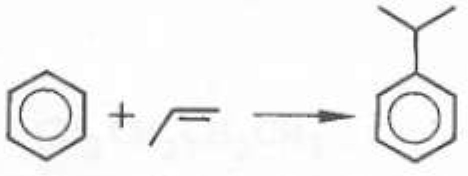
4 New routes to aromatics

paraffins,
olefins,
alcohols,
etc.



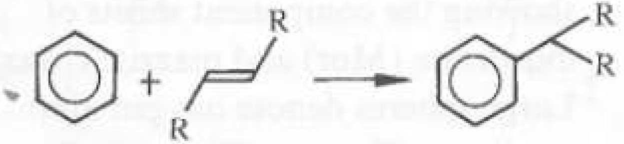
ZSM-5

5 Cumene from benzene plus propylene



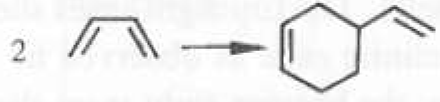
H-Y

6 Detergent alkylate from benzene plus C_6–C_{14} olefins



RE-Y or
H-morde

7 Vinylcyclohexene from butadiene



Cu-Y

Shape Selective Catalysis

(1) Reactant selectivity,
(2) Product selectivity, and
(3) Restricted transition-state selectivity

All these are examples of zeolites acting as selective catalysts in **ACID CATALYSED** reactions

Reactant Selectivity -

reactant molecules too large to enter cavities.

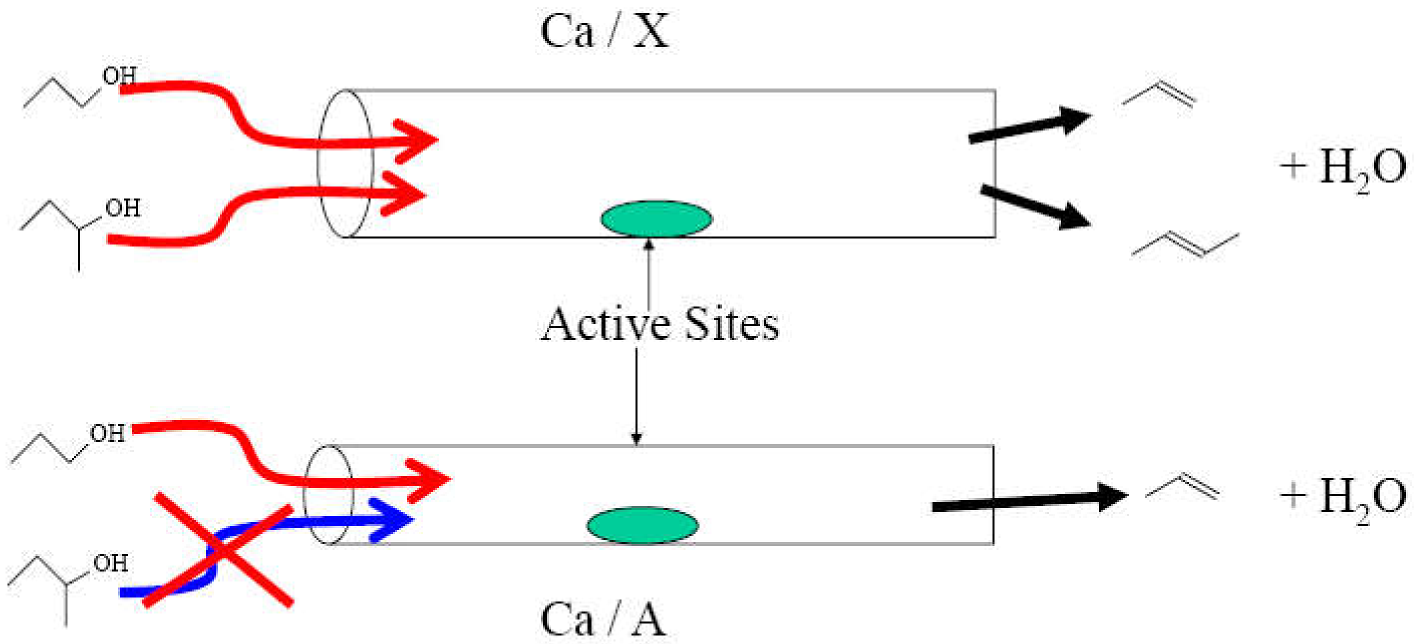
e.g. Ca / A and Ca / X as catalysts for R-OH $\rightarrow$ H_2O + alkene

1° and 2° alcohols dehydrate on Ca/X

only 1° alcohols dehydrate of Ca/A

(2° alcohols too large to get into the pores of zeolite A to the active Ca sites)

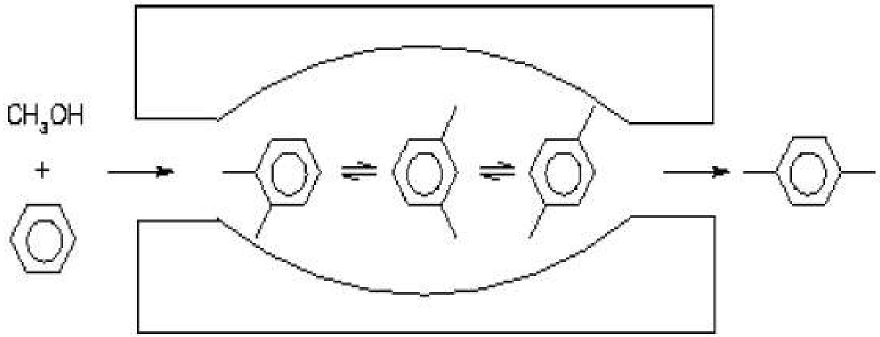
Reactant Shape Selectivity



Product Shape Selectivity;

benzene + methanol = xylene

Para-xylene is far more valuable than ortho or meta xylene - used in polyester manufacture

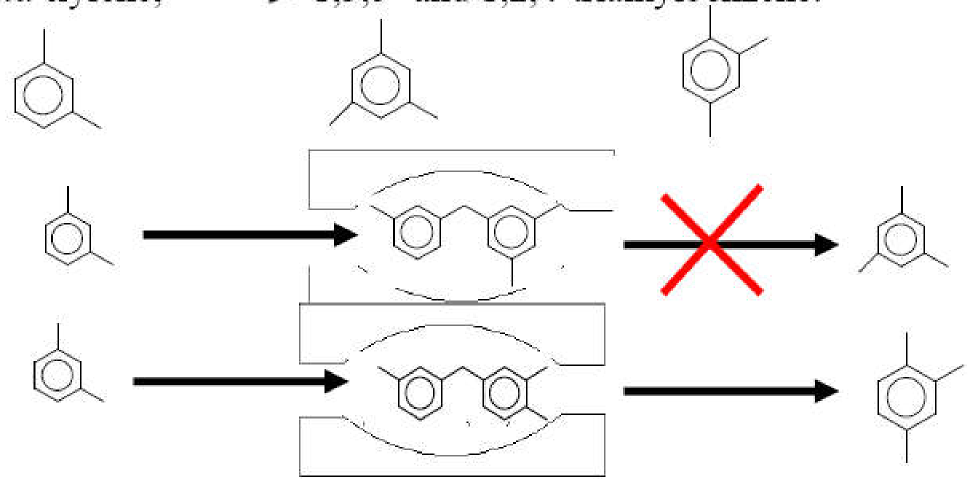


Only para xylene can diffuse out of the ZSM-5 channel pores

Transition State Shape Selectivity,
some transition-state intermediates are too large to be accommodated within the pores/cavities of the zeolites, even thoug diffusion of neither the reactants nor the products are restricted.

transalkylation of dialkylbenzenes

meta-xylene, → 1,3,5- and 1,2,4-trialkylbenzene.



ZSM-5 → Methanol → gasoline catalyst

ACTIVE Sites are zeolitic protons ACID catalysis

Two intersecting sets of channels.

Methanol diffuses in through one set of channels and gasoline diffuses out the second set, thereby avoiding "counter-diffusional" limitations in the reaction rate.

What about the "surface" of the zeolitic particle, i.e. the external surface ?

Also has active sites - but no "space" constraints.

DURENE (unwanted C_{10} aromatic) formed on these external sites during MTG. This has been combated by

making larger zeolite particles (proportionately less external acid sites) or

Selectively poisoning external acid sites with bases too large to enter pores, e.g. tri-methyl phosphine

Baku Mosque –
Azerbaijan (1086)

ZSM-5 (Zeolite Synthesised
by Mobil Corp (1974)

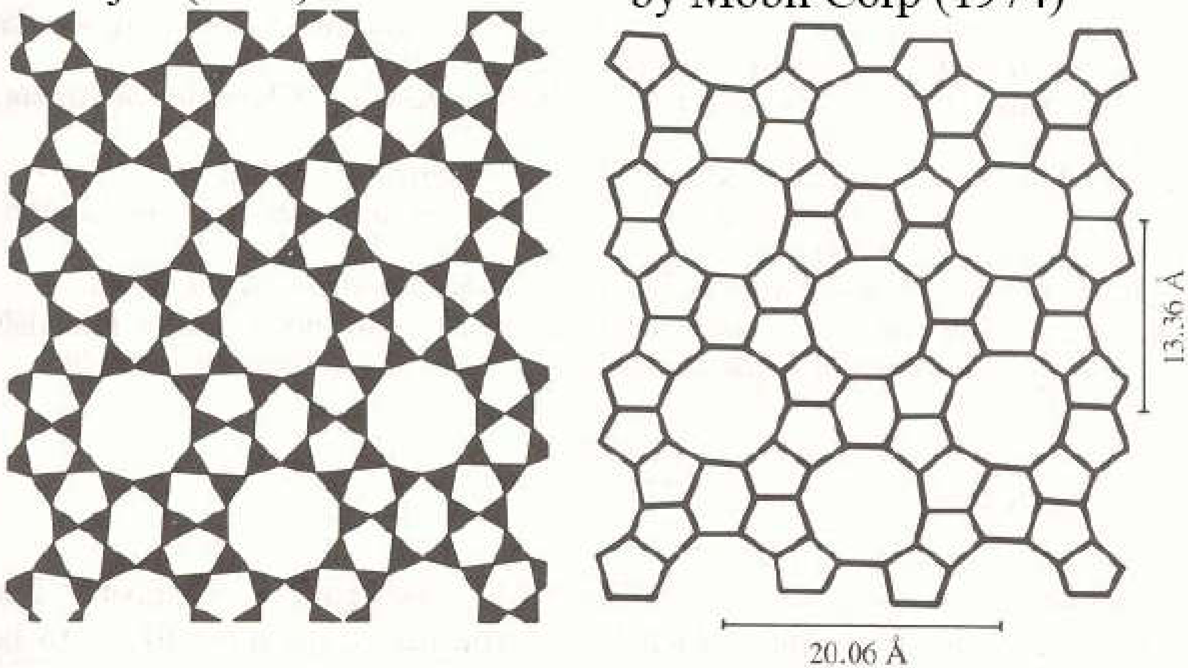


Figure 7.24 Around a central 10-memebered ring in the synthetic zeolite catalyst shown schematically on the right, there are eight five-membered and two six-membered rings. In exactly the same sequence, such five- and six-membered rings circumscribe a central 10-membered ring in the pattern on the left.